Spike-and-Slab Transfer Learning via Augmented Variables for Time-to-Event Analysis

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Abstract

High dimensional predictors, small samples, and censoring make survival prediction and variable selection challenging in many modern biomedical studies. Transfer learning (TL) has recently gained popularity as a technique that leverages related source datasets to improve inference in the target data. We propose spike-and-slab transfer learning (SSTL), a Bayesian TL method for variable selection and prediction in survival analysis via accelerated failure time models. We construct heavy-tailed discrete spike-and-slab (DSS) priors via novel latent Gaussian representations, including a new class of non-local priors. Leveraging this construction, we develop a tuning-free, scalable Monte Carlo algorithm for posterior inference based on slice sampling. To overcome source data heterogeneity, an Empirical Bayes approach implemented within the posterior sampler adaptively tunes the strength of information transfer from each source. We establish asymptotic properties and show that SSTL achieves a faster convergence rate and detects smaller signals than single-study methods when there are informative sources. SSTL is also robust to negative transfer and recovers the single-study minimax rates when no source is informative. Across several simulations and two cancer application studies, SSTL consistently outperforms existing approaches and remains robust when sources are uninformative.

Keywords: Accelerated Failure Time models, Bayesian Transfer Learning, Empirical Bayes, High-dimensional Data, Spike-and-slab Prior

1 Introduction

Many modern biomedical data science problems involve massive datasets, consisting of high-dimensional measurements collected on a small set of observations. For instance, oncology studies typically collect thousands of variables in molecular profiles, tumor scan summaries, and patient health records from a relatively small number of patients. With such small sample sizes, models fit to a single study are typically unstable, producing unreliable predictions and variable selection. This can lead to poor scientific decision-making, especially in rare disease studies. Such underpowered analyses are particularly prevalent in survival analysis, where structural missingness of outcome measurements due to censoring further reduces the available information in the data.

Recent efforts in reproducibility and data-sharing have led to the creation of several open-source data platforms [35, 52, 18, 51, 43, 48]. These resources have enabled researchers to access supplemental source datasets from populations that are related, though not necessarily identical, to the target population. Meta-analysis and hierarchical modeling [4, 14] are standard approaches for combining information across studies, either by constructing population-averaged estimators or by shrinking study-specific estimates towards a global population average. However, in many real-world applications, source studies can be highly heterogeneous and non-exchangeable, resulting in poor performance of such estimation approaches in the target study. Furthermore, both approaches treat all studies symmetrically and neither is designed to prioritize inference for the target study.

Transfer learning (TL) has recently emerged as a powerful tool that utilizes source data to directly optimize inference in the target population. The estimation and prediction accuracy of TL can improve substantially over single-study approaches when the outcome distributions of the source and target datasets align well. However, when cross-study het-

erogeneity exists between source and target outcome models, borrowing from all available source datasets can cause the TL model to perform worse than a target-only analysis, a phenomenon known as *negative transfer*.

Statistical TL methods frequently use variations of a two-step approach [36, 55, 37] that (i) obtain a pooled estimate and (ii) de-bias it towards the target population. However, this approach has limited ability to handle heterogeneous source datasets without additional source selection. Moreover, uncertainty quantification relies on large-sample procedures that may perform poorly in small, finite-sample target datasets and have no efficiency advantage over single-study approaches. Extensions of this paradigm to survival outcomes [38, 39, 46] inherit the first limitation and provide no uncertainty quantification. Recently, alternative TL approaches have been proposed to address these limitations in linear and generalized linear model settings [62, 23, 20, 26, 61]. However, none of them addresses all of the above shortcomings or is able to handle more general data such as censored outcomes.

We propose spike-and-slab transfer learning (SSTL), a Bayesian TL model for high-dimensional survival data. SSTL anchors the survival model on the target study with a discrete spike-and-slab (DSS) prior on the covariate effects on survival outcomes. A conditional prior is then specified for source- and covariate-specific bias parameters, so that each source study model is allowed to deviate partially or fully from the target outcome model. Our SSTL design allows information transfer to be determined individually for each predictor in each source study, and provides full posterior uncertainty quantification for both estimation and variable selection.

DSS priors achieve the optimal posterior contraction rate and model selection consistency in (generalized) linear models when the slab component has heavy tails [9, 8, 28, 27]. However, existing posterior inference relies on Gaussian conjugacy or model-specific like-

lihood augmentation, which does not scale to high-dimensional multi-study models. Motivated by this limitation, we introduce an augmented latent Gaussian representation of SSTL inspired by the neuronized priors [53] that enables scalable full posterior inference for non-Gaussian outcome models with structural missingness (i.e., censoring) via an efficient tuning-free *slice-within-Gibbs* algorithm. This representation allows us to design several novel DSS priors with desirable near-zero and tail properties for SSTL, including a new family of computationally efficient non-local slab distributions [29] that empirically performs the strongest in prediction and variable selection in our evaluations.

We establish minimax posterior contraction rates for all proposed priors and further show that SSTL achieves a faster convergence rate and detects smaller signals than single-study methods when at least one source is informative. The results also show that when no sources are informative, SSTL recovers the classical single-study contraction rate, thereby avoiding negative transfer. To address source-to-target heterogeneity in practice, we develop a data-driven empirical Bayes algorithm to adaptively calibrate the strength of information transfer from each source study. Using several simulations and two real data applications, we show that SSTL consistently outperforms alternative TL methods.

The rest of the paper is organized as follows. Section 2 introduces our setup and the SSTL model. We provide theoretical results for our SSTL model in high-dimensional settings in Section 3. Sections 4 and 5 describe adaptive source selection and posterior computation, respectively. Simulation studies and two real-data applications are provided in Sections 6 and 7. We conclude the paper with a discussion in Section 8.

Table 1: Notation frequently used throughout the manuscript.

Notation	Definition
$\mathcal{D}^{(k)}, \mathcal{D}_n$	target and source studies $\mathcal{D}^{(k)} = \{\mathbf{u}^{(k)}, \mathbf{c}^{(k)}, \mathbf{X}^{(k)}\}$, $k = T, 1, \ldots, K$ and the collection $\mathcal{D}_n = \{\mathcal{D}^{(k)}\}_{k=T,1,\ldots,K}$.
$y_i^{(k)}, r_i^{(k)}, c_i^{(k)}, u_i^{(k)}$	Survival and censoring time, censoring indicator, and the observed $u_i^{(k)} = \min\{y_i^{(k)}, r_i^{(k)}\}$ of observation i in study $k = T, 1, \ldots, K$.
$\mathbf{x}_i^{(k)}, \mathbf{X}^{(k)}$	Predictor and design matrix.
$\boldsymbol{\beta}, \boldsymbol{\beta}^{(k)}, \mathbf{b}^{(k)}$	Study-specific covariate-effects & the bias $\mathbf{b}^{(k)} = \boldsymbol{\beta}^{(k)} - \boldsymbol{\beta}$.
$\alpha, \alpha^{(k)}$; $\sigma, \sigma^{(k)}$	Target and source-specific intercepts and residual scale parameters.
$H_\psi(\cdot), W_\tau(\cdot)$	Activation and slab functions which define $\boldsymbol{\beta}$ and $\boldsymbol{b}^{(k)}$.
v_j, e_j; $v_j^{(k)}, e_j^{(k)}$	Independent standard normal latent variables representing target coefficients and source biases $\boldsymbol{\beta}$ and $\boldsymbol{b}^{(k)}$.
$\psi^{(T)}, \psi^{(k)}$	Activation thresholds controlling sparsity of $\boldsymbol{\beta}$ and $\boldsymbol{b}^{(k)}$.
$\tau^{(T)}, \tau^{(k)}$	Slab scale parameters for target coefficients and source biases.
$\omega^{(k)}, \lambda^{(k)}$	Prior inclusion probabilities and the hyperparameters in $\omega^{(k)} \sim \text{Beta}(1, \lambda^{(k)})$.
$\tau_\alpha^2, a_\sigma, c_\sigma$	Prior variance for the intercept, and inverse-gamma hyperparameter of the AFT residual scales $\sigma, \sigma^{(k)}$.

2 Methodology

2.1 Notation

Throughout the manuscript, matrices, vectors, and scalars are denoted respectively with bold capitalized letters ($\boldsymbol{X}$), bold lowercase letters ($\boldsymbol{x}$), and lowercase letters (x). We use $\boldsymbol{X}'$ to indicate the transpose of $\boldsymbol{X}$, and $\mathcal{I}(\cdot)$ denotes the indicator function. For two real-valued functions g and h, we use $g(t) \overset{c}{\sim} h(t)$ as $t \to t_0$ to indicate that $\lim_{t \to t_0} g(t)/h(t) = C > 0$ for some constant $C > 0$, and $g(t) = o(h(t))$ if $g(t)/h(t) \to 0$.

2.2 Settings

Target data. We consider a random sample $\mathcal{D}^{(T)} = \left\{\boldsymbol{u}^{(T)}, \boldsymbol{c}^{(T)}, \boldsymbol{X}^{(T)}\right\}$ of measurements from n_T observations in our target population. The matrix $\boldsymbol{X}^{(T)} = [\boldsymbol{x}_1^{(T)}, \ldots, \boldsymbol{x}_{n_T}^{(T)}]' \in \mathbb{R}^{n_T \times p}$ indicates the collection of covariates $\boldsymbol{x}_i^{(T)} \in \mathbb{R}^p$, $i = 1, \ldots, n_T$, and $\boldsymbol{y}^{(T)}, \boldsymbol{r}^{(T)} \in \mathbb{R}^{n_T}$ the underlying survival and censoring times. Let $\boldsymbol{u}^{(T)} \in \mathbb{R}^{n_T}$ indicate potentially right-censored observed outcomes and $\boldsymbol{c}^{(T)}$ denote the corresponding vector of censoring indicators. Specifically, $u_i^{(T)} = \min\{y_i^{(T)}, r_i^{(T)}\}$ and $c_i^{(T)} = \mathcal{I}(y_i^{(T)} \leq r_i^{(T)})$, i.e., if $c_i^{(T)} = 1$ then $u_i^{(T)} = y_i^{(T)}$ represents the observed survival time and $c_i^{(T)} = 0$ indicates that observation i was censored at time $u_i^{(T)} = r_i^{(T)}$. Table 1 summarizes notation and abbreviations used frequently throughout the manuscript.

We consider settings where the number of events in the target population is substantially smaller than the number of predictors p, and only a small subset of these covariates impacts the survival time. The aim is to identify a subset of the predictors $\boldsymbol{x}_i^{(T)}$ with non-zero effect on the survival time and perform inference on the magnitude of the corresponding non-zero covariate effects. In the following, $p_{\boldsymbol{\varrho}}(y|\boldsymbol{x})$ indicates the conditional survival distribution in the target population, where $\boldsymbol{\varrho} = \{\boldsymbol{\beta}, \boldsymbol{\rho}\}$ indicates the vector of covariate effects $\boldsymbol{\beta}$ in the target study and a nuisance parameter $\boldsymbol{\rho}$.

Source datasets. In addition to the target study, data from $K \geq 1$ additional source studies $\mathcal{D}^{(1)}, \ldots, \mathcal{D}^{(K)}$ are available from similar, but not necessarily identical populations, as the target population. For each source dataset $\mathcal{D}^{(k)} = \left\{\boldsymbol{u}^{(k)}, \boldsymbol{c}^{(k)}, \boldsymbol{X}^{(k)}\right\}$, we observe n_k possibly right-censored survival outcomes $\boldsymbol{u}^{(k)} \in \mathbb{R}^{n_k}$ from underlying survival and censoring times $\boldsymbol{y}^{(k)}, \boldsymbol{r}^{(k)} \in \mathbb{R}^{n_k}$, the corresponding censoring indicators $\boldsymbol{c}^{(k)}$, such that $u_i^{(k)} = \min\{y_i^{(k)}, r_i^{(k)}\}$ and $c_i^{(k)} = \mathcal{I}(y_i^{(k)} \leq r_i^{(k)})$, and the covariate matrix $\boldsymbol{X}^{(k)} = [\boldsymbol{x}_1^{(k)}, \ldots, \boldsymbol{x}_{n_k}^{(k)}]' \in \mathbb{R}^{n_k \times p}$. Survival times in each source population follow the conditional distribution $p_{\boldsymbol{\varrho}^{(k)}}(y^{(k)}|\boldsymbol{x}^{(k)})$ with

study-specific $\boldsymbol{\varrho}^{(k)} = \{\boldsymbol{\beta}^{(k)}, \boldsymbol{\rho}^{(k)}\}$, with covariate effect $\boldsymbol{\beta}^{(k)}$ and nuisance parameter $\boldsymbol{\rho}^{(k)}$. We assume that the same set of covariates is measured in each dataset.

The number of observed events in some of the source datasets can be substantially larger than the number of events in the target data. This motivates the use of auxiliary data to enhance the estimation accuracy of the target parameter $\boldsymbol{\beta}$ and improve inference on the set of non-zero coefficients, while accounting for potential biases $\boldsymbol{b}^{(k)} := \boldsymbol{\beta}^{(k)} - \boldsymbol{\beta}$.

2.3 Outcome model

Target Study. We focus on accelerated failure time (AFT) models [56] for the conditional survival distributions $p_{\boldsymbol{\varrho}}(y^{(T)}|\boldsymbol{x}^{(T)})$ in the target study. Given $\boldsymbol{x}_i^{(T)}$, the survival time of observation i equals

$$\log y_i^{(T)} = \alpha + \boldsymbol{x}_i^{(T)\prime}\boldsymbol{\beta} + \epsilon_i^{(T)} \text{ with } \epsilon_i^{(T)} \overset{iid}{\sim} p_\sigma, \tag{1}$$

with intercept α, time-acceleration factor $\boldsymbol{x}_i^{(T)\prime}\boldsymbol{\beta}$, covariate effect $\boldsymbol{\beta} \in \mathbb{R}^p$, and residual $\epsilon_i^{(T)}$. Different residual distributions can be considered, e.g. the extreme value distribution $p_\sigma(\epsilon) = \frac{1}{\sigma}\exp\left(\epsilon/\sigma - e^{\epsilon/\sigma}\right)$ for the Weibull AFT model or the logistic distribution $p_\sigma(\epsilon) = \frac{e^{-\epsilon/\sigma}}{\sigma[1+e^{-\epsilon/\sigma}]^2}$ for the log-logistic AFT model. We assume that p_σ has mode 0 and belongs to a scale family satisfying $\epsilon/\sigma \sim p_1(\epsilon)$ for any $\sigma > 0$. Together we have $\boldsymbol{\rho} = (\alpha, \sigma)$.

Different outcome models could be used, such as proportional hazards [12] or proportional odds [47] models. Posterior inference via our MCMC sampler (see Section 5) requires no modifications. We choose AFT models, as covariate effects are easier to interpret for medical investigators than the effect parameters in alternative models, such as proportional hazards models [56, 32, 31, 54]. Moreover, effect estimates in AFTs tend to be robust to the omission of covariates [32, 24] compared to proportional hazards models, and are relatively insensitive to the choice of the residual distribution [32, 34].

Source studies. The survival times of observations in each source dataset $k = 1, \ldots, K$ follow, similar to the target data, an AFT model of the form

$$\log y_i^{(k)} = \alpha^{(k)} + \boldsymbol{x}_i^{(k)\prime}\boldsymbol{\beta}^{(k)} + \epsilon_i^{(k)} \text{ with } \epsilon_i^{(k)} \overset{iid}{\sim} p_{\sigma^{(k)}}, \tag{2}$$

with source-specific intercept $\alpha^{(k)}$, covariate effect $\boldsymbol{\beta}^{(k)} \in \mathbb{R}^p$, and $\sigma^{(k)} > 0$, thus $\boldsymbol{\rho}^{(k)} = (\alpha^{(k)}, \sigma^{(k)})$. The residual distribution $p_{\sigma^{(k)}}$ belongs to the same parametric family as the corresponding target residual distribution p_σ, with study-specific $\sigma^{(k)}$.

2.4 The TL prior model

SSTL specifies a joint prior distribution for the parameters of the survival models for the target and source data. It encourages information transfer between source and target data through (i) a prior distribution for the target model $p_{\boldsymbol{\varrho}}$ and (ii) a conditional TL prior distribution for the parameters of the source studies $\boldsymbol{\varrho}^{(k)}$ given $\boldsymbol{\varrho}$. We first outline our prior distribution for $\boldsymbol{\varrho}$, and then introduce our Bayesian model to transfer information from the source data to inference on the parameter of interest $\boldsymbol{\beta}$.

Prior distribution for $\boldsymbol{\varrho}$. Several prior distributions have been proposed for inference and variable selection in high-dimensional settings [3, 17, 6, 45, 21]. The discrete spike-and-slab (DSS) prior is given by $\beta_j | A_j \overset{ind}{\sim} (1 - A_j) \times \delta_0(\beta_j) + A_j \times \pi_A(\beta_j)$, $A_j \overset{iid}{\sim} \text{Bern}(\omega^{(T)})$, with point-mass at 0 and continuous heavy-tailed slab distribution π_A for the non-zero components. This prior allows for exact variable selection and has been shown to have desirable optimality properties [9, 8, 16]. Inspired by the DSS and recent work on neuronized priors [53], we use the following augmented representation for the prior,

$$\beta_j = H_\psi(v_j) \times W_\tau(e_j), \quad v_j, e_j \overset{iid}{\sim} N(0,1) \text{ for } j = 1, \ldots, p, \tag{3}$$

where the *activation function* $H_\psi : \mathbb{R} \to [0, +\infty)$ controls the sparsity of $\boldsymbol{\beta}$ and the *slab-function* $W_\tau : \mathbb{R} \to \mathbb{R}$ controls the tails of the prior for the non-zero components of $\boldsymbol{\beta}$.

Spike-component. In the following, we focus on activation functions of the form $H_\psi(v) = \mathcal{I}(v > \psi)$ and the *Rectified Linear Unit* (ReLU) function $H_\psi(v) = \max\{0, v - \psi\}$. Under these activation functions, $\beta_j = 0$ when $v_j \leq \psi$, thereby recovering the discrete spike component of the DSS model. In this section, we write $\psi = \psi^{(T)}$.

Slab-component. To specify the slab-function $W_\tau(\cdot)$, we introduce three distinct formulations that induce different behaviors of the slab-distribution $\pi_A(\beta_j)$ at zero and in the tails. This includes a novel non-local prior distribution [29, 30, 49] which, unlike standard continuous shrinkage priors π_A, enforces a vanishing prior density $\pi_A(\beta_j) \to 0$ as $\beta_j \to 0$ to improve the identification of zero and non-zero coordinates of $\boldsymbol{\beta}$. We provide a derivation for each of the three specifications:

Example 2.1 (DSS with exponential slab)**.** In this implementation, we specify in the augmented representation (3) the ReLU activation function in combination with a linear slab function. Specifically, for a fixed $\tau > 0$,

$$H_\psi(v) = \max\{0, v - \psi\} \text{ and } W_\tau(e) = \tau \cdot e \tag{4}$$

As detailed in Supplemental Lemma S1, the corresponding DSS prior distribution $\pi(\boldsymbol{\beta})$ is then given by the hierarchical model,

$$\begin{aligned} A_j &\overset{iid}{\sim} Bern(\omega^{(T)}) \text{ with } \omega^{(T)} = 1 - \Phi(\psi), \\ \beta_j | A_j &\overset{ind}{\sim} (1 - A_j) \times \delta_0(\beta_j) + A_j \times \pi_A(\beta_j) \text{ where} \\ \pi_A(\beta_j) &= \frac{1}{2\pi\tau(1 - \Phi(\psi))} \int_0^{+\infty} u^{-1} e^{-\frac{(u+\psi)^2}{2} - \frac{\beta_j^2}{2\tau^2 u^2}} du \\ &= \frac{1}{\pi\tau} K_0\left(\frac{|\beta_j|}{\tau}\right) \quad \text{if } \psi = 0. \end{aligned}$$

Here $K_\nu(z) = z^\nu / 2 \int_0^\infty u^{-(\nu+1)} e^{-(u + z^2/u)/2} du$ indicates the modified Bessel function of the second kind [1]. By employing the linear slab function, we recovered the neuronized DSS prior from Shin et al. [53]. Using properties of Bessel functions (see Lemma S1), one can

show the slab distribution has a negative log pole at $\beta_j = 0$ and exponential tails

$$\pi_A(\beta_j) \overset{c}{\sim} \begin{cases} -\log\left(\frac{|\beta_j|}{2\tau}\right) & \text{as } \beta_j \to 0, \\ \left(\frac{|\beta_j|}{2\tau}\right)^{-1/2} \exp\{-\frac{|\beta_j|}{\tau} - \psi\sqrt{\frac{|\beta_j|}{\tau}}\} & \text{as } |\beta_j| \to \infty. \end{cases} \tag{5}$$

Example 2.2 (DSS with polynomial slab)**.** The second prior distribution we consider has a heavier-tailed slab density to avoid strong shrinkage of large signals compared to the previous slab distribution. Specifically, we specify the ReLU activation in combination with the exponential slab function,

$$H_\psi(v) = \max\{0, v - \psi\} \text{ and } W_\tau(e) = \tau \cdot \text{sign}(e) \exp(e^2/m) \text{ for } \tau > 0, m = 2, 3, \ldots. \tag{6}$$

The following lemma provides properties of the proposed DSS prior from this construction.

Lemma 2.1. *The prior distribution $\pi(\boldsymbol{\beta})$ for the covariate effect vector $\boldsymbol{\beta}$ in (3) with activation and slab functions specified as (6) correspond to the DSS distribution*

$$A_j \overset{iid}{\sim} Bern(\omega^{(T)}) \text{ with } \omega^{(T)} = 1 - \Phi(\psi),$$

$$\beta_j | A_j \sim (1 - A_j) \times \delta_0(\beta_j) + A_j \times \pi_A(\beta_j) \text{ where}$$

$$\pi_A(\beta_j) = \frac{\sqrt{m\tau^m}}{4\pi(1 - \Phi(\psi))} \times \int_\tau^\infty \frac{\exp\{-(|\beta_j|/u + \psi)^2/2\}}{u^{m/2+2}\sqrt{\log(|u|/\tau)}} du.$$

The above slab distribution is finite at $\beta_j = 0$ and has polynomial tails,

$$\pi_A(\beta_j) \overset{c}{\sim} \begin{cases} \frac{\exp(-\psi^2/2)}{2\sqrt{2\pi}\,\tau(1-\Phi(\psi))}\sqrt{\frac{m}{m+2}} & \text{as } |\beta_j| \to 0 \\ |\beta_j/\tau|^{-(m+2)/2} \times \log(|\beta_j|/\tau)^{-1/2} & \text{as } |\beta_j| \to \infty. \end{cases} \tag{7}$$

Example 2.3 (DSS with non-local slab)**.** Non-local prior distributions have been demonstrated to exhibit superior variable selection performance and faster posterior contraction rates compared to traditional "local" shrinkage priors [29, 30]. Posterior inference for non-local prior distributions is non-trivial outside the scope of normal outcome models [30, 49]. We introduce a non-local prior distribution via the latent variable representation (3) that enables straightforward posterior inference via our MCMC sampler introduced in Section

5. To construct a DSS prior distribution with non-local slab density π_A, we define the "fractional-power" transformation

$$H_\psi(v) = \mathcal{I}(v > \psi) \text{ and } W_\tau(e) = \tau \cdot \text{sign}(e) \left(\exp(me^2) - 1\right)^{1/(2m)}, \text{ for } m \geq 2, \tau > 0. \quad (8)$$

The next Lemma provides tail and near-zero properties of the proposed non-local prior.

Lemma 2.2. *The prior distribution $\pi(\boldsymbol{\beta})$ for the covariate effect vector $\boldsymbol{\beta}$ in (3) with activation and slab functions specified as (8) correspond to a DSS distribution of the form*

$$A_j \overset{iid}{\sim} Bern(\omega^{(T)}) \text{ with } \omega^{(T)} = 1 - \Phi(\psi),$$

$$\beta_j | A_j \sim (1 - A_j) \times \delta_0 + A_j \times \pi_A(\beta_j) \text{ where}$$

$$\pi_A(\beta_j) = \frac{1}{\sqrt{2\pi\tau^2}} \frac{|\beta_j/\tau|^{2m-1}}{((\beta_j/\tau)^{2m} + 1)^{(2m+1)/2m} \sqrt{\log((\beta_j/\tau)^{2m} + 1)/m}}$$

Moreover, the fractional-power transformation leads to a non-local slab distribution with polynomial tails

$$\pi_A(\beta_j) \overset{c}{\sim} \begin{cases} |\beta_j/\tau|^{m-1} & \text{as } |\beta_j| \to 0 \\ |\beta_j/\tau|^{-2} (\log(|\beta_j/\tau|))^{-1/2} & \text{as } |\beta_j| \to \infty. \end{cases}$$

The above lemma shows that the slab distribution of the proposed DSS prior possesses two critical properties: (i) a polynomial behavior with degree $m-1$ near zero, satisfying the non-local condition $f(\beta_j) \to 0$ as $\beta_j \to 0$, and (ii) Cauchy-like polynomial tails decaying at a rate of β_j^{-2} (up to a logarithmic factor), which is highly desirable in high dimensional estimation. In fact, the following lemma provides tight, simple upper and lower polynomial bounds for the slab density.

Lemma 2.3. *For fixed $m \geq 2$ and $\tau > 0$, consider the lower envelope*

$$f_L(\beta_j) = \begin{cases} c_L \times (|\beta_j|/\tau)^{m-1} & \text{if } 0 < |\beta_j| \leq \tau, \\ c_L \times (|\beta_j|/\tau)^{-2} \dfrac{1}{\sqrt{\log\{2(|\beta_j|/\tau)^{2m}\}}} & \text{if } |\beta_j| > \tau, \end{cases}$$

where $c_L = \frac{\sqrt{m}}{\sqrt{2\pi}\tau\, 2^{1+1/(2m)}}$. *Similarly, define the upper envelope*

$$f_U(\beta_j) = \begin{cases} c_U \times (|\beta_j|/\tau)^{m-1}, & \text{if } 0 < |\beta_j| \leq \tau, \\ c_U \times (|\beta_j|/\tau)^{-2} \dfrac{1}{\sqrt{\log\{1 + (|\beta_j|/\tau)^{2m}\}}}, & \text{if } |\beta_j| > \tau. \end{cases}$$

where $c_U = \frac{\sqrt{m}}{\sqrt{2\pi}\tau}$. *Then for all* $\beta_j \neq 0$, $f_L(\beta_j) \leq \pi_A(\beta_j) \leq f_U(\beta_j)$. *Moreover, the upper bound is sharp in the sense that* $\pi_A(\beta_j) \sim f_U(\beta_j)$ *as* $|\beta_j| \to 0$ *and* ∞.

The fractional-power transformation is algebraically tractable and possesses a simple closed-form inverse, enabling efficient posterior computation using our Monte Carlo algorithm for posterior inference which leverages the generic representation (3). Figure 1 visualizes the proposed slab densities and their standard counterparts. We denote each prior as *exp, poly, nlp* from examples 2.1–2.3.

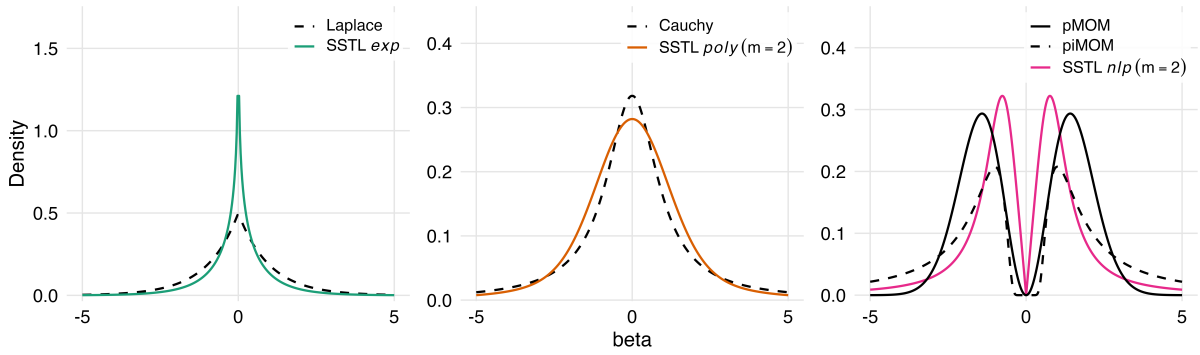


Figure 1: Comparison of prior densities of the augmented slab representations with existing standard counterparts.

Example 2.4 (DSS with global-local scale components)**.** In our last example we consider a small extension of the prior parameterization (3) using a latent variable representation of the form

$$\beta_j = H_\psi(v_j) \times W_{\tau_0}(e_j, \tau_j), \qquad \text{with } v_j, e_j, \tau_j \overset{iid}{\sim} N(0,1) \text{ for } j = 1, \ldots, p, \tag{9}$$

with global scale parameter τ_0 and local random scale parameters τ_j. For instance, specifying $H_\psi(v_j) = \mathcal{I}(v_j > \psi)$ and $W_{\tau_0}(e_j; \tau_j) = \tau_0 \times \tau_j \exp\{e_j^2/m\}$ for fixed $\tau_0 > 0$ and

$m = 2, 3, \ldots$, we recover the DSS model $\beta_j \overset{iid}{\sim} \Phi(\psi)\delta_0(\beta_j) + (1 - \Phi(\psi)) \times \pi_A(\beta_j)$ with

$$\pi_A(\beta_j) = \frac{\sqrt{m\tau_0^m}}{2\pi} \times \int_{\tau_0}^{\infty} \frac{\exp\{-(|\beta_j|/u)^2/2\}}{u^{m/2+2}\sqrt{\log(|u|/\tau_0)}} du. \tag{10}$$

which, from example 2.2, has polynomial tails. Specifying instead the slab component $W_{\tau_0}(e_j; \tau_j) = \tau_0 \times \tau_j e_j$, the corresponding DSS prior has, similar to Example 2.1, an exponentially-tailed slab distribution $\pi_A(\beta_j) = \frac{1}{\pi\tau_0} K_0\left(\frac{|\beta_j|}{\tau_0}\right)$.

Lastly, we complete the Bayesian model for the target data with a normal prior for the intercept and an inverse gamma prior for the residual scale parameter, i.e.

$$\alpha \sim N(0, \tau_\alpha^2) \quad \text{and} \quad \sigma^2 \sim IGam(a_\sigma, c_\sigma)$$

2.5 Information transferring prior

To transfer information on the covariate effects from source data to the target data, we parameterize the covariate effect vector of each source study in terms of a bias parameter and the covariate effect in the target study,

$$\boldsymbol{\beta}^{(k)} = \boldsymbol{\beta} + \boldsymbol{b}^{(k)}.$$

If source study k is highly similar to the target data, then the bias vector $\boldsymbol{b}^{(k)}$ should be sparse and small in magnitude.

Prior for the bias parameters. Similar to the target study, we leverage the augmented representation of the DSS prior to perform inference on the zero and non-zero components of the bias vectors. Specifically, for each source dataset $k = 1, \ldots, K$, the p components of the bias vector have an augmented representation

$$b_j^{(k)} = H_{\psi^{(k)}}\left(v_j^{(k)}\right) \times W_{\tau^{(k)}}\left(e_j^{(k)}\right), \quad v_j^{(k)}, e_j^{(k)} \overset{iid}{\sim} N(0,1) \text{ for } j = 1, \ldots, p. \tag{11}$$

The activation and slab component $H_{\psi^{(k)}}(\cdot)$ and $W_{\tau^{(k)}}(\cdot)$ are specified according to the parameterizations outlined in Examples 2.1-2.4. The prior (11) facilitates learning of

coordinate-specific zero and non-zero bias elements according to study-specific sparsity levels $1 - \Phi(\psi^{(k)})$,

$$b_j^{(k)} \overset{ind}{\sim} (1 - A_j^{(k)}) \times \delta_0(b_j^{(k)}) + A_j^{(k)} \times \pi_b(b_j^{(k)} \mid \psi^{(k)}, \tau^{(k)}), \quad A_j^{(k)} \overset{iid}{\sim} Bern(1 - \Phi(\psi^{(k)})). \quad (12)$$

We illustrate how to learn the sparsity level $\psi^{(k)}$ adaptively to facilitate informative source selection in Section 5.1. We complete the Bayesian model for the source data with prior distributions for the source-specific intercepts and the residual scale parameter

$$\alpha^{(k)} \mid \alpha \overset{iid}{\sim} N(\alpha, \tau_\alpha^2) \quad \text{and} \quad (\sigma^{(k)})^2 \overset{iid}{\sim} IGam(a_\sigma, c_\sigma).$$

2.6 Group-TL prior

In many instances, predictors are naturally organized into groups. For example, categorical variables are typically encoded as a set of dummy variables, and gene-expression measurements often cluster by biological pathway. Similarly, clinical lab measurements frequently form grouped effects because they provide complementary information on a physiological system. Standard penalized regression methods such as Lasso estimators do not perform well in such settings, and thus it is important to account for such group structures [58].

To account for group structures in predictors, we consider an extension of our TL prior. In the following, each variable $j = 1, \ldots, p$ is assumed to belong to one of $R \geq 1$ pre-defined groups $\mathcal{G}_r \subset \{1, \ldots, p\}, r = 1, \ldots, R$ of variables of sizes $g_r = |\mathcal{G}_r|$. The group Lasso penalty $\sum_r \lambda_r \|\boldsymbol{b}_{\mathcal{G}_r}\|_2$ [58] is a popular penalty function to introduce group-level regularization for $\boldsymbol{b}_{\mathcal{G}_r} = (b_j)_{j \in \mathcal{G}_r}$ into data analysis. For each group $\mathcal{G}_r$, the group Lasso estimator $\widehat{\boldsymbol{b}}_{GL}$ either sets $\hat{b}_j = 0$ for all $j \in \mathcal{G}_r$ or has $\hat{b}_j \neq 0$ for all $j \in \mathcal{G}_r$. A Bayesian version [33] of the group Lasso assumes independent multivariate Laplace distributions for each group of coefficients $\pi(\boldsymbol{b}) \propto \exp(\sum_{r=1}^R -\lambda \|\boldsymbol{b}_{\mathcal{G}_r}\|_2)$, which can be represented as a scale-mixture of a mean zero

normal distribution $\boldsymbol{b}_{\mathcal{G}_r}|z_r^2 \overset{ind}{\sim} N(\mathbf{0}_{g_r}, z_r^2\boldsymbol{I}_{g_r})$ with $z_r^2 \overset{iid}{\sim} Gam\left((g_r+1)/2, \lambda^2/2\right)$ such that

$$\exp\left(-\lambda\|\boldsymbol{b}_{\mathcal{G}_r}\|_2\right) \propto \int_0^\infty z_r^{-1} \exp\left(-\frac{\|\boldsymbol{b}_{\mathcal{G}_r}\|_2^2}{2z_r^2} - \frac{z_r^2\lambda^2}{2}\right) dz_r^2. \tag{13}$$

Bayesian inference under the group Lasso assigns, almost surely, zero posterior probability on the event $\boldsymbol{b}_{\mathcal{G}_r} = \mathbf{0}_{g_r}$. To remedy this, we adopt in the following lemma an augmented variable representation similar to (3) and (11) of a DSS distribution with a multivariate slab density that admits an approximate Bayesian group Lasso prior.

Lemma 2.4. *Consider random vectors $\boldsymbol{b}_{\mathcal{G}_r} = (b_j; j \in \mathcal{G}_r)$ defined by the augmented variable representation with activation and slab components of the form,*

$$b_j = H_\psi(v_r) \times W_{\tau_r}(e_j) \text{ for } j \in \mathcal{G}_r, r = 1, \ldots, R, \text{ where}$$

$$H_\psi(v_r) = \max(0, v_r - \psi) \text{ and } W_{\tau_r}(e_j) = \tau_r e_j, \tag{14}$$

$$e_j, v_r \overset{iid}{\sim} N(0,1) \text{ for } j = 1, \ldots, p, r = 1, \ldots, R. \tag{15}$$

Then, the prior distribution for $\boldsymbol{b}_{\mathcal{G}_r}$ corresponds to a grouped DSS distribution of the form

$$A_r \overset{iid}{\sim} Bern(\omega^{(T)}) \text{ with } \omega^{(T)} = 1 - \Phi(\psi),$$

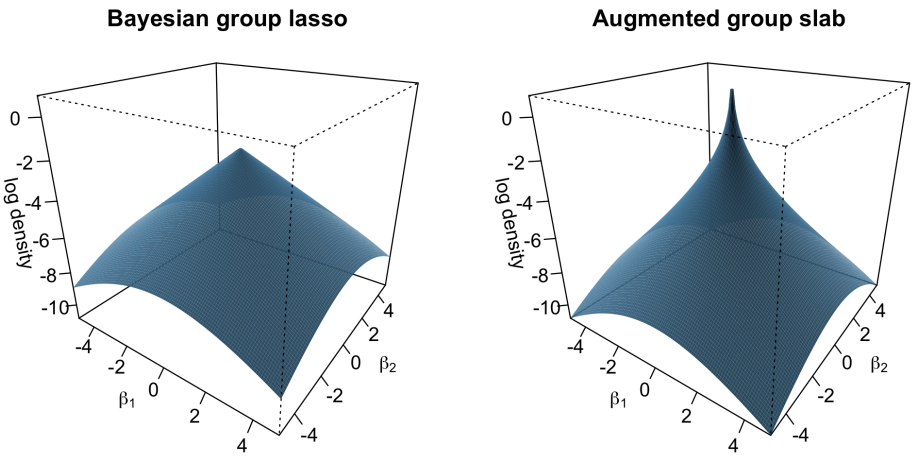
$$\pi(\boldsymbol{b}_{\mathcal{G}_r}) = (1 - A_r)\delta_{\mathbf{0}_{g_r}} + A_r \cdot \pi_A(\boldsymbol{b}_{\mathcal{G}_r}).$$

Moreover, the slab-distribution is given by

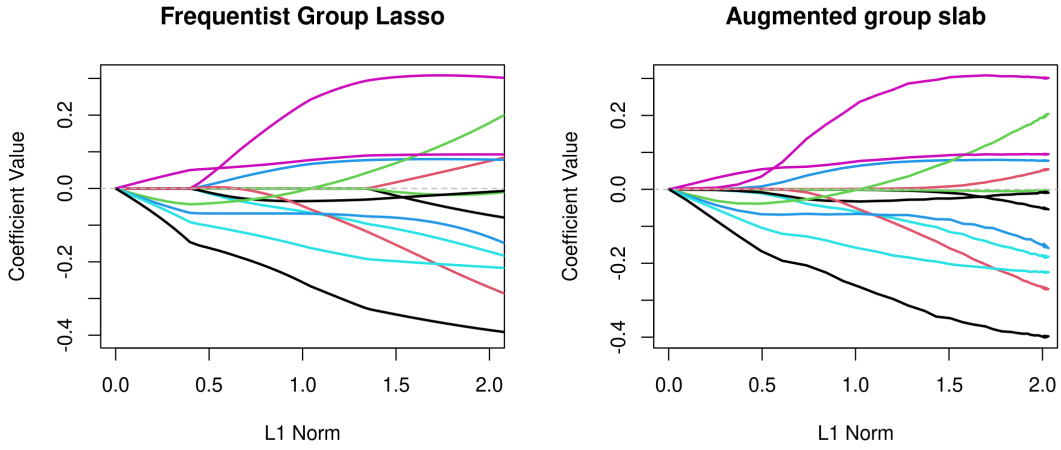
$$\begin{aligned}\pi_A(\boldsymbol{b}_{\mathcal{G}_r}) &= \frac{1}{(2\pi)^{(g_r+1)/2}\tau_r^{g_r}(1 - \Phi(\psi^{(T)}))} \int_0^\infty u^{-g_r} \exp\left(-\frac{\|\boldsymbol{b}_{\mathcal{G}_r}\|_2^2}{2z^2\tau_r^2} - \frac{(u+\psi)^2}{2}\right) du \\ &= \frac{2}{(2\pi)^{(g_r+1)/2}\ \tau_r^{(g_r+1)/2}} \times ||\boldsymbol{b}_{\mathcal{G}_r}||_2^{-\frac{g_r-1}{2}} K_{\frac{g_r-1}{2}}\left(\frac{||\boldsymbol{b}_{\mathcal{G}_r}||_2}{\tau_r}\right) \qquad \text{for } \psi = 0,\end{aligned} \tag{16}$$

For $g_r > 1$, the density has exponential tails

$$\pi_A(\boldsymbol{b}_{\mathcal{G}_r}) \overset{c}{\sim} \begin{cases} \|\boldsymbol{b}_{\mathcal{G}_r}\|_2^{1-g_r} & \text{as } \|\boldsymbol{b}_{\mathcal{G}_r}\|_2 \to 0 \\ \exp\left\{-\frac{\|\boldsymbol{b}_{\mathcal{G}_r}\|}{\tau_r} - \psi\sqrt{\frac{\|\boldsymbol{b}_{\mathcal{G}_r}\|_2}{\tau_r}}\right\} \times \|\boldsymbol{b}_{\mathcal{G}_r}\|_2^{-g_r/2} & \text{as } \|\boldsymbol{b}_{\mathcal{G}_r}\|_2 \to +\infty. \end{cases} \tag{17}$$



(a) Log density functions of the Bayesian group Lasso and the proposed group slab.



(b) Solution paths of the frequentist group Lasso and the proposed group slab.

Figure 2: Comparison of the augmented group slab density with the Bayesian group Lasso (Panel (a)) and the frequentist group Lasso (Panel (b)).

Figure 2a shows that the group slab distribution (16) is more spiked at zero and has slightly lighter tails than the Bayesian group lasso. Figure 2b compares the solution paths of the group Lasso to the proposed group slab for the analysis of the Boston housing dataset, which are nearly identical. Lemma 2.4 shows that the integrands that define the above group-slab prior distribution $\pi_A(\boldsymbol{b}_{\mathcal{G}_r})$ and the Bayesian group Lasso prior distribution (13) only differ by the term $z_r^{-g_r}$, and the tails of the two group-prior distributions deviate

by a factor $||\boldsymbol{b}_{\mathcal{G}_r}||^{-g_r/2}$ when $\psi = 0$. Crucially, the simple augmented parameterization (14) preserves the group-level shrinkage property of the group Lasso, as all coefficients b_j within group $j \in \mathcal{G}_r$ share the single scale parameter τ_r. Therefore small values of τ_r simultaneously shrink the entire vector $\boldsymbol{b}_{\mathcal{G}_r}$ toward the origin. The group-DSS prior can also be used to account for grouped-covariate structures for the covariate-effect vector $\boldsymbol{\beta}$ in the target data.

3 Large sample properties

Before we present our algorithms for posterior inference, we investigate large sample properties of SSTL in the high-dimensional setting with fixed (non-random) covariate matrices. To simplify the presentation, we focus on lognormal outcomes without right-censoring, with known variance $\sigma = \sigma^{(k)} = 1$ and zero intercepts $\alpha = \alpha^{(k)} = 0$ for $k = 1, \ldots, K$. We focus on the TL prior distributions outlined in Examples 2.3 and 2.4.

We first introduce some additional notation and assumptions on the TL prior and the unknown data-generating model that we require in the results below. Let $n = n_T + \sum_{k=1}^{K} n_k$ indicate the pooled sample size of all data sources, $\widetilde{\boldsymbol{X}} = \left[\boldsymbol{X}^{(T)\prime}, \boldsymbol{X}^{(1)\prime}, \ldots, \boldsymbol{X}^{(K)\prime}\right]'$ the stacked covariate matrix of all datasets, $\boldsymbol{Z} = [\boldsymbol{O}, \text{diag}\{\boldsymbol{X}^{(1)}, \ldots, \boldsymbol{X}^{(K)}\}']'$ the pooled source design matrix, and $\boldsymbol{b} = [\boldsymbol{b}^{(1)\prime}, \ldots, \boldsymbol{b}^{(K)\prime}]'$ the vector containing bias parameters of all K source studies. We also define $\|\widetilde{\boldsymbol{X}}\| = \max_{1 \leq j \leq p} \|\widetilde{\boldsymbol{X}}_{\cdot j}\|_2$ as the largest column norm of $\widetilde{\boldsymbol{X}}$. We assume K is fixed as n grows throughout this section.

For a positive integer m, write $[m] = \{1, \ldots, m\}$. We define $\mathcal{S} = \{j : \beta_j \neq 0\}$ and $s = \|\boldsymbol{\beta}\|_0$. Further, $\widetilde{\boldsymbol{X}}_{\mathcal{S}}$ denotes the submatrix of $\widetilde{\boldsymbol{X}}$ with columns indexed by $\mathcal{S}$. We index $\boldsymbol{b}$ and columns of $\boldsymbol{Z}$ by the pair $(k, j) \in [K] \times [p]$, where (k, j) denotes position $(k-1)p + j$, i.e. $b_j^{(k)}$ is coordinate (k, j) of $\boldsymbol{b}$. We use $\mathcal{R} \subset [K] \times [p]$ to index subsets of $\boldsymbol{b}, \boldsymbol{Z}$ in the same

fashion as for $\boldsymbol{\beta}, \widetilde{\boldsymbol{X}}$, and set $r = \|\boldsymbol{b}\|_0$, $r_k = \|\boldsymbol{b}^{(k)}\|_0$. Let $\boldsymbol{\beta}_0$ and $\boldsymbol{b}_0$ indicate the unknown true regression and bias parameters. Their supports are written as $\mathcal{S}_0$ and $\mathcal{R}_0$ respectively, each with size s_0, r_0. For each source, let $\mathcal{R}_{0k} = \{j : (k,j) \in \mathcal{R}_0\}$ and $r_{0k} = |\mathcal{R}_{0k}|$. We assume $s_0 \geq 1$, and set $s_\star = \max\{s_0, r_0\}$.

Let $\mathcal{D}_n = \{\mathcal{D}^{(T)}, \mathcal{D}^{(1)}, \ldots \mathcal{D}^{(K)}\}$, $\Pi(\cdot)$ denote the joint prior distribution, and $\Pi(\cdot \mid \mathcal{D}_n)$ the corresponding posterior. Probability and expectation under the true data-generating distribution are denoted by $\mathbb{P}_0$ and $\mathbb{E}_0$. Throughout the large-sample results and their proofs, $c, C \in (0, \infty)$ denote constants independent of n, p, allowed to change from line to line, and $a_n \lesssim b_n$ stands for $a_n \leq C b_n$. Lastly, as in Castillo et al. [8], the uniform compatibility number and the smallest scaled singular value of $\widetilde{\boldsymbol{X}}$ are

$$\phi_1(t) = \inf_{\boldsymbol{\beta}: 1 \leq s \leq t} \frac{\|\widetilde{\boldsymbol{X}}\boldsymbol{\beta}\|_2 s^{1/2}}{\|\widetilde{\boldsymbol{X}}\| \|\boldsymbol{\beta}\|_1}, \qquad \phi_2(t) = \inf_{\boldsymbol{\beta}: 1 \leq s \leq t} \frac{\|\widetilde{\boldsymbol{X}}\boldsymbol{\beta}\|_2}{\|\widetilde{\boldsymbol{X}}\| \|\boldsymbol{\beta}\|_2}.$$

Roadmap: We first discuss, in Section 3.1, posterior contraction results for the TL models in Examples 2.3 and 2.4, using *exp*, *poly* and *nlp* slab priors under mild regularity conditions on the priors and outcome models. We then consider in Section 3.2 additional results that explicitly quantify the added benefits of SSTL compared to single-study analysis in terms of variable selection consistency and posterior contraction around $\boldsymbol{\beta}_0$.

3.1 General contraction theory

Throughout this section, we make the following assumption on the TL hyperparameters:

(P-A1) $\omega^{(k)} = 1 - \Phi(\psi^{(k)}) \overset{ind}{\sim} Beta(1, p^{a_k})$ with $a_k > 1$ for $k = T, 1, \ldots, K$, with $\psi^{(k)}$ defined in (11) and (3).

(P-A2) For each of the three slab distributions *exp*, *poly*, *nlp* in Examples 2.3 and 2.4, assume:

(*exp*) For each data-source $k = T, 1, \ldots, K$ the scale parameter $\tau^{(k)}$ in (4) satisfy

$$\tau^{(k)} \in \left[\sqrt{n}/(L_0\|\boldsymbol{X}^{(k)}\|),\ p^{L_1}/\|\boldsymbol{X}^{(k)}\|\right] \text{ for some constants } L_0, L_1 > 0.$$

(poly), (nlp) For each data-source $k = T, 1, \ldots, K$, $\tau^{(k)} \leq p^{L_2}/\|\boldsymbol{X}^{(k)}\|$ for fixed $L_2 > 0$.

Condition (P-A1) imposes a Beta prior for the prior inclusion probability $1 - \Phi(\psi)$ in the spike function H_ψ in (11) and (3). This is a standard condition for sparse regression with DSS priors [8] that ensures the model introduces sufficient sparsity on the supports $\mathcal{S}$ and $\mathcal{R}$. Condition (P-A2) is also standard for *exp*-type prior [8, 28]. The order of the hyper-parameter $\tau^{(k)}$ in (P-A2) determines the bound on the signal strength of the true $\boldsymbol{\beta}_0$ below. Since L_2 can be set sufficiently large, the condition is very mild on *poly* and *nlp*.

We now state assumptions on the experimental design and the unknown $\boldsymbol{\beta}_0$ and $\boldsymbol{b}_0$ which are consistent with standard assumptions in the literature [9, 8, 16, 28].

(A1) (Sparsity) For each dataset $k = T, 1, \ldots, K$, we assume $n_k = o(p)$ as $n \to +\infty$, and $s_0 \log(p) = o(n_T)$ and $r_{0k} \log p = o(n_k)$ for all source studies $k \geq 1$.

(A2) (Signal size) For each of the three slab distributions (*exp*, *poly*, *nlp*) assume:

$$(exp)\ \max_{1\leq j\leq p} |\beta_{0j}| \lesssim \tau^{(T)} \log(p) \text{ and } \max_{1\leq k\leq K, 1\leq j\leq p} |b_{0j}^{(k)}| \lesssim \min_k \tau^{(k)} \log(p).$$

$$(poly),\ (nlp)\ \max_{1\leq j\leq p} |\beta_{0j}| \lesssim \tau^{(T)} p^{L_\beta} \text{ and } \max_{1\leq k\leq K, 1\leq j\leq p} |b_{0j}^{(k)}| \lesssim \min_k \tau^{(k)} p^{L_b} \text{ for fixed } L_\beta, L_b > 0.$$

(A3) (Design matrix) There exists $K_{\text{sep}} \in (0, 1]$ such that for sufficiently large n, for all supports $\mathcal{S}$ and $\mathcal{R}$ satisfying $|\mathcal{S}| \leq C_\beta s_\star$, $|\mathcal{R}| \leq C_b s_\star$ for sufficiently large C_β, C_b, and every $\boldsymbol{u} \in \mathbb{R}^{|\mathcal{S}|}$, $\boldsymbol{v} \in \mathbb{R}^{|\mathcal{R}|}$, the design matrix $\widetilde{\boldsymbol{X}}$ and $\boldsymbol{Z}$ satisfies

$$\|\widetilde{\boldsymbol{X}}_{\mathcal{S}}\boldsymbol{u} + \boldsymbol{Z}_{\mathcal{R}}\boldsymbol{v}\|_2^2 \geq K_{\text{sep}}(\|\widetilde{\boldsymbol{X}}_{\mathcal{S}}\boldsymbol{u}\|_2^2 + \|\boldsymbol{Z}_{\mathcal{R}}\boldsymbol{v}\|_2^2). \tag{18}$$

Condition (A2) is required in the general approach to establish posterior contraction rates for $\boldsymbol{\beta}$ involving the additional bias $\boldsymbol{b}$. It ensures sufficient prior mass of the Kullback-Leibler balls around $\boldsymbol{\beta}_0$ and $\boldsymbol{b}_0$. Condition (A3) ensures that the individual effects of $\boldsymbol{\beta}_0$

and $\boldsymbol{b}_0$ are identifiable. The following theorem shows that, as n becomes large, the TL posterior distribution of $\boldsymbol{\beta}$ given $\mathcal{D}_n$ places vanishing mass on models $\mathcal{S}$ that exceed size $s_\star = \max\{s_0, r_0\}$ by more than a constant factor.

Theorem 3.1 (Target Dimension Control)**.** *Consider the priors from Examples 2.3 and 2.4 with hyperparameters satisfying (P-A1), (P-A2), and conditions (A1) – (A3) are satisfied. There exists a constant $M_1 > 0$ such that*

$$\mathbb{E}_0\Big[\Pi\Big(\boldsymbol{\beta} : s > M_1 s_\star \mid \mathcal{D}_n\Big)\Big] = o(1) \qquad \text{as } n \to +\infty.$$

Moreover, if $r_0 \lesssim 1$, then the result holds with $s_\star$ replaced by s_0.

The next theorem establishes contraction rates for the posterior distribution of $\boldsymbol{\beta}$ and the mean log-survival times around the true values with respect to the ℓ_1 and ℓ_2 norms.

Theorem 3.2 (General Contraction)**.** *Suppose the same conditions as in Theorem 3.1 hold.*

(i) There exist constants $M_1, M_2 > 0$ such that, as $n \to +\infty$,

$$\begin{aligned}
\mathbb{E}_0\Big[\Pi\Big(\|\boldsymbol{\beta} - \boldsymbol{\beta}_0\|_2 > \frac{M_2\sqrt{s_\star \log p}}{\phi_2(M_1 s_\star + s_0)\|\widetilde{\boldsymbol{X}}\|} \mid \mathcal{D}_n\Big)\Big] &= o(1), \\
\mathbb{E}_0\Big[\Pi\Big(\|\boldsymbol{\beta} - \boldsymbol{\beta}_0\|_1 > \frac{M_2 s_\star\sqrt{\log p}}{\phi_1(M_1 s_\star + s_0)\|\widetilde{\boldsymbol{X}}\|} \mid \mathcal{D}_n\Big)\Big] &= o(1), \\
\mathbb{E}_0\Big[\Pi\Big(\|\widetilde{\boldsymbol{X}}(\boldsymbol{\beta} - \boldsymbol{\beta}_0)\|_2 > M_2\sqrt{s_\star \log p} \mid \mathcal{D}_n\Big)\Big] &= o(1) \text{ and} \\
\mathbb{E}_0\Big[\Pi\Big(\|\boldsymbol{Z}(\boldsymbol{b} - \boldsymbol{b}_0)\|_2 > M_2\sqrt{s_\star \log p} \mid \mathcal{D}_n\Big)\Big] &= o(1).
\end{aligned} \tag{19}$$

(ii) If $r_0 \lesssim 1$, then the result in (i) holds with $s_\star$ replaced by s_0.

These results can be regarded as an extension of the results in Castillo et al. [8] to the TL setting. Compared to Castillo et al., who focused on the Laplace slab prior which has similar properties to the *exp* slab, we also establish these results for the *poly* and *nlp* slabs, under weaker signal size assumptions than the *exp* slab.

3.2 Transfer Efficient Contraction and Consistency

We now show, under a set of modified assumptions, that SSTL can achieve faster posterior contraction than target-data only analysis. To keep the presentation compact, we focus on the *exp* slab model in Example 2.4 in this setting. Similar results can also be obtained (with some modifications to the regularity conditions) for the *poly* and *nlp* slabs.

We first introduce some additional notation. Let $\|\widetilde{\boldsymbol{X}}\|_\star = \max_{k\in\{T,1,\ldots,K\}} \|\boldsymbol{X}^{(k)}\|$, and for each $j \in [p]$, $n_j^{\text{eff}} = n_T + \sum_{k:(k,j)\notin\mathcal{R}_0} n_k$ denotes the coordinate-wise effective sample size. In this subsection, we assume a common slab scale parameter $\tau^{(k)} = \tau_B$ for the source studies $k = 1, \ldots, K$. Let $C_\ell > 0$ be the universal constant guaranteed by Lemma S6. We make the following minor modifications to the hyperparameters of our TL prior:

(P-B1) $\omega^{(T)} = 1 - \Phi(\psi^{(T)}) \sim \text{Beta}(1, c_T p^{a_T})$ and $\omega^{(k)} = 1 - \Phi(\psi^{(k)}) \overset{ind}{\sim} Beta(1, c_B p^{a_B})$ for $k = 1, \ldots, K$ for any constants $c_T, c_B > 0$, $a_T > 1 + C_\ell$, and $a_B > a_T + 2C_\ell + 1$.

(P-B2) $\tau^{(T)} = p^{L_3}/(c_\tau \|\widetilde{\boldsymbol{X}}\|_\star)$ and $\tau_B = p^{L_3}/\|\widetilde{\boldsymbol{X}}\|_\star$ for some fixed $c_\tau > 0, L_3 \geq 1$.

(P-B1) is a technical condition to prove Theorem 3.3 below, which is used to bound the marginal probabilities of models $\mathcal{S}, \mathcal{R}$ that are subsets or supersets of the true model $\mathcal{S}_0, \mathcal{R}_0$. (P-B2) fixes the scale parameters proportional to the upper bound of the range in (P-A2). We next state the set of conditions required to evaluate the efficiency of SSTL compared to standard single–study DSS models.

(B1) (Sparsity) In addition to Condition (A1), we assume $s_\star \log\log p = o(\log p)$.

(B2) (Sparse Riesz condition) There exist constants $0 < \underline{\kappa} \leq \overline{\kappa} < \infty$ such that for each $C > 0$, there is an $N_C \in \mathbb{N}$ such that for $n \geq N_C$ and every $\mathcal{A} \subset [p]$ with $|\mathcal{A}| \leq C s_\star$,

$$\underline{\kappa} n_k \|\boldsymbol{u}\|_2^2 \leq \|\boldsymbol{X}_{\mathcal{A}}^{(k)} \boldsymbol{u}\|_2^2 \leq \overline{\kappa} n_k \|\boldsymbol{u}\|_2^2 \quad \forall\, \boldsymbol{u} \in R^{|\mathcal{A}|}, k = T, 1, \ldots, K. \tag{20}$$

(B3) (Signal Size) The true bias and effects satisfy $\|\boldsymbol{\beta}_{0,\mathcal{S}_0}\|_1/\tau^{(T)} + \sum_{k=1}^{K} \|\boldsymbol{b}^{(k)}_{0,\mathcal{R}_{0k}}\|_1/\tau_B = o(\log p)$. Furthermore, there are fixed constants B_β and B_b, with $B_\beta > 4(K+1)(a_T + C_\ell + L_3 - 1)/\underline{\kappa}$ and $B_b > 4(K+1)(a_B + C_\ell + L_3 - 1)/\underline{\kappa}$, such that $\min_{j \in \mathcal{S}_0} n_j^{\text{eff}} \beta_{0j}^2 \geq B_\beta \log p$ and $\min_{(k,j)\in\mathcal{R}_0} \frac{n_T n_k}{n_T + n_k} (b^{(k)}_{0j})^2 \geq B_b \log p$.

(B1) is required to prove variable selection consistency, using bounds on the ratio of marginal likelihoods. The $\log\log p$ term stems from the pole of the *exp* slab density in (5) around zero. This restricts the sparsity in polynomial regimes $p \asymp n^c, c > 0$. In ultra-high dimensional settings such as $p \asymp \exp(n^c), 1/2 < c < 1$, (B1) doesn't impose any additional restriction over (A1). Condition (B2) is an asymptotic sparse eigenvalue condition commonly considered in the high-dimensional literature [60, 59, 50]. Condition (B3) imposes both upper and lower limits on the true signal size. Together with (P-B2), the upper bound required is $c_\tau \|\boldsymbol{\beta}_0\|_1 + \sum_{k=1}^{K} \|\boldsymbol{b}_0^{(k)}\|_1 = o(p^{L_3} \log p/\sqrt{n_{\max}})$, which is very mild if L_3 is sufficiently large. The lower bounds are common beta-min conditions required for variable selection consistency. Notably, each β_{0j} only needs to be of order $\sqrt{\log p/n_j^{\text{eff}}}$, which allows the identification of smaller signals compared to the classical order $\sqrt{\log p/n_T}$ in the single study literature [5]. The nonzero bias $\boldsymbol{b}_{0j}^{(k)}$ for each source study k in (B3) should be at least of order $\sqrt{\log p(n_k^{-1} + n_T^{-1})}$, which is comparable to conditions for single study Bayesian DSS methods when $n_k \gtrsim n_T$. The following theorem shows that the support of the TL posterior contracts (in probability) around the true model as n diverges.

Theorem 3.3 (Support Recovery)**.** *Consider the exp TL prior with hyperparameters satisfying (P-B1)–(P-B2). Suppose that conditions (B1) – (B3) hold, then the posterior distribution satisfies*

$$\mathbb{E}_0 \Pi((\mathcal{S}, \mathcal{R}) \neq (\mathcal{S}_0, \mathcal{R}_0) \mid \mathcal{D}_n) = o(1) \text{ as } n \to +\infty. \tag{21}$$

The next theorem expresses the contraction rate of SSTL in terms of efficiency gains (n^{eff})

compared to the optimal rate of a target–only analysis with DSS priors for $\boldsymbol{\beta}$.

Theorem 3.4 (TL–efficient Contraction Rates)**.** *Suppose the same conditions as in Theorem 3.3 hold. Then there exists a constant M_3 such that*

$$\mathbb{E}_0\Pi\left(\sum_{j\in\mathcal{S}_0} n_j^{\text{eff}}(\beta_j-\beta_{0j})^2 > M_3 s_0 \log p \;\middle|\; \mathcal{D}_n\right) = o(1) \text{ as } n\to+\infty. \tag{22}$$

Consequently, as $n\to+\infty$

$$\begin{aligned}
\mathbb{E}_0\Pi\left\{\|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|_2 > \sqrt{\frac{M_3 s_0 \log p}{\min_{j\in\mathcal{S}_0} n_j^{\text{eff}}}} \;\middle|\; \mathcal{D}_n\right\} &= o(1),\\
\mathbb{E}_0\Pi\left\{\|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|_1 > \sqrt{\frac{M_3 s_0^2 \log p}{\min_{j\in\mathcal{S}_0} n_j^{\text{eff}}}} \;\middle|\; \mathcal{D}_n\right\} &= o(1).
\end{aligned} \tag{23}$$

Equation (22) shows that the efficiency gain of SSTL for each covariate $j\in\mathcal{S}_0$ depends on the sample size of all source studies k with $b_{0j}^{(k)}=0$, i.e. the efficiency is proportional to the source-to-target homogeneity level. Since $n_j^{\text{eff}}\geq n_T$, Theorem 3.4 shows that the contraction rate of SSTL is never worse than the rate of a single-study analysis with DSS prior irrespective of $\mathcal{R}_0$, thus preventing negative information transfer. The rate in Theorem 3.4 differs from that of [61] who obtain (23) with $n_T+\sum_{k\in\mathcal{W}} n_k$ where $\mathcal{W}=\{k>0 \mid \exists j : \beta_{0j}\neq 0 \text{ and } b_{0j}^{(k)}=0\}$. Zhang et al. [61] work under the stronger assumption $n_j^{\text{eff}}\asymp n$. Under this assumption, the rates in (23) are substantially improved, replacing $\min_{j\in\mathcal{S}_0} n_j^{\text{eff}}$ by n.

4 Selection of informative sources

Within our augmented DSS framework (11), the global hyperparameter $\psi^{(k)}$ – assumed to follow a prior $1-\Phi(\psi^{(k)})\sim Beta(1,\lambda^{(k)})$ with $\lambda^{(k)}=c_k p^{a_k}$ – controls the sparsity of the source-specific bias vector $\boldsymbol{b}^{(k)}$ through the prior probability $\omega^{(k)}=\pi(b_j^{(k)}\neq 0)$ of a non-zero bias for covariate j in source study k. This probability $\omega^{(k)}$ quantifies the prior information on the overall similarity between source dataset k and the target data, as

$b_j^{(k)} = 0$ is equivalent to the j-th covariate having the same effect in both datasets. Large values of $\psi^{(k)}$ therefore correspond to strong prior beliefs that the covariate effects in source study k and the target population will be very similar, i.e. $\boldsymbol{\beta}^{(k)} \approx \boldsymbol{\beta}$, thereby maximizing information borrowing.

Since the true similarity between the covariate effects $\boldsymbol{\beta}$ and $\boldsymbol{\beta}^{(k)}$ of the target and source populations is *a priori* unknown, the pre-specification of the parameters of the Beta prior distributions can be undesirable in practice and may lead to suboptimal or potentially negative information transfer. To address this issue, we directly estimate the hyperparameters $\boldsymbol{\lambda} = (\lambda^{(T)}, \lambda^{(1)}, \ldots, \lambda^{(K)})$ via Empirical Bayes (EB), maximizing the marginal likelihood (see Section 5.1 for computational details) given by

$$\widehat{\boldsymbol{\lambda}} \in \arg\max_{\boldsymbol{\lambda}>0} p_{\boldsymbol{\lambda}}(\mathcal{D}_n), \qquad \text{where} \tag{24}$$
$$p_{\boldsymbol{\lambda}}(\mathcal{D}_n) = \int p(\mathcal{D}_n \mid \boldsymbol{\omega}) \prod_{k\in\{T,1,\ldots,K\}} \lambda^{(k)}(1-\omega^{(k)})^{\lambda^{(k)}-1}\, d\boldsymbol{\omega},$$

where $p(\mathcal{D}_n \mid \boldsymbol{\omega})$ is the marginal likelihood with other model parameters integrated out. This approach can be viewed as a data-driven source selection procedure by allowing the data to determine the sparsity level for each bias term $\boldsymbol{b}^{(k)}$. The model adaptively scales the degree of information borrowing based on the similarity of each source to the target.

5 Posterior inference

Markov Chain Monte Carlo (MCMC) is used to approximate the posterior distribution of our Bayesian TL model. We use the latent Gaussian representations (3) and (11) for our DSS TL prior to construct a tuning-free MCMC algorithm that is applicable with any (potentially non-Gaussian) residual distribution $p_\sigma(\cdot)$ in (1). As our algorithm utilizes slice sampling, we will briefly review the closely related elliptical slice sampling (ESS) [40].

Algorithm 1 Single iteration of the Elliptical Slice Sampler

Require: current state $\boldsymbol{\theta}^{(t)}$, log likelihood function $\log L(\cdot)$
1: Draw $\boldsymbol{\nu}^{(t)} \sim N(\mathbf{0}, \boldsymbol{\Sigma})$
2: Draw $U^{(t)} \sim Unif(0,1)$ and set $l_{thres} = \log L(\boldsymbol{\theta}^{(t)}) + \log U^{(t)}$
3: Generate an angle $\xi \sim Unif(0, 2\pi)$ and define $[\xi_{\min}, \xi_{\max}] = [\xi - 2\pi, \xi]$
4: Set the initial proposal to $\boldsymbol{\theta}^* = \cos(\xi)\boldsymbol{\theta}^{(t)} + \sin(\xi)\boldsymbol{\nu}^{(t)}$
5: **while** $\log L(\boldsymbol{\theta}^*) < l_{thres}$ **do**
6: Shrink the interval: if $\xi < 0$, set $\xi_{\min} = \xi$, else $\xi_{\max} = \xi$
7: Draw new angle $\xi \sim Unif(\xi_{\min}, \xi_{\max})$ and set $\boldsymbol{\theta}^* = \cos(\xi)\boldsymbol{\theta}^{(t)} + \sin(\xi)\boldsymbol{\nu}$
8: **end while**
9: **Return** the new state $\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^*$

The ESS algorithm can be used to sample from (posterior) distributions of the form $g(\boldsymbol{\theta}) \propto L(\boldsymbol{\theta})N(\boldsymbol{\theta} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})$ with a normal prior and generic likelihood $L(\boldsymbol{\theta})$. We can introduce a latent variable U with joint distribution $g_{(\boldsymbol{\theta},U)}(\boldsymbol{\theta}, U) \propto \mathcal{I}\{U \leq L(\boldsymbol{\theta})\}N(\boldsymbol{\theta} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})$ and marginal distribution $g(\boldsymbol{\theta})$. Sampling from $g(\boldsymbol{\theta})$ can therefore be implemented via a vanilla Gibbs sampler on the augmented parameter space $(\boldsymbol{\theta}, U)$ with full conditionals

$$g(U|\boldsymbol{\theta}) = Unif(U \mid 0, L(\boldsymbol{\theta})) \quad \text{and } g(\boldsymbol{\theta} \mid U) \propto N(\boldsymbol{\theta} \mid \mu, \Sigma)\mathcal{I}(\boldsymbol{\theta} \in \mathcal{A}_U),$$

where $\mathcal{A}_U = \{\boldsymbol{\theta} \mid L(\boldsymbol{\theta}) \geq U\}$. Now, instead of drawing at each MCMC iteration t a new $\boldsymbol{\theta}^{(t)}$ from the prior restricted to $\mathcal{A}_{U^{(t)}}$ one can use the fact that for any angle $\xi \in [\xi_0, \xi_0 + 2\pi]$

$$\boldsymbol{\theta}^\star_\xi = \boldsymbol{\mu} + \cos(\xi) \times (\boldsymbol{\theta}^{(t-1)} - \boldsymbol{\mu}) + \sin(\xi) \times (\boldsymbol{\nu} - \boldsymbol{\mu}) \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \tag{25}$$

whenever both $\boldsymbol{\nu}, \boldsymbol{\theta}^{(t-1)} \overset{ind}{\sim} N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Selecting a (random) angle $\xi^{(t)}$ such that $\boldsymbol{\theta}^\star_\xi \in \mathcal{A}_{U^{(t)}}$ therefore yields a valid draw from $g(\boldsymbol{\theta} \mid U^{(t)})$ – and, since $U^{(t)} \leq L(\boldsymbol{\theta}^{(t-1)})$, at least one angle $\xi^{(t)} = 0$ exists such that $\boldsymbol{\theta}^\star_0 \in \mathcal{A}_{U^{(t)}}$. Drawing $\xi^{(t)}$ can be done efficiently via the shrinkage interval algorithm of [42]. The procedure is summarized in Algorithm 1.

Setting $\boldsymbol{\theta}^{(k)} = (\boldsymbol{e}^{(k)\prime}, \boldsymbol{v}^{(k)\prime})'$ for each $k = T, 1, \ldots, K$ and $\boldsymbol{\theta} = (\boldsymbol{\theta}^{(T)}, \boldsymbol{\theta}^{(1)}, \ldots, \boldsymbol{\theta}^{(K)})' \in \mathbb{R}^{2p(K+1)}$, with $\boldsymbol{\psi} = (\psi, \psi^{(1)}, \ldots, \psi^{(K)})$ and $\boldsymbol{\sigma} = (\sigma, \sigma^{(1)}, \ldots, \sigma^{(K)})$ fixed, we could directly apply ESS to obtain, for any residual distribution $p_\sigma(\epsilon)$, posterior samples for $\boldsymbol{\beta}, \boldsymbol{b}^{(1)}, \ldots, \boldsymbol{b}^{(K)}$ given $\boldsymbol{\psi}$ and $\boldsymbol{\sigma}$. In high dimensional models, values of $\theta^\star_\xi$ on the ellipse (25)

belonging to $\mathcal{A}_{U^{(t)}}$ are highly constrained towards angles close to $\xi^{(t)} = 0$ (i.e. $\boldsymbol{\theta}^{(t)} \approx \boldsymbol{\theta}^{(t-1)}$), which leads to high autocorrelation, poor mixing and slow exploration of the posterior support. To address this issue, we propose an MCMC algorithm that incorporates ESS within a Gibbs sampler. We first define a partition of the $2p(K+1)$ coordinates of $\boldsymbol{\theta}$ into m sets $\mathcal{A} = \{\mathcal{A}_1, \ldots, \mathcal{A}_m\}$, i.e., $\bigcup_{\ell=1}^{m} \mathcal{A}_\ell = \{1, 2, \ldots, 2p(K+1)\}$, $\mathcal{A}_j \bigcap \mathcal{A}_k = \varnothing$ for $j \neq k$. Let $\boldsymbol{\theta}_{\mathcal{A}_\ell} = (\theta_j : j \in \mathcal{A}_\ell)'$. Similarly, $\boldsymbol{\theta}_{\mathcal{A}_{-\ell}}$ correspond to the sub-vectors with coordinates belonging to sets $\mathcal{A}_j, j < \ell$ and $\boldsymbol{\theta}_{\mathcal{A}_{+\ell}}$ indicates the components of $\boldsymbol{\theta}$ with coordinates belonging to sets $\mathcal{A}_j, j > \ell$. The size of each set $\mathcal{A}_\ell$ is substantially lower than $2p(K+1)$, in the order of 1 to 10. We then consider a Gibbs algorithm that samples each parameter block $\boldsymbol{\theta}_{\mathcal{A}_\ell}$ from the full conditional distribution

$$\pi\left(\boldsymbol{\theta}_{\mathcal{A}_\ell} \mid \boldsymbol{\theta}_{\mathcal{A}_{-\ell}}, \boldsymbol{\theta}_{\mathcal{A}_{+\ell}}, \boldsymbol{\psi}, \mathcal{D}_n\right) \propto L\left(\boldsymbol{\theta}_{\mathcal{A}_\ell}; \boldsymbol{\theta}_{\mathcal{A}_{-\ell}}, \boldsymbol{\theta}_{\mathcal{A}_{+\ell}}, \boldsymbol{\psi}\right) N(\boldsymbol{\theta}_{\mathcal{A}_\ell} \mid \mathbf{0}_{|\mathcal{A}_\ell|}, I_{|\mathcal{A}_\ell|})$$

using a single ESS-step similar to Algorithm 1.

To take advantage of the block-wise ESS sampler, we use a latent Gaussian representation of the non-Gaussian variable $\boldsymbol{\psi}$. Specifically, if $\omega^{(k)} = 1 - \Phi(\psi^{(k)}) \sim Beta(1, \lambda^{(k)})$, then $\psi^{(k)}$ is equivalent in distribution to $\psi^{(k)} = \Phi^{-1}(\Phi(Z^{(k)})^{1/\lambda^{(k)}})$ where $Z^{(k)} \sim N(0,1)$. The full algorithm is described in Algorithm 2.

5.1 Empirical Bayes hyperparameter estimation

We describe the implementation of the empirical Bayes estimates (24) of the sparsity parameters $\boldsymbol{\lambda}$ that govern source data selection and information transfer outlined in Section 4. Our algorithm is motivated by the empirical Bayes procedure based on an augmented MCMC sampler in [7]. We use an EM algorithm, treating $\boldsymbol{\psi}$ as latent variables, to obtain the maximum likelihood estimate $\hat{\boldsymbol{\lambda}} = \arg\max p_\lambda(\mathcal{D}_n)$ by cycling at each iteration $t = 1, 2, \ldots$ of the algorithm through the usual E-step $Q(\boldsymbol{\lambda} \mid \boldsymbol{\lambda}_t) = E_{\pi_{\boldsymbol{\lambda}_t}(\boldsymbol{\psi}|\mathcal{D}_n)}[\log p_{\boldsymbol{\lambda}}(\mathcal{D}_n, \boldsymbol{\psi})]$

Algorithm 2 Full MCMC algorithm

Require: Target data $\mathcal{D}^{(T)}$, source data $\mathcal{D}^{(1)}, \ldots, \mathcal{D}^{(K)}$
Require: Partitions for the target and each source vector $\mathcal{A}^{(T)}, \mathcal{A}^{(1)}, \ldots, \mathcal{A}^{(K)}$
Require: Sparsity hyperparameter $\boldsymbol{\lambda}$

Initialize all latent parameters $\boldsymbol{\theta}^{(k)}, Z^{(k)}, \alpha^{(k)}, \sigma^{(k)}$ for all populations $k = T, 1, \ldots, K$.
for $c = 1, \ldots, C$ **do**
  for each $\mathcal{A}_\ell^{(T)}$ in $\mathcal{A}^{(T)}$ **do** ▷ Update the target vector $\boldsymbol{\theta}^{(T)}$
    Update $\boldsymbol{\theta}^{(T)}_{\mathcal{A}_\ell^{(T)}}$ given $\boldsymbol{\theta}^{(T)}_{\mathcal{A}_{-\ell}^{(T)}}, \boldsymbol{\theta}^{(T)}_{\mathcal{A}_{+\ell}^{(T)}}, \boldsymbol{\theta}^{(1)}, \ldots, \boldsymbol{\theta}^{(K)}$ via Algorithm 1.
  end for
  Update α via Algorithm 1.
  Update the residual variability σ via random-walk Metropolis-Hastings sampler.
  Sample $Z^{(T)}$ via Algorithm 1 and set $\psi^{(T)} = \Phi^{-1}(\Phi(Z^{(T)})^{1/\lambda^{(T)}})$.

  for $k = 1, \ldots, K$ **do**
    for $A_\ell^{(k)} \in \mathcal{A}^{(k)}$ **do** ▷ Update the source vector $\boldsymbol{\theta}^{(k)}$
      Update $\boldsymbol{\theta}^{(k)}_{\mathcal{A}_\ell^{(k)}}$ given $\boldsymbol{\theta}^{(k)}_{\mathcal{A}_{-\ell}^{(k)}}, \boldsymbol{\theta}^{(k)}_{\mathcal{A}_{+\ell}^{(k)}}, \boldsymbol{\theta}^{(T)}$ and $\boldsymbol{\theta}^{(k')}, k' \neq k$ via Algorithm 1.
    end for
    Update $\alpha^{(k)}$ via Algorithm 1.
    Update $\sigma^{(k)}$ via random-walk Metropolis-Hastings sampler.
    Sample $Z^{(k)}$ via Algorithm 1 and set $\psi^{(k)} = \Phi^{-1}(\Phi(Z^{(k)})^{1/\lambda^{(k)}})$.
  end for
end for

followed by the M-step $\boldsymbol{\lambda}_{t+1} = \arg\max_{\boldsymbol{\lambda}} Q(\boldsymbol{\lambda} \mid \boldsymbol{\lambda}_t)$. Since the expectation $Q(\boldsymbol{\lambda} \mid \boldsymbol{\lambda}_t)$ is not available in closed form, we use our Gibbs sampler to approximate the expectation in the *E-step* using a Monte Carlo estimator of $Q(\boldsymbol{\lambda}|\boldsymbol{\lambda}_t)$. Specifically, the MC-EM algorithm starts with an initial guess $\boldsymbol{\lambda}_0$, and then repeats the following steps until convergence:

1. Draw M samples $\boldsymbol{\psi}_1, \ldots, \boldsymbol{\psi}_M$ from $\pi(\boldsymbol{\psi}|\mathcal{D}_n, \boldsymbol{\lambda}_t)$ using Algorithm 2.

2. MC-E step: Define $Q(\boldsymbol{\lambda}|\boldsymbol{\lambda}_t) = \frac{1}{M}\sum_{m=1}^{M} \log p_{\boldsymbol{\lambda}}(\boldsymbol{\psi}_m, \mathcal{D}_n)$

3. M Step: set $\lambda_{t+1}^{(k)} = \underset{\lambda^{(k)}>0}{\arg\max}\, Q(\boldsymbol{\lambda}|\boldsymbol{\lambda}_t) = -M/\sum_{m=1}^{M} \log \Phi(\psi_m^{(k)})$ for $k = T, 1, \ldots, K$.

In our empirical evaluations we observe that a reliable performance of the above MC-EM algorithm requires a large number of samples M, as the Monte Carlo approximation error of the Q-function dominates the M-step. This can lead to convergence issues and oscillations of

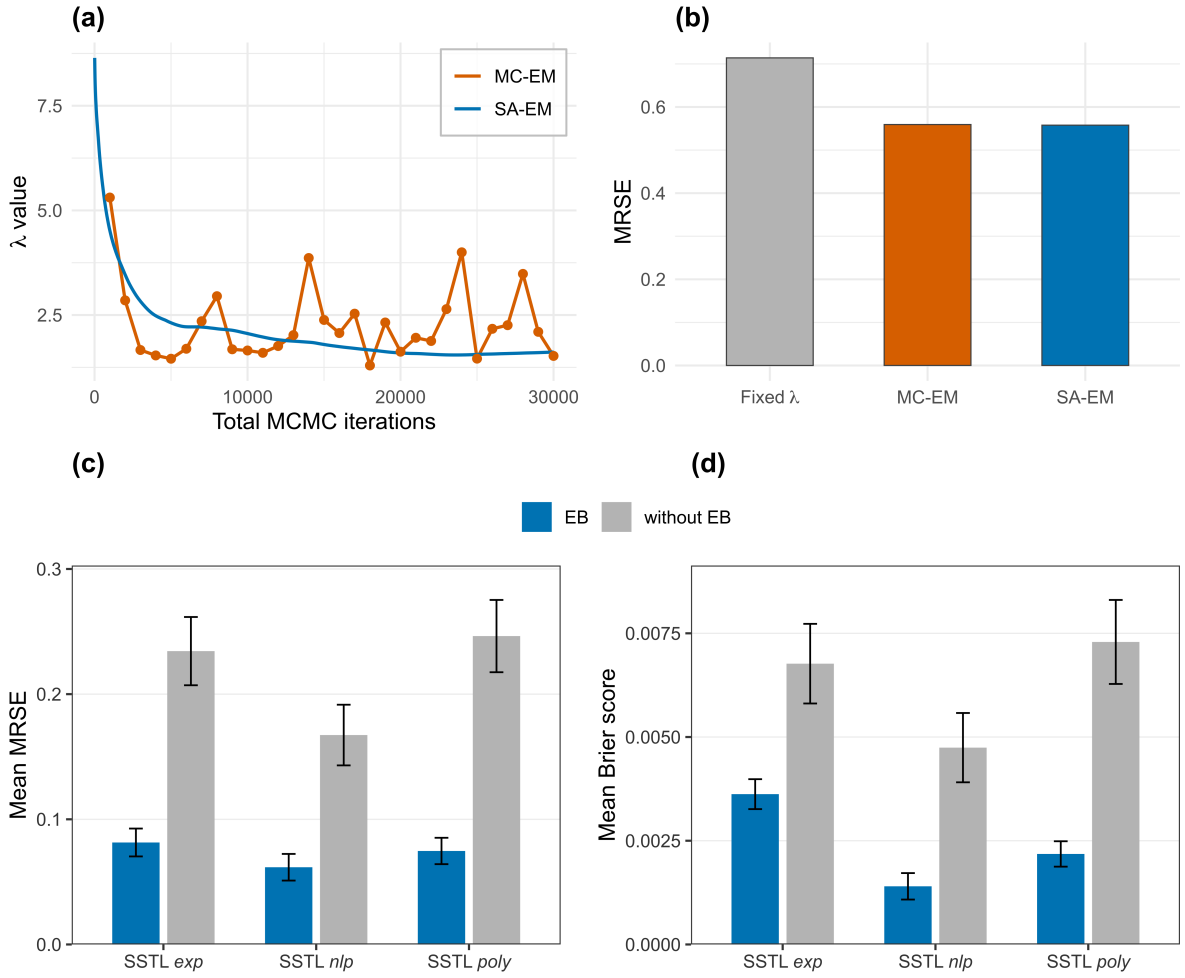


Figure 3: Comparison of SSTL with and without empirical Bayes source study selection. Panel (a) shows optimization trajectories of MC-EM and the proposed SA-EM algorithm for a simulated dataset ($K = 4, s_0 = 20, p = 200$ and *exp* slab). Panel (b) shows for the same dataset, the MRSE $= \mathbb{E}[\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_2^2 / \|\boldsymbol{\beta}_0\|_2^2]$ of the point estimator $\widehat{\boldsymbol{\beta}} = \mathbb{E}[\boldsymbol{\beta} \mid \mathcal{D}_n]$ using either a fixed $\boldsymbol{\lambda}$, MC-EM or SA-EM to estimate $\boldsymbol{\lambda}$. Panel (c) and (d) shows the average MRSE and the average Brier score for $\boldsymbol{\beta}$ across 100 simulations of $K = 4$ datasets.

the updated parameters $\boldsymbol{\lambda}_t$ around the optimum $\hat{\boldsymbol{\lambda}}$. Figure 3(a) visualizes this phenomenon, for a small *in silico* dataset with $n_T = 150$ observations and $p = 200$ potential predictors of survival, in which $s_0 = 20$ have a non-null effect on the outcome. To overcome this issue, we adapt a scheduled stochastic approximation EM (SA-EM) algorithm [13], which reduces the computational burden (i.e., large M) and has, according to our evaluations, better convergence behavior. The procedure is described in Algorithm 3.

Algorithm 3 Stochastic Approximation EM algorithm for EB estimation

Require: Initial values for $\boldsymbol{\lambda}_0$, $\boldsymbol{\theta}_0$, $\boldsymbol{\delta}_0$, set iteration index $t = 1$.
Require: Hyperparameters $\gamma \in (0.5, 1]$, $\alpha_{lr} \in (0, 1]$, $\phi > 0$.

while not converged **do**
  Draw M posterior samples conditional on $\boldsymbol{\lambda}_{t-1}$ via Algorithm 2
  for $k = T, 1, \ldots, K$ **do**
    Set $\delta_t^{(k)} = (1 - t^{-\gamma})\delta_{t-1}^{(k)} + t^{-\gamma} \frac{1}{M} \sum_{m=1}^{M} \log \Phi(\psi_m^{(k)})$
    Set $\bar{\lambda}^{(k)} = -1/\delta_t^{(k)}$,
    Set $\lambda_t^{(k)} = (1 - \alpha_{lr} \cdot t^{-\phi})\lambda_{t-1}^{(k)} + \alpha_{lr} \cdot t^{-\phi} \bar{\lambda}^{(k)}$
  end for
  $t = t + 1$
end while

5.2 Computational Demands

A direct implementation of Algorithm 2 can be computationally expensive due to multiple likelihood evaluations during the repeated angle-refinement step of the slice-sampling component. Naively, this would involve recomputing the linear predictor $\boldsymbol{\mu}_{(k)} = \alpha^{(k)} + \boldsymbol{X}^{(k)}\boldsymbol{\beta}^{(k)}$ and the residual $\boldsymbol{r}_{(k)} = \log \boldsymbol{y}^{(k)} - \boldsymbol{\mu}_{(k)}$ for each study $k = T, 1, \ldots, K$ at each likelihood evaluation. However, careful evaluations of the likelihood can avert unnecessary computations by storing and updating the residuals $\boldsymbol{r}_{(k)}$ efficiently. For simplicity, assume that a single Gibbs-step via ESS only updates parameters involving a subset $\mathcal{J} \subset \{1, \ldots, p\}$ of the regression coefficients in a single dataset $k, k \geq 1$. Each proposed parameter update $\boldsymbol{\beta}_{\mathcal{J}}^{(k,t-1)} \to \boldsymbol{\beta}_{\mathcal{J}}^{(k,t)}$ only requires an update of the form $\boldsymbol{r}_{(k)}^{(t)} = \boldsymbol{r}_{(k)}^{(t-1)} + \Delta\boldsymbol{r}$, where $\Delta\boldsymbol{r} = \boldsymbol{X}_{\mathcal{J}}^{(k)}(\boldsymbol{\beta}_{\mathcal{J}}^{(k,t-1)} - \boldsymbol{\beta}_{\mathcal{J}}^{(k,t)})$ for dataset k, which is significantly faster than computing the full product $\boldsymbol{X}^{(k)}\boldsymbol{\beta}^{(k)}$ for all datasets when $|\mathcal{J}|$ is small. In our numerical studies, larger blocks did not improve runtime meaningfully, whereas singleton blocks gave the best mixing. By default, we therefore set the size of each block $\mathcal{A}_\ell$ to 1. An R package is available at https://github.com/andyljk/SSTL.

6 Simulation studies

We report results of several simulation studies to evaluate the empirical performance of our TL model from Examples 2.1–2.3. We first compare the different parameterizations of SSTL under a range of simulation settings, then compare our models with existing alternative methods.

6.1 Comparison of different TL slab functions

Data Generation. We set the size of datasets to $n_T = 100$ and $n_k = 200$ for each source. Covariates are generated following $\boldsymbol{x}_i^{(k)} \overset{iid}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma})$ for all $k = T, 1, \ldots, K$ and $p = 500$, with the Toeplitz covariance matrix $\Sigma_{jj'} = 0.7^{|j-j'|}$. Given $\boldsymbol{x}_i^{(k)}$, the log-survival times are generated from the AFT models (1) and (2), under the extreme value residual distribution with scale parameters $\sigma = \sigma^{(1)} = \cdots = \sigma^{(K)} = 1$. Censoring times are iid exponentially distributed, and the mean is selected to achieve a 20% censoring rate.

Scenarios. We vary several outcome parameters: (a) The true target vector $\boldsymbol{\beta}_0$ has either $s_0 \in \{5, 20, 50\}$ nonzero entries. Each non-zero coefficient is randomly set to ± 0.5. (b) The number of source datasets equals $K = 1, \ldots, 10$. (c) We consider two settings of source data alignment: *Homogeneous setting*: The true bias $\boldsymbol{b}_0^{(k)} = \mathbf{0}_p$ for all k. *Heterogeneous setting:* $\boldsymbol{b}_0^{(k)} = \mathbf{0}_p$ for the first $\lfloor K/2 \rfloor$ sources. For the remaining studies, we set $b_{0j}^{(k)} = 0$ for $j = 1, \ldots, \lfloor s_0/2 \rfloor$, and $b_{0j}^{(k)} = \pm 0.5$ randomly for $j = \lfloor s_0/2 \rfloor + 1, \ldots, 2s_0$. For each scenario, we generate 100 repetitions of target and source datasets.

Metrics and benchmarking. We evaluate the estimation and variable selection performance of our TL models. For a fixed scenario, we use the Mean Relative Squared Error, $\text{MRSE} = \mathbb{E}[\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_2^2 / \|\boldsymbol{\beta}_0\|_2^2]$ to evaluate the estimation accuracy of $\widehat{\boldsymbol{\beta}} = \mathbb{E}[\boldsymbol{\beta} \mid \mathcal{D}_n]$. Variable selection performance is measured by the Brier score $E[\hat{f}_B]$, where $\hat{f}_B = \frac{1}{p}\sum_{j=1}^{p}(\hat{p}_j -$

$o_j)^2$ with $o_j = \mathcal{I}(\beta_{0j} \neq 0)$ and $\hat{p}_j = \pi(\beta_j \neq 0|\mathcal{D}_n)$. Lower scores reflect better variable selection accuracy. For initial benchmarking, we consider a *Target only* reference model, which used the same likelihood (1) and prior (3) without using the source datasets, and a *Pooled model* that pools all datasets and applies the *Target only* model. For illustration, we consider the *nlp* model in comparisons with these two models.

Results. Here we show results for $s_0 = 20$. Results for $s_0 = 5$ and 50 are shown in Supplementary Figure S1. Figure 4(a) shows the MRSE for SSTL when $K = 1, 2, \ldots,$ and 10. As expected, for all TL models the estimation and variable selection accuracy increases greatly with K. For instance, in the *homogeneous* setting, when $K = 1$, we observe an MRSE of approximately 0.4. In comparison, when $K = 2, 4, 6, 8$ or 10 the MRSE decreases to 0.05, 0.014, 0.011, 0.008, and 0.006. Across all K, the *nlp* model outperforms the remaining two slab models. We observe similar patterns in heterogeneous settings.

Figure 4(b) shows the variable selection performance of SSTL. We again observe substantial improvements in the identification of predictive covariates as K increases. We also observe that the *nlp* model outperforms the remaining two slab models by a substantial margin. For instance, in the homogeneous setting the *nlp* model with $K = 2$ source studies has a Brier score of 0.001. To achieve a comparable Brier score the remaining two TL models require at least $K = 4$ (*poly*) and $K = 8$ (*exp*) source studies.

In the *homogeneous* setting, simple pooling of all datasets into a single dataset (*Pooled* model) is optimal. Figure 4(c) shows that in this setting the TL model, which accounts for potential differences between the target and source datasets, has approximately the same efficiency as the *Pooled* model. In the *heterogeneous* setting (Figure 4(c)), the TL model significantly outperforms the *Pooled* model, highlighting the robustness to negative transfer via the empirical Bayes source selection procedure (see also Panels (c) and (d) in Figure 3).

For instance, when $K = 2$ or $K = 10$, the *Pooled* model had similar MRSE as the *Target only* model. In contrast, the TL models show reductions in MRSE of $\approx 62\%$ ($K = 2$) and $\approx 98\%$ ($K = 10$) compared to the *Target only* model.

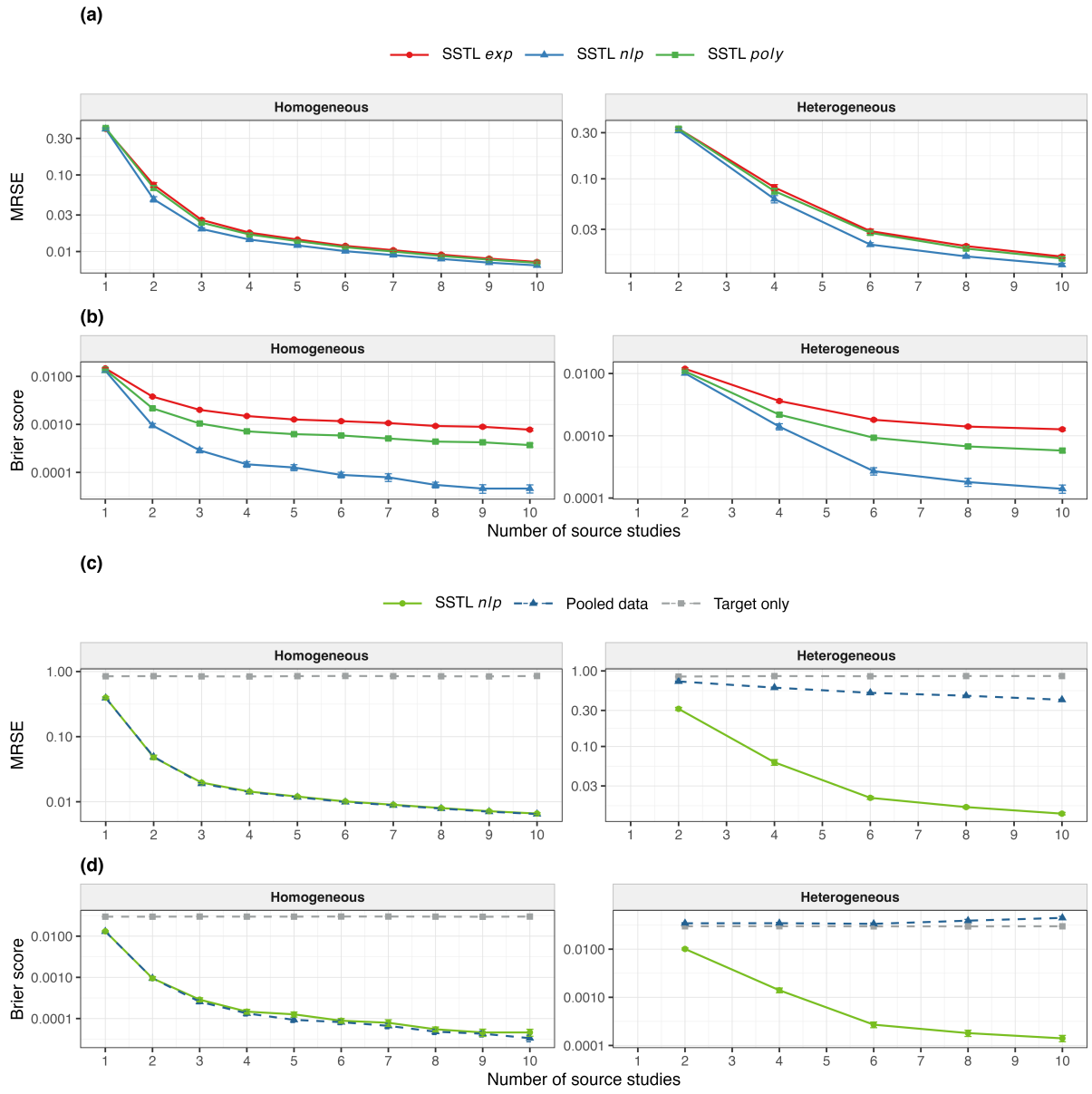


Figure 4: Performance of SSTL under *homogeneous* and *heterogeneous* source designs with $s_0 = 20$ and $p = 500$ across 100 simulations. Panels (a) and (b) show MRSE and Brier scores for the three proposed slab models. Panels (c) and (d) show MRSEs and Brier scores for SSTL *nlp*, pooled, and target-only analyses.

6.2 Comparison with alternative data analysis

To enable a comparison of SSTL with the closest existing TL models, we initially consider stylized settings without censoring.

Our **initial scenarios** specify $K = 2$ and 8, $p = 500$ and $s_0 = 10$. For the target vector $\boldsymbol{\beta}_0$ we specified *two settings*: (1) *Uniform effects* – each active β_{0j} is randomly set to ± 0.5. (2) *Varying effects* – the $s_0 = 10$ active β_{0j} equal $\pm\{0.8, 0.8, 0.6, 0.6, 0.6, 0.4, 0.4, 0.4, 0.2, 0.2\}$ with random signs. Otherwise, the data-generating mechanism is identical to the *heterogeneous* design from Section 6.1. We provide **additional results** for (i) log-normal residual and (ii) active parameters fixed at $\beta_{0j} = +0.5$ in Supplemental Section S1.

We then specify an additional **higher-dimensional scenario** similar to [55]: In summary, $p = 1000$, $s_0 = 15$, $n_k = 100$ for $k = T, 1, \ldots, K$, $K \in \{2, 4, 6, 8, 10\}$, and heterogeneity levels $h \in \{10, 20\}$ (see below for the latter). Covariates in $\mathcal{D}^{(T)}$ are iid $N_p(\mathbf{0}, \boldsymbol{\Sigma})$, $\Sigma_{ij} = 0.9^{|i-j|}$, whereas source covariates $\boldsymbol{X}_i^{(k)}$ are iid t_4-distributed. Log-survival times follow a Gaussian AFT with unit error variance, intercept $\alpha = 0$, target coefficients $\beta_{0j} = 0.5$ for $j = 1, \ldots, s_0$ and $\beta_{0j} = 0$ otherwise. Source coefficients are set to $\beta_{0j}^{(k)} = 0.5 + (h/p)R_j^{(k)}$ for $j \in \mathcal{A}^{(k)}$ and $\beta_{0j}^{(k)} = (h/p)R_j^{(k)}$ for $j \notin \mathcal{A}^{(k)}$, where $R_j^{(k)} \in \{-1, 1\}$ are independent Rademacher random variables. For the first $K/2$ ('informative') source datasets, $\mathcal{A}^{(k)} = \{1, \ldots, s_0\}$. For each of the remaining $K/2$ ('noninformative') source datasets, $\mathcal{A}^{(k)} = \{s_0 + 1, \ldots, 2s_0\} \cup \mathcal{B}^{(k)}$, where $\mathcal{B}^{(k)}$ is a randomly selected subset of $\{2s_0 + 1, \ldots, p\}$ with size $|\mathcal{B}^{(k)}| = s_0$. Furthermore, an intercept $\alpha^{(k)} = 0.5$ is added to each noninformative source, while $\alpha^{(k)} = 0$ for the remaining source studies.

Alternative methods. We consider the TL methods (i) BLAST [26], (ii) GLMtrans [55], (iii) CONCERT [61], (iv) Trans Lasso [36], (v) Trans-AFT [46], (vi) Transfer Cox [38]. The first four methods cannot handle censored data (they can estimate AFT models

without censoring) and thus aren't applicable to survival analysis with censoring.

Operating characteristics. In addition to MRSE and Brier score, we consider (iii) the average F1 score, $\mathbb{E}[\widehat{F1}]$, where $\widehat{F1} = \frac{2|\mathcal{S}_0 \cap \hat{\mathcal{S}}|}{|\mathcal{S}_0|+|\hat{\mathcal{S}}|}$ and $\hat{\mathcal{S}} = \{j : \pi(\beta_j \neq 0 \mid \mathcal{D}_n) > 0.5\}$, (iv/v) the coverage of the 95% posterior intervals for the non-zero (Cov_1) and zero (Cov_0) coefficients, (vi/vii) the average length of these two intervals (Len_1 and Len_0), and the expected average interval scores $\text{Int}_\ell = \mathbb{E}\left[\sum_{j\in\mathcal{I}_\ell} S_{j,\text{Int}}/|\mathcal{I}_\ell|\right]$ across the non-zero ($\ell = 1, \mathcal{I}_1 = \{1, \ldots, s_0\}$), the zero ($\ell = 0, \mathcal{I}_0 = \{s_0 + 1, \ldots, p\}$), and all p coefficients ($\ell = a, \mathcal{I}_a = \{1, \ldots, p\}$). The interval score $S_{j,\text{Int}}$ [19] for each variable $j = 1, \ldots, p$ is defined as $S_{j,\text{Int}} = (u_j - l_j) + 2/\alpha(l_j - \beta_{0j})\mathbb{1}(\beta_{0j} < l_j) + 2/\alpha(\beta_{0j} - u_j)\mathbb{1}(\beta_{0j} > u_j)$, where l_j, u_j denote the estimated lower and upper ends of the $(1-\alpha)$-interval, respectively.

Results. Table 2 summarizes operating characteristics for all TL models when $p = 500$. SSTL substantially outperform all competing methods in both estimation and variable selection accuracy. The differences become more pronounced with larger K. Our TL methods also yield high coverage for inactive coefficients with very narrow intervals, reflecting the exact sparsity induced by the discrete spike component. The coverage of SSTL for active coefficients improves with larger K, and the average 95% posterior interval lengths of active coordinates decrease substantially. Figure 5 shows a similar relative performance of SSTL compared to the alternative models in **higher-dimensional scenarios** with $p = 1,000$.

Scenarios with censoring. We now compare SSTL to existing TL methods for survival analysis with censoring. The outcome scenarios we consider here are nearly identical to our **initial scenarios** above, i.e. $p = 500$ predictors, $s_0 = 10$ non-zero covariate effects, and $K \in \{2, 8\}$ source studies, and the residuals follow an extreme value distribution. We tuned the mean of the exponential censoring times so that 30%, 50% or 70% of the outcomes in each dataset are right-censored.

Table 2: Comparison of SSTL with existing transfer-learning methods under two signal configurations in the *heterogeneous* design.

Method	K	MRSE	$10^2\times$ Brier	F1	Cov_1	Cov_0	Len_1	Len_0	Int_1	Int_0	Int_{all}
Panel A: Uniform Effect Size											
SSTL *exp*	2	0.139	0.283	0.932	0.856	**1.000**	0.401	0.009	**1.361**	0.009	0.036
SSTL *nlp*	2	**0.126**	**0.217**	0.934	0.858	**1.000**	**0.340**	**0.003**	1.616	**0.003**	0.035
SSTL *poly*	2	0.129	0.226	**0.939**	**0.863**	**1.000**	0.385	0.005	1.420	0.005	**0.034**
BLAST	2	0.679	1.289	0.354	0.548	**1.000**	0.658	0.132	5.647	0.132	0.242
GLMtrans	2	0.671	13.459	0.191	0.667	0.870	0.531	0.537	2.052	0.996	1.017
CONCERT	2	0.657	1.222	0.576	–	–	–	–	–	–	–
Trans Lasso	2	0.593	8.185	0.408	–	–	–	–	–	–	–
Trans-AFT	2	0.608	1.741	0.572	–	–	–	–	–	–	–
Transfer Cox	2	0.797	3.029	0.380	–	–	–	–	–	–	–
SSTL *exp*	8	0.012	0.069	0.987	0.929	**1.000**	0.191	0.003	0.240	0.003	0.007
SSTL *nlp*	8	**0.010**	**0.014**	**0.996**	**0.942**	**1.000**	**0.183**	**0.000**	**0.226**	**0.000**	**0.005**
SSTL *poly*	8	0.012	0.046	0.988	0.938	**1.000**	0.189	0.002	0.236	0.002	0.006
BLAST	8	0.546	1.023	0.455	0.696	**1.000**	0.731	0.124	3.437	0.124	0.190
GLMtrans	8	0.561	13.137	0.211	0.757	0.872	0.551	0.557	1.607	1.018	1.030
CONCERT	8	0.290	0.670	0.834	–	–	–	–	–	–	–
Trans Lasso	8	0.343	14.878	0.498	–	–	–	–	–	–	–
Trans-AFT	8	0.410	1.661	0.678	–	–	–	–	–	–	–
Transfer Cox	8	0.803	2.785	0.398	–	–	–	–	–	–	–
Panel B: Varying Effect Size											
SSTL *exp*	2	**0.111**	0.529	0.828	**0.757**	**1.000**	0.336	0.006	2.037	0.006	**0.047**
SSTL *nlp*	2	0.119	0.542	0.824	0.678	**1.000**	**0.283**	**0.002**	2.959	**0.002**	0.061
SSTL *poly*	2	**0.111**	**0.515**	**0.832**	0.738	**1.000**	0.321	0.004	2.298	0.004	0.050
BLAST	2	0.557	1.257	0.391	0.577	**1.000**	0.664	0.129	4.608	0.129	0.218
GLMtrans	2	0.583	13.806	0.183	0.682	0.866	0.532	0.537	**2.033**	1.007	1.027
CONCERT	2	0.415	1.135	0.614	–	–	–	–	–	–	–
Trans Lasso	2	0.479	6.918	0.443	–	–	–	–	–	–	–
Trans-AFT	2	0.444	1.949	0.562	–	–	–	–	–	–	–
Transfer Cox	2	0.770	2.408	0.416	–	–	–	–	–	–	–
SSTL *exp*	8	**0.022**	0.206	0.944	**0.843**	**1.000**	0.191	0.002	**0.642**	0.002	**0.015**
SSTL *nlp*	8	0.023	**0.183**	**0.949**	0.818	**1.000**	**0.176**	**0.000**	0.881	**0.000**	0.018
SSTL *poly*	8	0.023	0.189	0.946	0.834	**1.000**	0.187	0.001	0.754	0.001	0.016
BLAST	8	0.328	0.952	0.557	0.749	**1.000**	0.736	0.122	2.318	0.122	0.166
GLMtrans	8	0.415	13.179	0.196	0.772	0.872	0.552	0.558	1.538	1.025	1.035
CONCERT	8	0.190	0.916	0.759	–	–	–	–	–	–	–
Trans Lasso	8	0.239	12.027	0.483	–	–	–	–	–	–	–
Trans-AFT	8	0.229	1.743	0.641	–	–	–	–	–	–	–
Transfer Cox	8	0.775	2.181	0.411	–	–	–	–	–	–	–

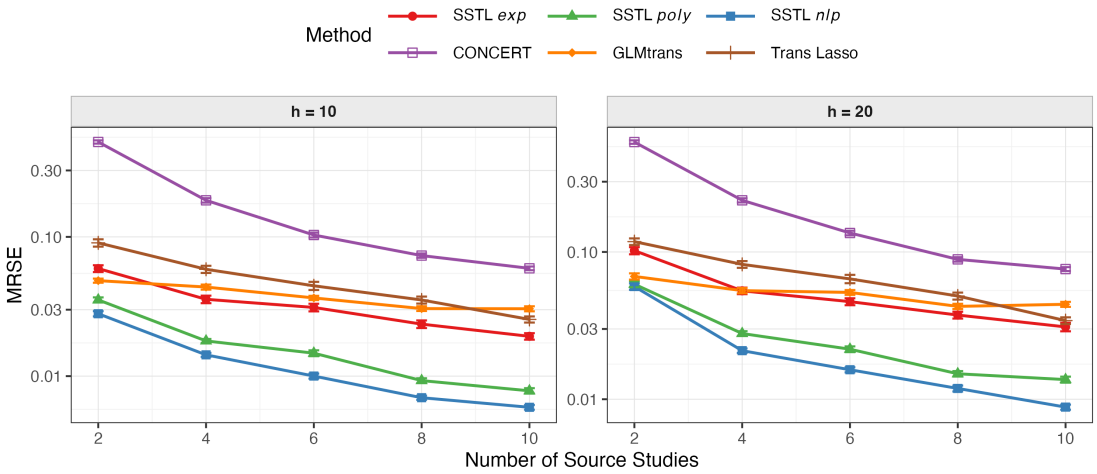


Figure 5: MRSE for higher-dimensional scenario $p = 1{,}000$ and $s_0 = 15$, with source heterogeneity level $h \in \{10, 20\}$.

Alternative methods. Among the above alternative methods, only *Trans-AFT* and *Transfer Cox* are applicable to censored survival data. Since these two methods do not provide uncertainty quantification for their estimates, we focus on the MRSE, Brier and F1 scores. We also attempted to use the recently proposed *CoxTL* [39] – which unfortunately could not handle more than 10 covariates or more than one source study (recall $p = 500$ and $K \in \{2, 8\}$) – and therefore excluded *CoxTL* from our analyses.

Results. Table 3 shows that, under each censoring level, SSTL performs substantially better in estimation. For instance when 50% of the observed survival times are right-censored, SSTL *nlp* has an MRSE of 0.10 when $K = 2$ source datasets are available, compared to 0.80 for *Trans-AFT* and 0.75 for *Transfer Cox*, respectively. These performance differences are more pronounced as K increases, i.e. 0.01 for SSTL *nlp* compared to 0.65 and 0.75 for *Trans-AFT* and *Transfer Cox* when $K = 8$ source datasets are leveraged for information transfer. Similarly, we observe significantly better variable selection accuracy for SSTL compared to *Trans-AFT* and *Transfer Cox*. For instance, at 50% censoring, *nlp* TL has a Brier score of 0.002 for $K = 2$, compared to 0.032 and 0.024 for *Trans-AFT* and

Table 3: Comparison of SSTL with censored AFT transfer-learning methods.

		30% censoring			*50% censoring*			*70% censoring*		
Method	K	MRSE	$10^2\times$ Brier	F1	MRSE	$10^2\times$ Brier	F1	MRSE	$10^2\times$ Brier	F1
SSTL *exp*	2	0.135	0.271	0.930	0.128	0.262	0.935	0.085	0.193	0.958
SSTL *nlp*	2	**0.121**	**0.194**	**0.937**	**0.104**	**0.164**	**0.953**	0.077	**0.113**	**0.965**
SSTL *poly*	2	0.137	0.231	0.924	0.120	0.207	0.939	**0.076**	0.132	**0.965**
Trans-AFT	2	0.722	2.337	0.455	0.804	3.246	0.332	0.879	5.137	0.242
Transfer Cox	2	0.781	2.451	0.419	0.755	2.448	0.469	0.730	2.288	0.513
SSTL *exp*	8	0.012	0.062	0.990	0.013	0.082	0.983	**0.010**	0.054	0.991
SSTL *nlp*	8	0.012	**0.014**	**0.997**	**0.012**	**0.034**	**0.991**	**0.010**	**0.011**	**0.997**
SSTL *poly*	8	**0.011**	0.036	0.992	**0.012**	0.051	0.988	**0.010**	0.054	0.994
Trans-AFT	8	0.578	2.248	0.576	0.649	4.206	0.435	0.776	7.886	0.293
Transfer Cox	8	0.786	2.937	0.408	0.750	2.303	0.482	0.731	2.160	0.515

Transfer Cox, respectively. These differences again become substantially more pronounced when a larger number of source datasets are available (i.e., < 0.001 vs 0.042 and 0.023 for SSTL *nlp*, *Trans-AFT* and *Transfer Cox* when $K = 8$).

7 Real Data Applications

7.1 Predicting survival in ovarian cancer

Ovarian cancer is associated with substantial mortality and exhibits significant heterogeneity in prognosis. Approximately 1 out of every 91 women in the U.S. will be diagnosed during their lifetime [2, 41], and the 5-year survival rate of 52% renders it the 5th leading cause of cancer-related death among women [44]. Common treatment options include combinations of surgery and chemotherapy, with additional approaches selected according to tumor and patient characteristics [10]. Tailoring survival predictions to specific populations is therefore pivotal for personalized treatments, which motivates survival models that transfer information across studies while remaining specific to a target group.

Data and pre-processing. We seek to predict overall survival using demographic and

gene expression data for newly diagnosed ovarian cancer patients at a target site leveraging supplemental source data from the *curatedOvarianData* database [15]. The database provides survival, demographic, and transcriptomic data from 23 studies containing a total of 2970 patients. To evaluate SSTL, we select nine sufficiently large transcriptomic studies (see Table 4) with a total of 1637 patients and 850 events. Observations with negative survival time, missing event status, or incomplete covariate information are removed. We use RNA expression measurements of genes present in all nine datasets as potential predictors. Finally, covariates are standardized separately in each dataset before model fitting.

Each of the nine datasets is used as the target study and the remaining as source studies. For each target analysis, following [25], we perform target-specific feature screening by first retaining genes whose 1st quartile $Q1$ of expression values exceed the inter-quartile range (i.e., $Q1 > IQR$). We then fit separate univariate Cox models for each retained gene and select the 500 genes with the smallest Wald p-values. This set is used in the target and source datasets to fit SSTL.

Alternative methods. We compare SSTL with (i) *Lasso-AFT*, a target-only analysis, and the two survival TL methods from the simulation studies, (ii) *Trans-AFT* and (iii) *Transfer Cox*.

Performance Metrics. We consider the integrated Brier score (IBS) and the cross-validated C-index [22] for each data-analysis approach. The C-index is the proportion of pairs of observations whose predicted and observed survival orderings agree; values of 1 and 0.5 indicate perfect and at-random discrimination. IBS is the time-averaged squared error between predicted survival probabilities and observed survival status, inversely weighted by the censoring probability. Lower IBS values indicate better predictive accuracy.

Results. For each target-dataset, we use 5-fold cross validation (CV) to evaluate the

Table 4: Five-fold cross-validation performance on Ovarian Cancer using Cox p-value gene screening. SSTL reports the *nlp* slab and the Weibull outcome model. Standard deviations across validation folds are shown in parentheses.

		Lasso-AFT		Trans-AFT		Transfer Cox		SSTL	
Target Study	Samples (events)	IBS	C-index	IBS	C-index	IBS	C-index	IBS	C-index
GSE14764	78 (21)	0.40 (0.20)	0.37 (0.11)	0.27 (0.08)	0.48 (0.14)	0.21 (0.07)	0.67 (0.16)	**0.13** (0.03)	**0.83** (0.08)
GSE18520	53 (41)	0.21 (0.07)	0.50 (0.00)	0.22 (0.07)	0.59 (0.16)	0.23 (0.10)	0.65 (0.13)	**0.18** (0.03)	**0.73** (0.09)
GSE26712	185 (129)	0.23 (0.05)	0.52 (0.03)	0.23 (0.05)	0.61 (0.10)	0.21 (0.05)	0.66 (0.06)	**0.17** (0.02)	**0.70** (0.04)
GSE30161	58 (36)	0.39 (0.19)	0.47 (0.05)	0.31 (0.09)	0.58 (0.16)	0.28 (0.12)	**0.76** (0.07)	**0.22** (0.05)	0.64 (0.11)
GSE32062	179 (82)	0.27 (0.09)	0.59 (0.08)	0.24 (0.10)	0.56 (0.09)	0.20 (0.09)	0.68 (0.04)	**0.16** (0.05)	**0.71** (0.05)
GSE49997	194 (57)	0.37 (0.07)	0.62 (0.08)	0.35 (0.03)	0.55 (0.08)	0.20 (0.04)	**0.74** (0.06)	**0.14** (0.02)	**0.74** (0.07)
GSE51088	122 (93)	0.24 (0.03)	0.55 (0.11)	0.25 (0.02)	0.54 (0.06)	0.20 (0.03)	**0.70** (0.04)	**0.17** (0.02)	**0.70** (0.05)
GSE9891	258 (113)	0.23 (0.05)	0.53 (0.04)	0.23 (0.04)	0.50 (0.00)	0.19 (0.04)	0.70 (0.05)	**0.16** (0.02)	**0.75** (0.05)
TCGA	510 (278)	0.18 (0.03)	0.50 (0.00)	0.18 (0.03)	0.59 (0.07)	0.15 (0.03)	0.66 (0.03)	**0.14** (0.02)	**0.69** (0.04)

TL model performance, splitting the target data into training and validation sets, while keeping the full source datasets across CV splits. Table 4 shows the full results. In most target datasets, the TL models outperform the target-only Lasso-AFT model, indicating that at least some sources are informative towards prediction in the target study. Across the nine target studies, SSTL achieves the lowest IBS in every analysis and the highest C-index in eight. We also observe negative transfer in GSE51088 for Trans-AFT and GSE18520 for Transfer Cox, which could suggest that some sources are uninformative in these target analyses. Meanwhile, SSTL remains robust and achieves improved C-index and IBS on both studies.

7.2 Pan-cancer transfer learning using the TCGA database

Background. Cancer treatment development increasingly focuses on therapeutics that target biological mechanisms shared across multiple cancer types. Accordingly, clinical investigations have shifted toward biomarker-driven approaches that match therapies to cancer types most likely to benefit from a given treatment [11]. The pan-cancer analysis pioneered by The Cancer Genome Atlas (TCGA) program integrates data across multiple tumor types and sites to improve our understanding of tumor evolution and to enhance outcome prediction [57]. TCGA provides, among other data types, transcriptomic measurements and survival outcomes for a broad range of cancers. Here, we consider a pan-cancer setting in which different TCGA tumor types that are biologically related to a target cancer are leveraged to improve survival prediction beyond single tumor type predictions. However, direct pooling of these cancer-specific datasets may be inappropriate because of heterogeneity in tumor-specific characteristics, clinical settings, patient populations, and censoring patterns. We therefore applied SSTL to pan-cancer analysis of TCGA datasets.

Data, pre-processing and alternative methods. We selected 9 datasets from the TCGA database of various cancer types (see Table 5) with available survival times, event indicators and transcriptomic data, resulting in a total of $2,246$ patients and 609 events. Our data pre-processing steps and the alternative survival analysis methods we considered are identical to Section 7.1.

Results. Table 5 shows, for each survival data analysis approach, the cross-validated IBS and C-index scores for each of the nine target tumor types (target TCGA datasets). SSTL attains the lowest IBS-score in eight of the nine tumor types. Similarly for the C-index, we observe that SSTL has in seven of the nine cancer types the best predictive performance.

Table 5: Five-fold cross-validation performance on TCGA using Cox p-value gene screening. SSTL reports the *nlp* slab and the Weibull outcome model. Standard deviations across validation folds are shown in parentheses.

		Lasso-AFT		Trans-AFT		Transfer Cox		SSTL	
Study	Samples (events)	IBS	C-index	IBS	C-index	IBS	C-index	IBS	C-index
BLCA	125 (48)	0.32 (0.11)	0.56 (0.09)	0.30 (0.12)	0.50 (0.02)	0.21 (0.05)	0.75 (0.08)	**0.15** (0.03)	**0.77** (0.10)
HNSC	159 (69)	0.28 (0.09)	0.56 (0.07)	0.28 (0.10)	0.61 (0.12)	0.24 (0.13)	0.67 (0.04)	**0.21** (0.03)	**0.71** (0.10)
KIRC	415 (141)	0.31 (0.06)	0.51 (0.02)	0.31 (0.06)	0.54 (0.09)	0.17 (0.03)	0.68 (0.04)	**0.16** (0.02)	**0.70** (0.04)
LAML	172 (113)	0.32 (0.08)	0.53 (0.05)	0.32 (0.08)	0.53 (0.05)	**0.18** (0.03)	0.68 (0.03)	0.20 (0.02)	**0.69** (0.02)
LUAD	203 (62)	0.24 (0.13)	0.56 (0.08)	0.23 (0.13)	0.63 (0.07)	**0.18** (0.05)	0.73 (0.09)	**0.18** (0.06)	**0.74** (0.06)
SARC	206 (78)	0.32 (0.11)	0.59 (0.09)	0.31 (0.10)	0.59 (0.06)	0.20 (0.05)	0.68 (0.05)	**0.18** (0.03)	**0.72** (0.06)
STAD	239 (46)	0.41 (0.26)	0.58 (0.11)	0.22 (0.12)	0.55 (0.13)	**0.17** (0.08)	**0.71** (0.15)	**0.17** (0.07)	0.68 (0.14)
THCA	396 (14)	0.43 (0.33)	0.56 (0.06)	0.31 (0.12)	0.57 (0.23)	**0.09** (0.02)	0.82 (0.16)	**0.09** (0.03)	**0.87** (0.10)
UCEC	331 (38)	0.24 (0.11)	0.64 (0.15)	0.22 (0.11)	0.61 (0.15)	0.14 (0.09)	**0.73** (0.09)	**0.13** (0.04)	0.72 (0.10)

Abbreviations: BLCA, Bladder Urothelial Carcinoma; HNSC, Head and Neck Squamous Cell Carcinoma; KIRC, Kidney Renal Clear Cell Carcinoma; LAML, Acute Myeloid Leukemia; LUAD, Lung Adenocarcinoma; SARC, Sarcoma; STAD, Stomach Adenocarcinoma; THCA, Thyroid Carcinoma; UCEC, Uterine Corpus Endometrial Carcinoma.

8 Discussion

In this work, we propose a novel Bayesian transfer learning method for the analysis of high-dimensional survival data via accelerated failure time models. Using an augmented variable representation, we introduce flexible DSS prior distributions with heavy-tailed slab densities, including a novel non-local prior. Based on this framework, we build a TL prior for the joint distribution of the target and source study outcome parameters and carry out posterior inference via a scalable slice-within-Gibbs sampling algorithm for arbitrary AFT residual distributions. We show that when at least one informative source exists, SSTL achieves a faster contraction rate and detects smaller signals compared to single-

study methods with DSS priors. SSTL is also robust to negative transfer and recovers the single-study minimax rate when sources are uninformative. Source study-specific information transfer is controlled by a novel empirical Bayes approach, implemented efficiently using a stochastic approximation EM algorithm. Empirical evaluations on synthetic and real datasets demonstrate better prediction and variable selection performance of SSTL compared to state-of-the-art TL models.

SSTL and its posterior inference algorithm can be directly applied to alternative statistical models outside of survival analysis, such as generalized linear models. Directions of future work include hierarchical Bayesian modeling as an alternative to our Empirical Bayesian source-study inclusion algorithm, which can properly quantify a posterior distribution over similarities between source and target studies. Although our posterior sampling algorithm is computationally scalable, alternative posterior approximations such as variational approximations can also be explored.

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Supplementary Material for "Spike-and-Slab Transfer Learning via Augmented Variables for Time-to-Event Analysis"

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S1 Additional Simulation Experiments

This section reports additional results and comparisons to alternative data-analysis approaches, which could not be presented in the main document due to space limitations.

Additional Results for Section 6.1

Figure S1 reports comparisons of the three SSTL slab model *exp*, *nlp*, and *poly* under two additional sparsity settings (a) $s_0 = 5$ and (b) $s_0 = 50$. We observe similar relative performances with the $s_0 = 20$ scenario from Figure 4. With *nlp* typically outperforming the other two prior marginally in MRSE and substantially in Brier score.

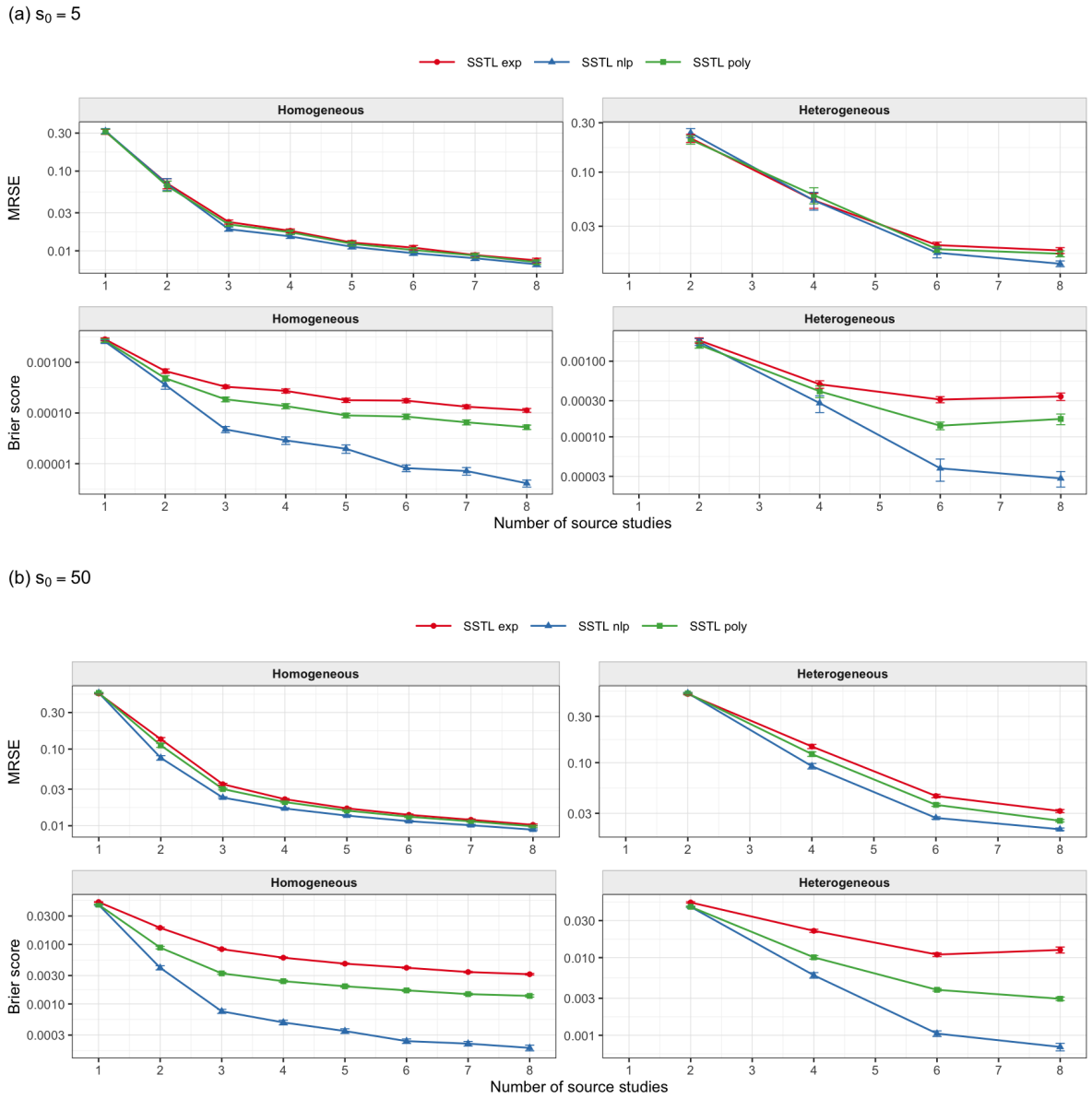


Figure S1: Performance of SSTL *exp*, *nlp*, and *poly* across 100 simulations with $p = 500$ and (a) $s_0 = 5$ or (b) $s_0 = 50$ active coefficients in the settings of Section 6.1.

Additional Results for Section 6.2

Table S1 reports comparisons of SSTL with *exp*, *nlp*, and *poly* slab priors to alternative TL models under the setting discussed in Section 6.2 with normal residual distributions $\epsilon_i^{(k)} \sim N(0, (\sigma^{(k)})^2)$ instead of extreme value residual distributions. Table S2 reports results with the active β_{0j} fixed at $+0.5$ under the extreme value residual distribution.

Table S1: Comparison with existing TL methods for log-normal outcomes without censoring under (A) random-signed and (B) uniform-signed β.

Method	K	MRSE	$10^2\times$ Brier	F1	Cov_1	Cov_0	Len_1	Len_0	Int_1	Int_0	Int_a
Panel A: $\beta_{0j} \in \{-0.5, +0.5\}$											
SSTL *exp*	2	0.085	0.185	0.959	**0.908**	1.000	0.371	0.008	**0.776**	0.008	**0.023**
SSTL *nlp*	2	0.089	**0.140**	0.958	0.897	1.000	**0.329**	**0.002**	1.109	**0.003**	0.025
SSTL *poly*	2	**0.081**	0.141	**0.965**	0.903	1.000	0.355	0.005	0.906	0.005	**0.023**
BLAST	2	0.568	1.059	0.486	0.637	1.000	0.689	0.115	4.515	0.115	0.203
GLMtrans	2	0.522	5.471	0.386	0.779	**0.949**	0.530	0.535	1.272	0.650	0.662
CONCERT	2	0.469	0.854	0.724	–	–	–	–	–	–	–
Trans Lasso	2	0.494	8.420	0.490	–	–	–	–	–	–	–
SSTL *exp*	8	0.011	0.065	0.987	0.944	1.000	0.182	0.002	0.217	0.002	0.007
SSTL *nlp*	8	**0.009**	**0.009**	**0.999**	**0.947**	1.000	**0.177**	**0.000**	**0.211**	**0.000**	**0.005**
SSTL *poly*	8	0.010	0.033	0.994	0.943	1.000	0.181	0.001	**0.211**	0.001	0.006
BLAST	8	0.403	0.714	0.637	0.820	1.000	0.748	0.111	1.963	0.111	0.148
GLMtrans	8	0.375	4.833	0.453	0.871	**0.954**	0.549	0.555	0.921	0.662	0.667
CONCERT	8	0.190	0.552	0.871	–	–	–	–	–	–	–
Trans Lasso	8	0.236	17.847	0.452	–	–	–	–	–	–	–
Panel B: $\beta_{0j} = +0.5$											
SSTL *exp*	2	0.037	0.116	0.986	**0.951**	1.000	0.352	0.008	0.400	0.008	0.015
SSTL *nlp*	2	**0.028**	**0.032**	**0.994**	0.962	1.000	**0.328**	**0.002**	**0.362**	**0.002**	**0.009**
SSTL *poly*	2	0.036	0.064	0.988	0.954	1.000	0.351	0.004	0.399	0.004	0.012
BLAST	2	0.266	0.480	0.825	0.981	1.000	1.114	0.148	1.175	0.148	0.169
GLMtrans	2	0.063	2.993	0.591	0.937	**0.970**	0.527	0.533	0.713	0.586	0.589
CONCERT	2	0.256	0.378	0.894	–	–	–	–	–	–	–
Trans Lasso	2	0.065	6.587	0.607	–	–	–	–	–	–	–
SSTL *exp*	8	0.009	0.055	0.991	0.954	1.000	0.182	0.002	0.211	0.002	0.007
SSTL *nlp*	8	**0.008**	**0.008**	**0.998**	0.958	1.000	**0.177**	**0.000**	0.216	**0.000**	**0.005**
SSTL *poly*	8	0.009	0.029	0.995	0.959	1.000	0.181	0.001	**0.210**	0.001	**0.005**
BLAST	8	0.126	0.247	0.956	1.000	1.000	1.037	0.130	1.037	0.130	0.149
GLMtrans	8	0.031	3.071	0.583	**0.951**	**0.969**	0.546	0.550	0.668	0.608	0.609
CONCERT	8	0.128	0.427	0.900	–	–	–	–	–	–	–
Trans Lasso	8	0.037	15.620	0.422	–	–	–	–	–	–	–

Notes: MRSE denotes mean relative squared error. Brier and F1 evaluate variable-selection accuracy. Cov_1 and Cov_0 denote coverage for active and inactive coefficients, respectively. Best performing is considered closest to nominal value. Len_1 and Len_0 denote the corresponding average interval lengths. Int_1, Int_0, and Int_a denote interval scores for active, inactive, and all coefficients. Best-performing values within each panel and source-size setting are bolded; ties are bolded jointly.

Table S1 shows that the performance gains of SSTL in the main documents are robust to variations in the residual distribution. Again we observe slight performance advantages of the *nlp* slab compare to the remaining SSTL slab distributions. Table S2 also shows that

the observe the relative performance gains of SSTL compared to alternative TL methods persist under alternative effect coefficient design.

Table S2: Additional comparison with existing TL methods under fixed active $\beta_{0j} = +0.5$.

Method	K	MRSE	$10^2\times$ Brier	F1	Cov_1	Cov_0	Len_1	Len_0	Int_1	Int_0	Int_a
SSTL *exp*	2	0.053	0.151	0.976	0.942	**1.000**	0.384	0.009	0.499	0.009	0.019
SSTL *nlp*	2	**0.049**	**0.060**	**0.983**	**0.945**	**1.000**	**0.345**	**0.002**	0.518	**0.002**	**0.013**
SSTL *poly*	2	0.054	0.091	0.980	0.936	**1.000**	0.379	0.005	**0.494**	0.005	0.015
BLAST	2	0.382	0.710	0.715	0.960	**1.000**	1.204	0.169	1.401	0.169	0.194
GLMtrans	2	0.165	10.800	0.295	0.794	0.891	0.531	0.534	1.413	0.874	0.884
CONCERT	2	0.466	0.693	0.796	–	–	–	–	–	–	–
Trans Lasso	2	0.130	6.642	0.599	–	–	–	–	–	–	–
Trans-AFT	2	0.273	0.884	0.805	–	–	–	–	–	–	–
Transfer Cox	2	0.342	1.895	0.772	–	–	–	–	–	–	–
SSTL *exp*	8	0.010	0.051	0.992	0.945	**1.000**	0.191	0.002	0.229	0.002	0.007
SSTL *nlp*	8	**0.009**	**0.008**	**0.998**	**0.949**	**1.000**	**0.184**	**0.000**	**0.218**	**0.000**	**0.005**
SSTL *poly*	8	0.014	0.032	0.994	0.935	**1.000**	0.190	0.001	0.306	0.001	0.007
BLAST	8	0.224	0.477	0.851	0.995	**1.000**	1.186	0.156	1.201	0.156	0.177
GLMtrans	8	0.105	10.537	0.298	0.856	0.894	0.548	0.553	1.140	0.902	0.906
CONCERT	8	0.263	0.665	0.837	–	–	–	–	–	–	–
Trans Lasso	8	0.069	17.002	0.459	–	–	–	–	–	–	–
Trans-AFT	8	0.199	0.667	0.853	–	–	–	–	–	–	–
Transfer Cox	8	0.339	1.398	0.804	–	–	–	–	–	–	–

Note: MRSE denotes mean relative squared error. Brier and F1 evaluate variable-selection accuracy. Cov_1 and Cov_0 denote coverage for active and inactive coefficients, respectively. Best performing is considered closest to nominal value. Len_1 and Len_0 denote the corresponding average interval lengths. Int_1, Int_0, and Int_a denote interval scores for active, inactive, and all coefficients. Best-performing values within each panel and source-size setting are bolded; ties are bolded jointly.

Lastly, Table S3 reports the full results for the survival analysis settings with 30%, 50%, and 70% censoring discussed in the main text. In addition to the MRSE, Brier and F1 score, we also report the accuracy of the estimated posterior confidence intervals (Cov_1, Cov_0, Len_1, Len_0, Int_1, Int_0, Int_a). The latter quantities are not available for the alternative TL methods (Trans-AFT and Trans-Cox).

Table S3: Comparison with censored AFT transfer-learning methods.

Method	K	MRSE	Brier	F1	Cov_1	Cov_0	Len_1	Len_0	Int_1	Int_0	Int_a
Panel A: 30% censoring											
SSTL *exp*	2	0.135	0.003	0.930	0.846	**1.000**	0.381	0.008	**1.408**	0.008	0.036
SSTL *nlp*	2	**0.121**	**0.002**	**0.937**	**0.857**	**1.000**	**0.332**	**0.002**	1.656	**0.002**	**0.035**
SSTL *poly*	2	0.137	**0.002**	0.924	0.845	**1.000**	0.371	0.005	1.499	0.005	**0.035**
Trans-AFT	2	0.722	0.023	0.455	–	–	–	–	–	–	–
Transfer Cox	2	0.781	0.025	0.419	–	–	–	–	–	–	–
SSTL *exp*	8	0.012	0.001	0.990	**0.929**	**1.000**	0.188	0.002	0.249	0.002	0.007
SSTL *nlp*	8	0.012	**0.000**	**0.997**	0.922	**1.000**	**0.183**	**0.000**	0.268	**0.001**	**0.006**
SSTL *poly*	8	**0.011**	**0.000**	0.992	0.928	**1.000**	0.187	0.001	**0.239**	**0.001**	**0.006**
Trans-AFT	8	0.578	0.022	0.576	–	–	–	–	–	–	–
Transfer Cox	8	0.786	0.029	0.408	–	–	–	–	–	–	–
Panel B: 50% censoring											
SSTL *exp*	2	0.128	0.003	0.935	0.855	**1.000**	0.378	0.008	1.221	0.008	0.032
SSTL *nlp*	2	**0.104**	**0.002**	**0.953**	**0.874**	**1.000**	**0.331**	**0.002**	1.375	**0.002**	0.030
SSTL *poly*	2	0.120	**0.002**	0.939	0.862	**1.000**	0.365	0.005	**1.203**	0.005	**0.029**
Trans-AFT	2	0.804	0.032	0.332	–	–	–	–	–	–	–
Transfer Cox	2	0.755	0.024	0.469	–	–	–	–	–	–	–
SSTL *exp*	8	0.013	0.001	0.983	0.927	**1.000**	0.185	0.003	**0.237**	0.003	0.007
SSTL *nlp*	8	**0.012**	**0.000**	**0.991**	0.923	**1.000**	**0.180**	**0.000**	0.243	**0.001**	**0.006**
SSTL *poly*	8	**0.012**	0.001	0.988	**0.930**	**1.000**	0.184	0.002	0.241	0.002	**0.006**
Trans-AFT	8	0.649	0.042	0.435	–	–	–	–	–	–	–
Transfer Cox	8	0.750	0.023	0.482	–	–	–	–	–	–	–
Panel C: 70% censoring											
SSTL *exp*	2	0.085	0.002	0.958	0.901	**1.000**	0.363	0.007	0.921	0.007	0.026
SSTL *nlp*	2	0.077	**0.001**	**0.965**	0.908	**1.000**	**0.327**	**0.002**	1.000	**0.002**	0.022
SSTL *poly*	2	**0.076**	**0.001**	**0.965**	**0.914**	**1.000**	0.359	0.005	**0.772**	0.005	**0.020**
Trans-AFT	2	0.879	0.051	0.242	–	–	–	–	–	–	–
Transfer Cox	2	0.730	0.023	0.513	–	–	–	–	–	–	–
SSTL *exp*	8	**0.010**	0.001	0.991	0.947	**1.000**	0.182	0.002	0.226	0.002	0.007
SSTL *nlp*	8	**0.010**	**0.000**	**0.997**	0.939	**1.000**	**0.177**	**0.000**	0.246	**0.000**	**0.005**
SSTL *poly*	8	**0.010**	0.001	0.994	**0.948**	**1.000**	0.181	0.001	**0.223**	0.001	0.006
Trans-AFT	8	0.776	0.079	0.293	–	–	–	–	–	–	–
Transfer Cox	8	0.731	0.022	0.515	–	–	–	–	–	–	–

Notes: MRSE denotes mean relative squared error. Brier and F1 evaluate variable-selection accuracy. Cov_1 and Cov_0 denote coverage for active and inactive coefficients, respectively. Best performing is considered closest to nominal value. Len_1 and Len_0 denote the corresponding average interval lengths. Int_1, Int_0, and Int_a denote interval scores for active, inactive, and all coefficients. Best-performing values within each panel and source-size setting are bolded; ties are bolded jointly.

S2 Proofs of results in Section 2

Lemma S1. *Consider the ReLU activation function in combination with a linear slab function (4) in the augmented representation for the covariate effects (3).*
(i) the DSS prior distribution $\pi(\beta)$ is then given by the hierarchical model,

$$\begin{aligned} A_j &\overset{iid}{\sim} Bern(\omega_0) \text{ with } \omega_0 = 1 - \Phi(\psi), \\ \beta_j | A_j &\overset{ind}{\sim} (1 - A_j) \times \delta_0 + A_j \times \pi_A(\beta_j) \text{ where, when } \psi = 0, \\ \pi_A(\beta_j) &= \frac{1}{\pi\tau} K_0\left(\frac{|\beta_j|}{\tau}\right). \end{aligned}$$

(ii) Let $\gamma \approx 0.5772156$ indicate Euler's constant [1], as $|\beta_j| \to 0$,

$$\pi_A(\beta) = \frac{1}{\pi\tau}\left[-\log\left(\frac{|\beta|}{2\tau}\right) - \gamma + O\left\{\frac{\beta^2}{\tau^2}\left|\log\left(\frac{|\beta|}{\tau}\right)\right|\right\}\right].$$

(iii) As $|\beta| \to \infty$,

$$\pi_A(\beta) = \frac{\exp(-|\beta|/\tau)}{\sqrt{2\pi\tau|\beta|}}\left[1 - \frac{\tau}{8|\beta|} + O\left(\frac{\tau^2}{\beta^2}\right)\right].$$

Proof. For simplicity, we will drop the subscript j in the proofs of this section. (i) Clearly $\beta = 0$ if $A = I(v \geq 0) = 0$. Whereas, when $A = 1$, then $\beta | v = \tau e v \sim N(0, \tau^2 v^2)$. Using the integral representation of the modified Bessel function of the second kind [3, Eq. 10.32.10],

$$K_\nu(z) = \int_0^\infty \exp\{-z\cosh(u)\}\cosh(\nu u)\, du = \frac{1}{2} z^\nu \int_0^\infty u^{-(\nu+1)} \exp\left\{-\frac{1}{2}\left(u + \frac{z^2}{u}\right)\right\} du$$

the marginal distribution equals

$$\begin{aligned} \pi_A(\beta) &= \frac{1}{\pi\tau}\int_0^\infty \frac{1}{v}\exp\left\{-\frac{1}{2}\left(v^2 + \frac{\beta^2}{\tau^2 v^2}\right)\right\} dv \qquad u = v^2/2 \;\;\&\;\; z = |\beta|/\tau. \\ &= \frac{1}{\pi\tau}\frac{1}{2}\int_0^\infty u^{-1}\exp\left\{-\left(u + \frac{z^2}{4u}\right)\right\} du \\ &= \frac{1}{\pi\tau} K_0\left(\frac{|\beta|}{\tau}\right). \end{aligned}$$

(ii) Using $K_0(z) = \sum_{m=0}^\infty k_m(z) = \sum_{m=0}^\infty \frac{(z^2/4)^m}{(m!)^2}\left[\psi(m+1) - \log\left(\frac{z}{2}\right)\right]$, where ψ is the digamma function. We can therefore write $K_0(z) = k_0(z) + r(z) = -\log\left(\frac{z}{2}\right) - \gamma + r(z)$, where $r(z) = \sum_{m=1}^\infty \frac{(z^2/4)^m}{(m!)^2}\left[\psi(m+1) - \log\left(\frac{z}{2}\right)\right]$. Observe that, since $\psi(m+1) = H_m - \gamma$, where H_m is the mth harmonic number, the coefficients in the above expansion grow at most logarithmically in m, with leading term of order z^2. Thus $r(z) = O\{z^2 |\log z|\}$, and setting $z = |\beta|/\tau$ shows the result.
(iii) The standard large-argument expansion [3, Eq. 10.40.2] gives

$$K_0(z) = \sqrt{\frac{\pi}{2z}} e^{-z}\left\{1 - \frac{1}{8z} + O(z^{-2})\right\}, \qquad z \to \infty.$$

Substituting $z = |\beta|/\tau$ into $\pi_A(\beta) = (\pi\tau)^{-1} K_0(|\beta|/\tau)$ yields the stated tail expansion. □

Proof for Lemma 2.1.

(i) The slab density π_A. If $A = I(v > \psi) = 0$, then $\beta = 0$. Whereas $\beta = w(v - \psi)$ with $w = \tau \cdot \text{sign}(e)\exp(e^2/m)$ when $A = 1$. Note that $|w| > \tau$. Using the Jacobian $\left|\frac{de}{dw}\right| = \frac{\sqrt{m}}{2|w|\sqrt{\log(|w|/\tau)}}$, w of the inverse $e = \text{sign}(w)\sqrt{m\log(|w|/\tau)}$, the density of w equals

$$f_W(w) = \frac{\sqrt{m}}{2\sqrt{2\pi}\tau} \times \frac{1}{(|w|/\tau)^{m/2+1}\sqrt{\log(|w|/\tau)}} \text{ for } |w| \geq \tau.$$

which is symmetric around 0. Finally, $\beta|A = 1$ has density

$$\begin{aligned}\pi_A(\beta) = p_{w(v-\psi)|v>\psi}(\beta) &= \int_\tau^\infty \frac{\phi(|\beta|/w + \psi)}{1 - \Phi(\psi)} f_W(w)\frac{1}{w}\, dw \\ &= \frac{\tau^{m/2}\sqrt{m}}{4\pi(1 - \Phi(\psi))}\int_\tau^\infty \frac{e^{-0.5(|\beta|/w+\psi)^2}}{|w|^{m/2+2}\sqrt{\log(|w|/\tau)}} dw. \qquad \text{(S1)}\end{aligned}$$

(ii) Behavior of the slab-distribution π_A at $\beta = 0$. Since the integrand in (S1) is uniformly bounded above by the integrable function $\frac{1}{|w|^{m/2+2}\sqrt{\log(|w|/\tau)}}$ dominated convergence gives

$$\begin{aligned}\lim_{\beta\to 0}\pi_A(\beta) &= \frac{\tau^{m/2}\sqrt{m}}{4\pi(1 - \Phi(\psi))}\int_\tau^\infty \frac{e^{-0.5\psi^2}}{|w|^{m/2+2}\sqrt{\log(|w|/\tau)}} dw \\ &= \frac{\tau^{m/2}\sqrt{m}}{4\pi(1 - \Phi(\psi))} \times e^{-0.5\psi^2} \times \frac{\sqrt{2\pi}}{\tau^{(m+2)/2}\sqrt{2+m}}.\end{aligned}$$

(iii) Tail-behavior of the slab-distribution π_A. Set $u = \beta/w$ and define $L = \log(\beta/\tau)$. It suffices to show that $\int_0^{e^L} \frac{u^{m/2}\exp(-(u+\psi)^2/2)}{\sqrt{L-\log u}} du = C_m \cdot L^{-1/2}(1 + o(1))$ as $L \to \infty$.
Let $J(L) = \int_0^{e^L} \frac{u^{m/2}\exp(-(u+\psi)^2/2)}{\sqrt{1-\log(u)/L}} du$, and decompose the function into two parts:

$$J(L) = \int_0^\infty \frac{u^{m/2}\exp(-(u+\psi)^2/2)}{\sqrt{1-\log(u)/L}} \cdot I(u \in (0, e^{L/2}])du + \int_{e^{L/2}}^{e^L} \frac{u^{m/2}\exp(-(u+\psi)^2/2)}{\sqrt{1-\log(u)/L}} du$$

Note that in the first term, $u \leq e^{L/2}$, which implies $1 - \log(u)/L \geq 1/2$, so

$$\frac{u^{m/2}\exp(-(u+\psi)^2/2)}{\sqrt{1-\log(u)/L}} \cdot I(u \in (0, e^{L/2}]) \leq \sqrt{2}u^{m/2}\exp(-(u+\psi)^2/2)$$

which is integrable on $(0, \infty)$. By the Dominated Convergence Theorem,

$$\begin{aligned}&\lim_{L\to\infty}\int_0^\infty \frac{u^{m/2}\exp(-(u+\psi)^2/2)}{\sqrt{1-\log(u)/L}} \cdot I(u \in (0, e^{L/2}])du \\ &= \int_0^\infty \lim_{L\to\infty}\frac{u^{m/2}\exp(-(u+\psi)^2/2)}{\sqrt{1-\log(u)/L}} \cdot I(u \in (0, e^{L/2}])du \qquad \text{(S2)} \\ &= \int_0^\infty u^{m/2}\exp(-(u+\psi)^2/2)du = C_m\end{aligned}$$

as the final integral is a moment of a truncated Gaussian random variable.

For the second term, let $t = L - \log u$, the integral can be transformed to

$$\begin{aligned}
&\int_0^{L/2} \exp(\frac{m+2}{2}(L-t)) \exp(-\frac{1}{2}(e^{L-t}+\psi)^2)\sqrt{L/t}\, dt \\
&\leq \int_0^{L/2} \exp(\frac{m+2}{2}L - \frac{1}{2}(e^{L/2}+\psi)^2)\sqrt{L/t}\, dt \\
&= \exp(\frac{m+2}{2}L - \frac{1}{2}(e^{L/2}+\psi)^2)\sqrt{2}L = o(1)
\end{aligned} \tag{S3}$$

as $L \to \infty$, since $t \leq L/2$ implies $L - t \geq L/2$ and $\exp(L-t) \leq \exp(L)$. The conclusion follows by combining all terms. □

Proof of Lemma 2.2.

(i) The slab density. Note that $w = \tau \operatorname{sign}(e) \left\{\exp(me^2) - 1\right\}^{1/(2m)}$, for $m \in \mathbb{N}$ with $m \geq 2$, is symmetric around 0. Moreover, $|e_w| = \left[\log\left\{1 + (|w|/\tau)^{2m}\right\}/m\right]^{1/2}$ and

$$\left|\frac{de_w}{dw}\right| = \frac{\sqrt{m}}{\tau} \frac{(|w|/\tau)^{2m-1}}{\left\{1 + (|w|/\tau)^{2m}\right\}\sqrt{\log\left\{1 + (|w|/\tau)^{2m}\right\}}}.$$

The prior density of $w = \beta | A_j = 1$ for $w \neq 0$ is therefore given by

$$\begin{aligned}
\pi_A(w) &= \phi(e_w) \times \left|\frac{de_w}{dw}\right| \\
&= \frac{1}{\sqrt{2\pi}}\left\{1 + (|w|/\tau)^{2m}\right\}^{-1/(2m)} \times \frac{\sqrt{m}}{\tau} \frac{(|w|/\tau)^{2m-1}}{\left\{1 + (|w|/\tau)^{2m}\right\}\sqrt{\log\left\{1 + (|w|/\tau)^{2m}\right\}}} \\
&= \frac{1}{\sqrt{2\pi\tau^2}} \times \frac{(|w|/\tau)^{2m-1}}{\left\{1 + (|w|/\tau)^{2m}\right\}^{(2m+1)/(2m)}\sqrt{\log\left\{1 + (|w|/\tau)^{2m}\right\}/m}}
\end{aligned}$$

ii) Slab prior when $\beta_j \to 0$: Let $u = \beta_j/\tau$, then $\pi_A \propto \frac{|u|^{2m-1}}{h(u)\cdot l(u)}$ where, $h(u) = \sqrt{\log(1+u^{2m})/m}$ and $l(u) = (u^{2m}+1)^{\frac{2m+1}{2m}}$. Using $\log(1+x) = x + o(x)$ and $(1+x)^a = 1 + ax + o(x)$ as $x \to 0$.

$$\begin{aligned}
h(u) &= \sqrt{(u^{2m} + o(u^{2m}))/m} = \frac{|u|^m}{\sqrt{m}}(1 + o(1)) = |u|^m/\sqrt{m} + o(|u|^m) \\
l(u) &= 1 + o(1)
\end{aligned} \tag{S4}$$

Therefore, $h(u) \cdot l(u) = \frac{|u|^m}{\sqrt{m}}(1 + o(1))$. Since $u^{2m} = o(1)$, $\pi_A(u) \propto \frac{|u|^{2m-1}}{|u|^m(1+o(1))} = |u|^{m-1} + o(|u|^{m-1})$

iii) Slab prior when $\beta_j \to +\infty$: Let $u = \frac{|w|}{\tau}$, using $\log(1+u^{2m}) \sim 2m \log u$ and $(1 + u^{2m})^{1+1/(2m)} \sim u^{2m+1}$, we have

$$f_W(w) \sim \frac{1}{\tau\sqrt{2\pi}} \frac{u^{2m-1}}{u^{2m+1}\sqrt{2\log u}} = \frac{1}{\tau\sqrt{4\pi\log(|w|/\tau)}} \left(\frac{\tau}{|w|}\right)^2$$

□

Proof of Lemma 2.3.
Lower envelope. Let $x = |\beta|/\tau$ and $a = \frac{2m+1}{2m} = 1 + \frac{1}{2m}$. Then

$$\pi(\beta) = \frac{1}{\sqrt{2\pi}\tau} \frac{x^{2m-1}}{(1+x^{2m})^a \sqrt{\log(1+x^{2m})/m}}.$$

If $0 < x \le 1$, then since $1 + x^{2m} \le 2$ and $\log(1 + x^{2m}) \le x^{2m}$, we have $(1 + x^{2m})^a \le 2^a$ and $\sqrt{\frac{\log(1+x^{2m})}{m}} \le \frac{x^m}{\sqrt{m}}$. Therefore, $\pi(\beta) \ge \frac{1}{\sqrt{2\pi}\tau} \frac{x^{2m-1}}{2^a x^m/\sqrt{m}} = \frac{\sqrt{m}}{\sqrt{2\pi}\tau\, 2^a} x^{m-1}$.

Whereas, if $x \ge 1$, then $1 + x^{2m} \le 2x^{2m}$, therefore $(1 + x^{2m})^a \le (2x^{2m})^a = 2^a x^{2m+1}$. and $\log(1 + x^{2m}) \le \log(2x^{2m})$. Thus $\sqrt{\frac{\log(1+x^{2m})}{m}} \le \sqrt{\frac{\log(2x^{2m})}{m}}$. and combining these bounds gives

$$\pi(\beta) \ge \frac{1}{\sqrt{2\pi}\tau} \frac{x^{2m-1}}{2^a x^{2m+1} \sqrt{\log(2x^{2m})/m}} = \frac{1}{\sqrt{2\pi}\tau\, 2^a} x^{-2} \sqrt{\frac{m}{\log(2x^{2m})}}.$$

Upper envelope. For $0 < x \le 1$, using the inequality $\log(1+y) \ge \frac{y}{1+y}$, $y \ge 0$. with $y = x^{2m}$, we get $\sqrt{\frac{\log(1+x^{2m})}{m}} \ge \frac{x^m}{\sqrt{m}\sqrt{1+x^{2m}}}$. Therefore, $(1+x^{2m})^a \sqrt{\frac{\log(1+x^{2m})}{m}} \ge \frac{x^m}{\sqrt{m}}(1+x^{2m})^{a-1/2}$., and since $a - 1/2 > 0$, we have $(1 + x^{2m})^{a-1/2} \ge 1$. Hence

$$\pi(\beta) \le \frac{1}{\sqrt{2\pi}\tau} \frac{x^{2m-1}}{x^m/\sqrt{m}} = \frac{\sqrt{m}}{\sqrt{2\pi}\tau} x^{m-1}.$$

Finally, for $x \ge 1$, using $(1 + x^{2m})^a \ge x^{2ma} = x^{2m+1}$, we obtain

$$\pi(\beta) \le \frac{1}{\sqrt{2\pi}\tau} \frac{x^{2m-1}}{x^{2m+1} \sqrt{\log(1+x^{2m})/m}} = \frac{1}{\sqrt{2\pi}\tau} x^{-2} \sqrt{\frac{m}{\log(1+x^{2m})}}.$$

The upper envelope is sharp. Since $\log(1 + x^{2m}) \sim x^{2m}$, as $x \downarrow 0$,

$$f_U(\beta) \sim \frac{1}{\sqrt{2\pi}\tau} \frac{x^{2m-1}}{x^m/\sqrt{m}} = \frac{\sqrt{m}}{\sqrt{2\pi}\tau} x^{m-1}.$$

Similarly, $\log(1 + x^{2m}) \sim \log(x^{2m})$, as $x \to \infty$, and hence

$$\pi(\beta) \sim \frac{1}{\sqrt{2\pi}\tau} x^{-2} \sqrt{\frac{m}{\log(x^{2m})}}.$$

□

Proof of Lemma 2.4. Denote g_r as g, $z = \|\boldsymbol{b}_{\mathcal{G}_r}\|_2/\tau_r$, and define

$$h_g(z) = \int_0^\infty u^{-g} \exp\left\{-\frac{z^2}{2u^2} - \frac{(u+\psi)^2}{2}\right\} du.$$

Conditional on $A_r = 1$ and $u = v_r - \psi > 0$, $\boldsymbol{b}_{\mathcal{G}_r} \sim N(\boldsymbol{0}, \tau_r^2 u^2 \boldsymbol{I}_g)$; integrating against $\phi(u+\psi)/(1-\Phi(\psi))$ gives the stated slab density. For $\psi = 0$, the substitution $t = u^2$ and the Bessel identity used in Lemma S1 give the closed form. For $g > 1$, substituting $t = z/u$ and applying the Dominated Convergence Theorem gives

$$h_g(z) \sim e^{-\psi^2/2} 2^{(g-3)/2} \Gamma\left(\frac{g-1}{2}\right) z^{1-g}, \qquad z \to 0.$$

For fixed ψ, Laplace's method around $u = \sqrt{z} - \psi/4 + o(1)$ gives

$$h_g(z) \sim \sqrt{\frac{\pi}{2}} z^{-g/2} \exp\left\{-z - \psi\sqrt{z} - \frac{3\psi^2}{8}\right\}, \qquad z \to \infty,$$

which proves both asymptotic statements. □

S3 Proofs of results in Section 3

S3.1 Proof of Theorem 3.1 & 3.2

Our proof uses large sample theory for sparse regression models with general nuisance parameter structures in [5]. The setup in [5] considers models of the form

$$Y_i = X_i\boldsymbol{\beta} + \xi_{\eta,i} + \varepsilon_i, \quad \varepsilon \overset{ind}{\sim} N(0, \Delta_{\eta,i}), \quad i = 1, \ldots, n \tag{S5}$$

where $\xi_{\eta,i}$ and $\Delta_{\eta,i}$ are parameterized by a (finite or infinite dimensional) $\boldsymbol{\eta} \in \mathbb{H}$ with values in a parameter space $\mathbb{H}$ and product prior on $\mathbb{R}^p \times \mathcal{H}$ (where $\mathcal{H}$ is a subspace of $\mathbb{H}$)

$$\begin{aligned} &(\mathcal{S}, \boldsymbol{\beta}) \to \pi(\mathcal{S})\pi_{\mathcal{S}}(\boldsymbol{\beta}) \quad \text{with} \\ &\pi(\mathcal{S}) = \frac{\pi_p(|\mathcal{S}|)}{\binom{p}{|\mathcal{S}|}} \quad \text{and} \quad \pi_{\mathcal{S}}(\boldsymbol{\beta}) = \delta_{\mathbf{0}_{|\mathcal{S}^c|}}(\boldsymbol{\beta}_{\mathcal{S}^c}) \times \prod_{j\in\mathcal{S}} \frac{\lambda}{2} e^{-\lambda|\beta_j|} \\ &\boldsymbol{\eta} \sim \pi_{\boldsymbol{\eta}}(\boldsymbol{\eta}). \end{aligned} \tag{S6}$$

where π_p indicates a probability mass function on $\{0, \ldots, p\}$. Sufficient conditions on the experimental design, the prior model (S6), and the true $\boldsymbol{\beta}_0$ and $\boldsymbol{\eta}_0$ are provided in [5] to obtain posterior contraction rates for $\boldsymbol{\beta}_0$ and $\boldsymbol{\eta}_0$ under model (S5).

We recast our TL model in the form (S5) by setting $\boldsymbol{\eta} = \boldsymbol{b}$, and defining for each observation $i = 1, \ldots, n = n_T + \sum_k n_k$, the nuisance as $\xi_{\eta,i} = \boldsymbol{z}_i'\boldsymbol{b}$, where $\boldsymbol{z}_i = (\mathbf{0}_p, \ldots, \mathbf{0}_p, \boldsymbol{x}_i^{(k)\top}, \mathbf{0}_p, \ldots, \mathbf{0}_p)'$ and $\Delta_{\eta,i} = {\sigma^{(k)}}^2$ if observation i belongs to source study k, whereas $\boldsymbol{z}_i = \mathbf{0}_{Kp}$ (i.e. $\xi_{\eta,i} = 0$) and $\Delta_{\eta,i} = \sigma^2$ if observation i belongs to the target data.

Our TL prior can be recast into a form similar to (S6) with prior for $(\mathcal{S}, \boldsymbol{\beta})$

$$\pi_p(s_\beta) = \text{Beta-Binom}(s_\beta \mid p, 1, \lambda^{(T)}) \quad \text{and } \pi_{\mathcal{S}}(\boldsymbol{\beta}) = \delta_{\mathbf{0}}(\boldsymbol{\beta}_{\mathcal{S}^c}) \times \prod_{j\in\mathcal{S}} \pi_A(\beta_j) \tag{S7}$$

with slab distribution π_A specified in Examples 2.3-2.4 in the main manuscript instead of the Laplace prior studied by [5]. The prior for $\boldsymbol{b} = \boldsymbol{\eta}$ in the main manuscript can be written as

$$\begin{aligned} &(\mathcal{R}, \boldsymbol{b}) \to \pi(\mathcal{R})\pi_{\mathcal{R}}(\boldsymbol{b}) \quad \text{where} \\ &\pi(\mathcal{R}) = \prod_{k=1}^{K} \pi(\mathcal{R}_k) = \prod_{k=1}^{K} \frac{\text{Beta-Binom}(r_k \mid p, 1, \lambda^{(k)})}{\binom{p}{r_k}} \\ &\pi_{\mathcal{R}}(\boldsymbol{b}) = \delta_0(\boldsymbol{b}_{\mathcal{R}^C}) \prod_{(k,j)\in\mathcal{R}} \pi_A^{(k)}(b_j^{(k)}) \end{aligned} \tag{S8}$$

Below, we will use the the following pseudo-distance

$$d_n(\boldsymbol{b}, \boldsymbol{b}_0) = \|\boldsymbol{Z}(\boldsymbol{b} - \boldsymbol{b}_0)\|_2/\sqrt{n}.$$

We will verify that conditions JG-C1-JG-C5* and JG-C6 in [5] hold for our TL model with prior (S7) and (S8). Conditions, JG-C1, JG-C3 and JG-C4 hold, using respectively $a_n = 1$ and $\mathcal{H} = \mathbb{H} = \mathbb{R}^{pK}$ in our setting, Condition (A2), and fixed noise variance. Below, we verify in detail conditions JG-C2 and JG-C5*.

As outlined in prior Condition (P-A1), $\pi_p(s_\beta)$ in (S7) satisfies

$$\frac{A_1}{p^{a_1}} \leq \frac{\pi_p(s)}{\pi_p(s-1)} \leq \frac{A_2}{p^{a_2}} \qquad \forall s = 1, \ldots, p \tag{S9}$$

for some positive constants. Similarly, for each $k = 1, \ldots, K$, the prior $\pi_{k,p}(s_{\boldsymbol{b}^{(k)}}) = \text{Beta-Binom}(s_{\boldsymbol{b}^{(k)}} \mid p, 1, \lambda^{(k)})$ in (S7) for the support of $\boldsymbol{b}^{(k)}$ satisfies

$$\frac{A_3}{p^{a_3}} \leq \frac{\pi_{k,p}(r)}{\pi_{k,p}(r-1)} \leq \frac{A_4}{p^{a_4}} \qquad \forall r = 1, \ldots, p \tag{S10}$$

uniformly in k, for some positive constants. The following lemmas give the properties of the slab densities in Examples 2.3 and 2.4 used below.

S3.1.1 Supporting lemmas

Lemma S2 (Local mass of the *nlp* and *poly* slabs)**.** *Fix $m \geq 2$. Let $g_{m,\tau}$ denote the nlp or poly slab density with scale $\tau > 0$, and define*

$$J_{m,\tau}(x) = \log\left\{1 + (|x|/\tau)^{2m}\right\}. \tag{S11}$$

There exist constants $c_m, C_m > 0$, depending only on m, such that for every $x \in \mathbb{R}$ and $0 < h \leq 1$,

$$\int_{x-h\tau}^{x+h\tau} g_{m,\tau}(u)\, du \geq c_m h^m \exp\{-C_m J_{m,\tau}(x)\}. \tag{S12}$$

For the poly slab in Example 2.4, the bound also holds with h in place of h^m.

Proof for the nlp slab. Since the *nlp* density has the scale form $g_{m,\tau}(u) = \tau^{-1} g_{m,1}(u/\tau)$ and is symmetric around 0, it suffices to consider $\tau = 1$ and $x \geq 0$. Consider the integral over $(x, x+h)$, we divide into three cases:

If $x + h \leq 1$, Lemma 2.3 gives $g_{m,1}(u) \geq c_m u^{m-1}$ on $[x, x+h]$. Therefore,

$$\int_x^{x+h} g_{m,1}(u)\, du \geq c_m\{(x+h)^m - x^m\} \geq c_m h^m,$$

where the last inequality follows from $(x+h)^m \geq x^m + h^m$ for $m \geq 2$. Since $\exp\{-C_m J_{m,1}(x)\} \leq 1$, the desired bound follows in this case.

If $x < 1 < x + h$, then $J_{m,1}(x) \leq \log 2$. Further, if $x \geq 1/2$, then $[x, x+h] \subset [1/2, 2]$. Let $c'_m := \inf_{1/2 \leq u \leq 2} g_{m,1}(u) > 0$, we have that

$$\int_x^{x+h} g_{m,1}(u)\, du \geq c'_m h \geq c_m h^m \exp\{-C_m J_{m,1}(x)\}.$$

If $x < 1/2$, the condition $x + h > 1$ implies that $h > 1/2$ and $[1/2, 1] \subset [x, x+h]$. We then have that

$$\int_x^{x+h} g_{m,1}(u)\, du \geq \int_{1/2}^{1} g_{m,1}(u)\, du \geq c'_m/2 \geq c_m h^m \exp\{-C_m J_{m,1}(x)\}.$$

Finally, suppose $x \geq 1$. For $u \in [x, x+h]$, $x \leq u \leq x+1 \leq 2x$, and thus the lower envelope in Lemma 2.3 gives that

$$g_{m,1}(u) \geq c_m \frac{u^{-2}}{\sqrt{\log(2u^{2m})}} \geq c_m x^{-3},$$

since $u^2 \leq 4x^2$ and $\log(2u^{2m}) \leq C'_m x^2$. Consequently,

$$\int_x^{x+h} g_{m,1}(u)\, du \geq c_m h x^{-3} \geq c_m h^m x^{-3}.$$

Setting $C_m = 3/(2m)$, we have that $\exp\{-C_m J_{m,1}(x)\} = (1+x^{2m})^{-3/(2m)} \leq x^{-3}$ which proves the case. Combining all cases, the result follows as $\int_{x-h}^{x+h} g_{m,1}(u)\, du \geq \int_x^{x+h} g_{m,1}(u)\, du$. □

Proof for the poly slab. Similar to the *nlp* slab, since the *poly* density is a scale family and possess symmetry around 0, it suffices to consider $\tau = 1$ and $x \geq 0$. By the density formula in Example 2.4, for $u \geq 0$, restricting the integral to $[1+u, 2(1+u)]$ gives

$$\begin{aligned} g_{m,1}(u) &\geq \frac{\sqrt{m}}{2\pi} \int_{1+u}^{2(1+u)} \frac{\exp\{-u^2/(2v^2)\}}{v^{m/2+2}\sqrt{\log v}}\, dv \\ &\geq c_m \frac{(1+u)^{-(m+2)/2}}{\sqrt{\log\{2(1+u)\}}} \geq c_m (1+u)^{-(m+3)/2}, \end{aligned}$$

since $u/v \leq 1$ on this interval and $\log\{2(1+u)\} \leq 1+u$.

Consider the integral over $(x, x+h)$. For $u \in [x, x+h]$, we have $1+u \leq 2+x \leq 3(1+x^{2m})^{1/(2m)}$. Consequently, setting $C_m = (m+3)/(4m)$ and decreasing c_m if necessary gives

$$\int_x^{x+h} g_{m,1}(u)\, du \geq c_m h (1+x^{2m})^{-(m+3)/(4m)} = c_m h \exp\{-C_m J_{m,1}(x)\}.$$

The result follows as $\int_{x-h}^{x+h} g_{m,1}(u)\, du \geq \int_x^{x+h} g_{m,1}(u)\, du$. The weaker bound with h^m in place of h follows from $h \geq h^m$ for $0 < h \leq 1$. □

S3.1.2 Condition JG-C2

Condition JG-C2 states that for some sequence $\bar{\varepsilon}_n$ such that $a_n \bar{\varepsilon}_n^2 \to 0$ and $n\bar{\varepsilon}_n^2 \to \infty$ with a_n satisfying JG-C1,

$$\log \Pi_B\big(d_n(\boldsymbol{b}, \boldsymbol{b}_0) \leq \bar{\varepsilon}_n\big) \gtrsim -n\bar{\varepsilon}_n^2.$$

When $r_0 > 0$, by setting $\bar{\varepsilon}_n^2 = r_0 \log(p)/n$, we see that $\bar{\varepsilon}_n^2 \to 0$ and $n\bar{\varepsilon}_n^2 \to \infty$ under Condition (A1). We show that this condition holds for all three slab distributions. If $r_0 = 0$, take $\bar{\varepsilon}_n^2 = \log n/n$. Then $\bar{\varepsilon}_n^2 \to 0$, $n\bar{\varepsilon}_n^2 \to \infty$, and $\Pi_B(d_n(\boldsymbol{b}, \boldsymbol{0}) \leq \bar{\varepsilon}_n) \geq \Pi_B(\mathcal{R} = \varnothing) \to 1$, and JG-C2 holds. We focus on the case $r_0 \neq 0$ below. For convenience, we suppress the indexing $(k, j) \in \mathcal{R}_0$ to $j \in \mathcal{R}_0$ and imply that all $k = 1, \ldots, K$ are included.

Verification of JG-C2 for the *exp* slab. Since we aim to derive a lower bound, we first restrict to a smaller set. Notice that conditional on $\mathcal{R} = \mathcal{R}_0$, the triangle inequality gives

$$d_n(\boldsymbol{b}, \boldsymbol{b}_0) \leq \frac{1}{\sqrt{n}} \sum_{j \in \mathcal{R}_0} \|\boldsymbol{Z}_j\|_2 |b_j - b_{0j}|.$$

Thus it suffices to consider the set $\left\{\mathcal{R}=\mathcal{R}_0, \frac{1}{\sqrt{n}}\sum_{j\in\mathcal{R}_0}\|\boldsymbol{Z}_j\|_2|b_j-b_{0j}|\le\bar{\varepsilon}_n\right\}$. Let $\tau_j=\tau^{(k)}$ for a variable belonging to source study k, and $h=\min\left\{1,\frac{\sqrt{n}\,\bar{\varepsilon}_n}{\sum_{j\in\mathcal{R}_0}\tau_j\|\boldsymbol{Z}_j\|_2}\right\}$. By Lemma S4, we have that

$$\begin{aligned}
&\Pi_B\left(\frac{1}{\sqrt{n}}\sum_{j\in\mathcal{R}_0}\|\boldsymbol{Z}_j\|_2|b_j-b_{0j}|\le\bar{\varepsilon}_n\,\middle|\,\mathcal{R}=\mathcal{R}_0\right)\\
&\ge\Pi_B\left(|b_j-b_{0j}|\le h\tau_j,\ j\in\mathcal{R}_0\,|\,\mathcal{R}=\mathcal{R}_0\right)\\
&=\prod_{j\in\mathcal{R}_0}\int_{b_{0j}-h\tau_j}^{b_{0j}+h\tau_j}\pi_{A,\tau_j}(u)\,du\\
&\ge(c_1h)^{r_0}\exp\{-C_1\sum_{j\in\mathcal{R}_0}|b_{0j}|/\tau_j\},
\end{aligned}$$

It follows that $\Pi_B\{d_n(\boldsymbol{b},\boldsymbol{b}_0)\le\bar{\varepsilon}_n\}\ge\pi(\mathcal{R}_0)(c_1h)^{r_0}\exp\left\{-C_1\sum_{j\in\mathcal{R}_0}|b_{0j}|/\tau_j\right\}$. Since the specification of $\lambda^{(k)}$ in (P-A1) gives $\pi(\mathcal{R}_0)\ge\exp\{-C_Rr_0\log p\}$ for some $C_R>0$, taking negative logarithms yields

$$-\log\Pi_B\{d_n(\boldsymbol{b},\boldsymbol{b}_0)\le\bar{\varepsilon}_n\}\le C_Rr_0\log p+r_0\log(1/h)-r_0\log c_1+C_1\sum_{j\in\mathcal{R}_0}|b_{0j}|/\tau_j.$$

By (P-A2), $\max_{1\le j\le Kp}\tau_j\|\boldsymbol{Z}_j\|_2=\max_{1\le k\le K}\tau^{(k)}\|\boldsymbol{X}^{(k)}\|\le p^{L_1}$. Consequently,

$$\begin{aligned}
\frac{1}{h}&=\max\left\{1,\frac{\sum_{j\in\mathcal{R}_0}\tau_j\|\boldsymbol{Z}_j\|_2}{\sqrt{n}\,\bar{\varepsilon}_n}\right\}\\
&\le\max\left\{1,\frac{r_0p^{L_1}}{\sqrt{n}\,\bar{\varepsilon}_n}\right\}=\max\left\{1,p^{L_1}\sqrt{\frac{r_0}{\log p}}\right\}.
\end{aligned}$$

Since $r_0\le Kp$ and K is fixed, $\log(1/h)\lesssim\log p$. Together with Condition (A2), combining these bounds verifies the result.

Verification of JG-C2 for the *poly* and *nlp* slab. Similar to the *exp* slab model, we consider the same set $\{\mathcal{R}=\mathcal{R}_0,\frac{1}{\sqrt{n}}\sum_{j\in\mathcal{R}_0}\|\boldsymbol{Z}_j\|_2|b_j-b_{0j}|\le\bar{\varepsilon}_n\}$ and $h=\min\{1,\frac{\sqrt{n}\bar{\varepsilon}_n}{\sum_{j\in\mathcal{R}_0}\|\boldsymbol{Z}_j\|\tau_j}\}$. With a different local mass lower bound, Lemma S2, we have that

$$\begin{aligned}
&\Pi_B\left(\frac{1}{\sqrt{n}}\sum_{j\in\mathcal{R}_0}\|\boldsymbol{Z}_j\|_2|b_j-b_{0j}|\le\bar{\varepsilon}_n\,\middle|\,\mathcal{R}=\mathcal{R}_0\right)\\
&\ge\prod_{j\in\mathcal{R}_0}\int_{b_{0j}-h\tau_j}^{b_{0j}+h\tau_j}g_{m,\tau_j}(u)\,du\\
&\ge c_m^{r_0}h^{mr_0}\exp\left\{-C_m\sum_{j\in\mathcal{R}_0}J_{m,\tau_j}(b_{0j})\right\},
\end{aligned}$$

Again using $\Pi_B(\mathcal{R}_0)\ge\exp\{-C_Rr_0\log p\}$, it follows that

$$-\log\Pi_B\{d_n(\boldsymbol{b},\boldsymbol{b}_0)\le\bar{\varepsilon}_n\}\le C_Rr_0\log p-r_0\log c_m+mr_0\log(1/h)+C_m\sum_{j\in\mathcal{R}_0}J_{m,\tau_j}(b_{0j}).$$

By previous arguments, $\log(1/h) \lesssim \log p$. Moreover, Condition (A2) implies

$$J_{m,\tau_j}(b_{0j}) = \log\left\{1 + \left(\frac{|b_{0j}|}{\tau_j}\right)^{2m}\right\} \leq \log(1 + p^{2mL_b}) \lesssim \log p, \qquad j \in \mathcal{R}_0.$$

Combining these bounds verifies condition JG-C2.

S3.1.3 Condition JG-C5*

This condition requires the existence of a sieve $\mathcal{H}_n \subset \mathcal{H}$ and a sequence γ_n such that

$$\min_{1\leq i\leq n} \inf_{B\in\mathcal{H}_n} \rho_{\min}(\Delta_{B,i}) \geq \gamma_n^{-1} \tag{S13}$$

$$\log N\left(\frac{1}{6\bar{m}\gamma_n n^{3/2}}, \mathcal{H}_n, d_n\right) \lesssim n\varepsilon_n^2, \tag{S14}$$

and

$$e^{C_5 s_\star \log p}\Pi_B(\mathcal{H} \setminus \mathcal{H}_n) \to 0. \tag{S15}$$

Where $\varepsilon_n^2 = \frac{s_\star \log p}{n}$, $\log \gamma_n \lesssim \log p$, $a_n\varepsilon_n^2 \to 0$ where a_n is the sequence from JG-C1. Since we assume the noise variance are fixed, the first condition (S13) is trivially satisfied and we take $\gamma_n = 1$. It remains to construct a sieve that satisfies (S14) and (S15). Again, we suppress the indexing $(k, j) \in \mathcal{R}$ to $j \in \mathcal{R}$ and imply that all $k = 1, \ldots, K$ are included.

Proof. Let $\bar{r}_n := M_r s_\star$ for a sufficiently large constant $M_r > 0$, and define the sieve $\mathcal{H}_n = \{\boldsymbol{b} : r \leq \bar{r}_n, \max_j |b_j|/\tau_j \leq L_n\}$, where L_n is a sequence satisfying $\log L_n \lesssim \log p$ to be specified later. Let $M_n := \max_{1\leq j\leq Kp} \tau_j \|Z_j\|_2 = \max_{1\leq k\leq K} \tau^{(k)} \|\boldsymbol{X}^{(k)}\|$. By Condition (P-A2) $M_n \leq p^{L_1}$. Our strategy to find a cover for $\mathcal{H}_n$ is to combine covers for all subsets of the form $\mathcal{H}_\mathcal{R} = \left\{\boldsymbol{b} \in \mathbb{R}^{Kp} : \boldsymbol{b}_{\mathcal{R}^c} = 0, \max_j |b_j|/\tau_j \leq L_n\right\}$.
For fixed $\mathcal{R}$, note that for any $\boldsymbol{b}, \boldsymbol{b}' \in \mathcal{H}_\mathcal{R}$,

$$d_n(\boldsymbol{b}, \boldsymbol{b}') = \frac{1}{\sqrt{n}}\|\boldsymbol{Z}_\mathcal{R}(\boldsymbol{b}_\mathcal{R} - \boldsymbol{b}'_\mathcal{R})\|_2 \leq \frac{1}{\sqrt{n}} \sum_{j\in\mathcal{R}} |b_j - b'_j| \|Z_j\|_2 \leq \frac{M_n r}{\sqrt{n}} \max_{j\in\mathcal{R}} \frac{|b_j - b'_j|}{\tau_j}.$$

Consider a Cartesian grid on the set $\prod_{j\in\mathcal{R}}[-\tau_j L_n, \tau_j L_n]$ separated by distance $\tau_j u$ in each coordinate j. The number of grid points is bounded by $\left(1 + \frac{2L_n}{u}\right)^r$. For any $\boldsymbol{b} \in \mathcal{H}_\mathcal{R}$, there exists a grid point $\widetilde{\boldsymbol{b}}$ such that $\max_{j\in\mathcal{R}} |b_j - \widetilde{b}_j|/\tau_j \leq u$, which implies $d_n(\boldsymbol{b}, \widetilde{\boldsymbol{b}}) \leq M_n r u/\sqrt{n}$. Therefore, for any radius $\delta_n > 0$, choosing $u = \delta_n\sqrt{n}/(M_n r)$ gives a d_n-cover of $\mathcal{H}_\mathcal{R}$. Hence

$$N(\delta_n, \mathcal{H}_\mathcal{R}, d_n) \leq \left(1 + \frac{2L_n M_n r}{\delta_n\sqrt{n}}\right)^r.$$

Now we sum over all possible $\mathcal{R}$. Since there are $\binom{Kp}{r}$ supports of size r,

$$\begin{aligned} N(\delta_n, \mathcal{H}_n, d_n) &\leq \sum_{r=0}^{\bar{r}_n} \binom{Kp}{r} \left(1 + \frac{2L_n M_n r}{\delta_n\sqrt{n}}\right)^r \\ &\leq (\bar{r}_n + 1)\left(\frac{eKp}{\bar{r}_n}\right)^{\bar{r}_n} \left(1 + \frac{2L_n M_n \bar{r}_n}{\delta_n\sqrt{n}}\right)^{\bar{r}_n}. \end{aligned}$$

Thus

$$\log N(\delta_n, \mathcal{H}_n, d_n) \le \log(\bar{r}_n + 1) + \bar{r}_n \log\left(\frac{eKp}{\bar{r}_n}\right) + \bar{r}_n \log\left(1 + \frac{2L_n M_n \bar{r}_n}{\delta_n \sqrt{n}}\right).$$

By replacing $\delta_n = 1/(6n^{3/2})$, we have

$$\begin{aligned}\log N\left(\frac{1}{6n^{3/2}}, \mathcal{H}_n, d_n\right) &\lesssim \bar{r}_n(\log p + \log L_n) \\ &\lesssim s_\star \log p = n\varepsilon_n^2.\end{aligned}$$

since $M_n \le p^{L_1}$, $n = o(p)$, K is fixed, $\bar{r}_n = M_r s_\star$ and $\log L_n \lesssim \log p$. This verifies (S14). Now we verify (S15). By (P-A1) the prior on $\boldsymbol{b}$ given in (11) satisfies the upper bound $\Pi_B\{r > t\} \le \exp\{-c_r t \log(Kp)\}$. The *exp* slab satisfies the tail bound $\pi_{A,\tau_j}(|b_j|/\tau_j > t) \le e^{-t}$, for all $t \ge 0$. This follows from the fact that if $X, Y \sim N(0,1)$ are independent, then $P(|XY| > t) \le P(X^2 + Y^2 > 2t) = e^{-t}$. Then

$$\Pi_B(\mathcal{H}_n^c) = \Pi_B\{r > \bar{r}_n\} + \Pi_B\{r \le \bar{r}_n,\ \max_j |b_j|/\tau_j > L_n\}.$$

For the first term, we have $\Pi_B\{r > \bar{r}_n\} \le \exp\{-c_r \bar{r}_n \log(Kp)\} \le \exp\{-(C_5+1)s_\star \log p\}$. by choosing M_r sufficiently large so that $c_r M_r \log(Kp) \ge (C_5+1)\log p$. For the second term, we have that

$$\Pi_B\{r \le \bar{r}_n,\ \max_j |b_j|/\tau_j > L_n\} \le \sum_{\mathcal{R}:|\mathcal{R}|\le \bar{r}_n} \pi(\mathcal{R}) \sum_{j\in\mathcal{R}} \Pi_{A,\tau_j}(|b_j|/\tau_j > L_n) \le \bar{r}_n e^{-L_n}.$$

By choosing $L_n = (C_5+1)s_\star \log p + \log \bar{r}_n$, we have $\bar{r}_n e^{-L_n} \le \exp\{-(C_5+1)s_\star \log p\}$. Note that the choice L_n is consistent with the previous specification, as $\log L_n \lesssim \log(s_\star \log p) \lesssim \log p$. Combining inequalities for the two terms verifies the result. □

Verification of JG-C5* for the *nlp* and *poly* slabs. Let $J_{m,\tau}(x)$ be defined as in (S11), and $\bar{r}_n = M_r s_\star$, $T_n = M_J s_\star \log p$, where $M_r, M_J > 0$ are sufficiently large constants. Define the common sieve

$$\mathcal{H}_n^J = \left\{\boldsymbol{b} \in \mathbb{R}^{Kp} : r \le \bar{r}_n,\ \sum_{j\in\mathcal{R}} J_{m,\tau_j}(b_j) \le T_n\right\}, \tag{S16}$$

where τ_j denotes the scale corresponding to coordinate j. As above, the fixed noise variances give $\gamma_n = 1$, so it remains to verify (S14) and (S15).

Proof. nlp slab: For any $j \in \mathcal{R}$, an *nlp* coefficient can be represented as

$$b_j = \tau_j \operatorname{sign}(E_j)\{\exp(mE_j^2) - 1\}^{1/(2m)}, \qquad E_j \overset{\text{iid}}{\sim} N(0,1).$$

Consequently, $J_{m,\tau_j}(B_j) = mE_j^2$ and conditional on any support $\mathcal{R}$, we have that $\sum_{j\in\mathcal{R}} J_{m,\tau_j}(B_j) \sim m\chi_r^2$. Thus, Markov's inequality and the chi-square MGF give

$$\Pr\left\{\sum_{j\in\mathcal{R}} J_{m,\tau_j}(B_j) > T_n \mid \mathcal{R}\right\} \le \exp\left\{-\frac{T_n}{4m} + \frac{r}{2}\log 2\right\}.$$

poly slab: For any $j \in \mathcal{R}$, a *poly* coefficient can be represented as

$$b_j = \tau_j G_j \exp(E_j^2/m), \qquad G_j, E_j \overset{\text{iid}}{\sim} N(0,1).$$

Since $m \geq 2$ and $1 + y^m \leq (1+y)^m$ for $y \geq 0$, we have that

$$\begin{aligned} J_{m,\tau_j}(B_j) &= \log\{1 + |G_j|^{2m} \exp(2E_j^2)\} \\ &\leq m \log(1 + G_j^2) + 2E_j^2 \leq m(G_j^2 + E_j^2). \end{aligned}$$

Consequently, conditional on any support $\mathcal{R}$, $\sum_{j\in\mathcal{R}} J_{m,\tau_j}(B_j)$ is bounded above by a random variable distributed as $m\chi^2_{2r}$. Thus, the same argument as above gives

$$\Pr\left\{\sum_{j\in\mathcal{R}} J_{m,\tau_j}(B_j) > T_n \mid \mathcal{R}\right\} \leq \exp\left\{-\frac{T_n}{4m} + r\log 2\right\}.$$

Using the model-size tail in the preceding proof,

$$\begin{aligned} \Pi_B\{(\mathcal{H}_n^J)^c\} &\leq \Pi_B(r > \bar{r}_n) + \Pi_B\left\{r \leq \bar{r}_n, \sum_{j\in\mathcal{R}} J_{m,\tau_j}(b_j) > T_n\right\} \\ &\leq C\exp\{-cM_r s_\star \log p\} + \exp\left\{-\frac{M_J}{4m} s_\star \log p + M_r s_\star \log 2\right\}. \end{aligned} \tag{S17}$$

By choosing M_r, M_J such that $cM_r > C_5 + 1$ and $M_J/(4m) > C_5 + 1$, this implies

$$e^{C_5 s_\star \log p}\Pi_B\{(\mathcal{H}_n^J)^c\} \longrightarrow 0,$$

which verifies (S15).

It remains to verify the entropy bound. For a fixed support $\mathcal{R}$, let

$$\mathcal{A}_\mathcal{R} = \left\{\boldsymbol{b} : \boldsymbol{b}_{\mathcal{R}^c} = 0,\ \sum_{j\in\mathcal{R}} J_{m,\tau_j}(b_j) \leq T_n\right\}.$$

For $\boldsymbol{b} \in \mathcal{A}_\mathcal{R}$, define $k_j = \lceil J_{m,\tau_j}(b_j)\rceil$ for $j \in \mathcal{R}$. Then $\sum_{j\in\mathcal{R}} k_j \leq T_n + r$, and $|b_j| \leq \tau_j(e^{k_j} - 1)^{1/(2m)} \leq \tau_j e^{k_j/(2m)}$. Thus for any $\boldsymbol{b} \in \mathcal{A}_\mathcal{R}$, $b_j \in [-\tau_j e^{k_j/(2m)}, \tau_j e^{k_j/(2m)}]$ for all j and the cover number with radius $\delta\tau_j$ is at most $\prod_{j\in\mathcal{R}}(\frac{2e^{k_j/2m}}{\delta} + 1)$, and we have that for a fixed $(k_j)_{j\in\mathcal{R}}$,

$$\log N_k(\delta) \leq r\log(1 + 2/\delta) + \frac{1}{2m}\sum_{j\in\mathcal{R}} k_j.$$

Now, to translate this into a cover number for d_n, notice that

$$\begin{aligned} d_n(\boldsymbol{b}, \widetilde{\boldsymbol{b}}) &= \frac{1}{\sqrt{n}}\left\|\sum_{j\in\mathcal{R}} \boldsymbol{Z}_j(b_j - \widetilde{b}_j)\right\|_2 \\ &\leq \frac{1}{\sqrt{n}}\sum_{j\in\mathcal{R}} \|\boldsymbol{Z}_j\|_2 |b_j - \widetilde{b}_j| \\ &\leq \frac{r}{\sqrt{n}} \max_{j\in\mathcal{R}} \frac{|b_j - \widetilde{b}_j|}{\tau_j} \max_{j\in\mathcal{R}} \tau_j \|\boldsymbol{Z}_j\|_2. \end{aligned}$$

Therefore, to obtain a d_n-cover with radius $\delta_n = 1/(6n^{3/2})$, the preceding value with $\delta = (6nr\max_{j\in\mathcal{R}}\tau_j\|\boldsymbol{Z}_j\|_2)^{-1}$ would suffice. Combined with the facts that $1 + ae^x \le e^x(1+a)$ for $a, x \ge 0$, and the number of nonnegative integer vectors satisfying $\sum_{j\in\mathcal{R}} k_j \le T_n + r$ is at most $\binom{\lceil T_n\rceil+2r}{r} \le \left\{\frac{e(\lceil T_n\rceil+2r)}{r}\right\}^r$,

$$\log N\{\delta_n, \mathcal{A}_\mathcal{R}, d_n\} \le r\log\left\{1 + \frac{2r}{\delta_n\sqrt{n}}\max_{j\in\mathcal{R}}\tau_j\|\boldsymbol{Z}_j\|_2\right\} + \frac{T_n + r}{2m} + r\log\left\{\frac{e(\lceil T_n\rceil + 2r)}{r}\right\}.$$

Finally, summing over the $\binom{Kp}{r}$ possible supports of size r and over $r \le \bar{r}_n$,

$$\log N\left\{\frac{1}{6n^{3/2}}, \mathcal{H}_n^J, d_n\right\} \lesssim \bar{r}_n\log\left(\frac{eKp}{\bar{r}_n}\right) + T_n + \bar{r}_n\log\left(1 + \frac{T_n}{\bar{r}_n}\right) + \bar{r}_n\log\left\{1 + 12n\bar{r}_n\max_{1\le j\le Kp}\tau_j\|\boldsymbol{Z}_j\|_2\right\}.$$

By the specification of τ_k, $k = 1, \ldots, K$ of the *nlp* or *poly* slab,

$$\max_{1\le j\le Kp}\tau_j\|\boldsymbol{Z}_j\|_2 = \max_{1\le k\le K}\tau^{(k)}\|\boldsymbol{X}^{(k)}\| \le p^{L_2}.$$

Since K is fixed, $n = o(p)$, $\bar{r}_n = M_r s_\star$, and $T_n = M_J s_\star \log p$, every term in the preceding entropy bound is $O(s_\star \log p)$. Hence

$$\log N\left\{\frac{1}{6n^{3/2}}, \mathcal{H}_n^J, d_n\right\} \lesssim s_\star \log p = n\varepsilon_n^2,$$

which verifies (S14). Together with (S17), $\gamma_n = 1$, and $a_n\varepsilon_n^2 = s_\star\log(p)/n \to 0$, this verifies condition JG-C5* for the *nlp* and *poly* slabs. □

S3.1.4 Condition JG-C6

Condition JG-C6 requires that for $s_\star = s_0 \vee r_0$, there exists $\boldsymbol{b}_*$ such that

$$\liminf_{n\ge1}\inf_{(\boldsymbol{\beta},\boldsymbol{b})\in\mathcal{A}_n}\frac{(\boldsymbol{\beta}-\boldsymbol{\beta}_0)'\widetilde{\boldsymbol{X}}'\boldsymbol{Z}(\boldsymbol{b}-\boldsymbol{b}_*)}{\|\widetilde{\boldsymbol{X}}(\boldsymbol{\beta}-\boldsymbol{\beta}_0)\|_2^2 + \|\boldsymbol{Z}(\boldsymbol{b}-\boldsymbol{b}_*)\|_2^2} > -\frac{1}{2}$$

$$\frac{1}{\sqrt{n}}\|\boldsymbol{Z}(\boldsymbol{b}_* - \boldsymbol{b}_0)\|_2 \lesssim \sqrt{\frac{s_\star\log p}{n}}$$

Where the set $\mathcal{A}_n \coloneqq \left\{(\boldsymbol{\beta},\boldsymbol{b}) : s \le K_1 s_\star, \frac{1}{n}\|\widetilde{\boldsymbol{X}}(\boldsymbol{\beta}-\boldsymbol{\beta}_0) + \boldsymbol{Z}(\boldsymbol{b}-\boldsymbol{b}_0)\|_2^2 \le M_1\varepsilon_n^2\right\}$.
For our model, we use the following modified

$$\mathcal{A}_n^* \coloneqq \left\{(\boldsymbol{\beta},\boldsymbol{b}) : s \le K_1 s_\star,\ r \le M_r s_\star,\ \frac{1}{n}\|\widetilde{\boldsymbol{X}}(\boldsymbol{\beta}-\boldsymbol{\beta}_0) + Z(\boldsymbol{b}-\boldsymbol{b}_0)\|_2^2 \le M_1\varepsilon_n^2\right\}$$

For the *nlp* and *poly* slabs, the denominator bound in Lemma 1 of [5], with its target-prior calculation replaced by Lemma S3 implies

$$\mathbb{E}_0[\Pi_n(\boldsymbol{b}\notin\mathcal{H}_n^J \mid \mathcal{D}^{(T)}, \{\mathcal{D}^{(k)}\}_{k=1}^K)] \le o(1) + e^{Cs_\star\log p}\Pi_B\{(\mathcal{H}_n^J)^c\} = o(1)$$

for a fixed $C > 0$, by (S17) with M_r, M_J sufficiently large. Since $\mathcal{A}_n \cap (\mathbb{R}^p \times \mathcal{H}_n^J) \subseteq \mathcal{A}_n^*$, the dimension and joint mean contraction bounds from their Theorems 1–2, with Lemma S3 and the β sieve bounds (S19) below, imply that $\mathcal{A}_n^*$ has posterior mass tending to one. The same argument applies to the *exp* slab using its sieve $\mathcal{H}_n$. Thus the recovery argument may be restricted to $\mathcal{A}_n^*$.
Next we show that condition JG-C6 holds under $\mathcal{A}_n^*$.

Proof. Take $\boldsymbol{b}^* = \boldsymbol{b}_0$, then the approximation part of JG-C6 is exact. Now fix any $(\boldsymbol{\beta}, \boldsymbol{b}) \in \mathcal{A}_n^*$ and set $u = \boldsymbol{\beta} - \boldsymbol{\beta}_0, v = \boldsymbol{b} - \boldsymbol{b}_0$. Denote $\tilde{\mathcal{S}} = \mathcal{S} \cup \mathcal{S}_0, \tilde{\mathcal{R}} = \mathcal{R} \cup \mathcal{R}_0$, then $\widetilde{\boldsymbol{X}}(\boldsymbol{\beta} - \boldsymbol{\beta}_0) = \widetilde{\boldsymbol{X}}_{\tilde{\mathcal{S}}} \boldsymbol{u}_{\tilde{\mathcal{S}}}$, $\boldsymbol{Z}(\boldsymbol{b} - \boldsymbol{b}_0) = \boldsymbol{Z}_{\tilde{\mathcal{R}}} \boldsymbol{v}_{\tilde{\mathcal{R}}}$. Since $|\tilde{\mathcal{S}}| \le (K_1 + 1)s_\star$ and $|\tilde{\mathcal{R}}| \le (M_r + 1)s_\star$, we take $C_\beta \ge K_1 + 1$ and $C_b \ge \max\{M_r + 1, K\}$, which also covers the requirements in Lemma S3. By Condition (18), we have

$$\|\widetilde{\boldsymbol{X}}_{\tilde{\mathcal{S}}} \boldsymbol{u}_{\tilde{\mathcal{S}}} + \boldsymbol{Z}_{\tilde{\mathcal{R}}} \boldsymbol{v}_{\tilde{\mathcal{R}}}\|_2^2 \ge K_{\text{sep}} \left(\|\widetilde{\boldsymbol{X}}_{\tilde{\mathcal{S}}} \boldsymbol{u}_{\tilde{\mathcal{S}}}\|_2^2 + \|\boldsymbol{Z}_{\tilde{\mathcal{R}}} \boldsymbol{v}_{\tilde{\mathcal{R}}}\|_2^2 \right)$$

$$\boldsymbol{u}'_{\tilde{\mathcal{S}}} \widetilde{\boldsymbol{X}}'_{\tilde{\mathcal{S}}} \boldsymbol{Z}_{\tilde{\mathcal{R}}} \boldsymbol{v}_{\tilde{\mathcal{R}}} \ge -\frac{1 - K_{\text{sep}}}{2} \left(\|\widetilde{\boldsymbol{X}}_{\tilde{\mathcal{S}}} \boldsymbol{u}_{\tilde{\mathcal{S}}}\|_2^2 + \|\boldsymbol{Z}_{\tilde{\mathcal{R}}} \boldsymbol{v}_{\tilde{\mathcal{R}}}\|_2^2 \right)$$

Equivalently,

$$\frac{(\boldsymbol{\beta} - \boldsymbol{\beta}_0)' \widetilde{\boldsymbol{X}}' \boldsymbol{Z} (\boldsymbol{b} - \boldsymbol{b}_0)}{\|\widetilde{\boldsymbol{X}}(\boldsymbol{\beta} - \boldsymbol{\beta}_0)\|_2^2 + \|\boldsymbol{Z}(\boldsymbol{b} - \boldsymbol{b}_0)\|_2^2} \ge -\frac{1 - K_{\text{sep}}}{2} > -\frac{1}{2}$$

□

S3.1.5 Justifications for the Slab distributions

We point out that the slab used in [5] is the exact Laplace distribution, which has the same tail property as the *exp* slab from our model. However, there are two places from their paper that exploits the distributional form of the Laplace distribution, a procedure that our proposed slab distributions do not straightforwardly follow. Lemma 1 from their paper proves a lower bound on a small ball integral of the slab distribution on the true $\mathcal{S}_0$, and a sieve set is constructed using explicit properties of the Laplace distribution in the proof of Theorem 2. The following small-ball bounds and β sieve verification supply the required replacements for all three slab models.

Lemma S3. *Let $\pi_{A,\mathcal{S}_0}(\boldsymbol{\beta}_{\mathcal{S}_0}) = \prod_{j \in \mathcal{S}_0} \pi_A(\beta_j)$ denote the product slab density on the active coordinates, where π_A can be the exp, poly, or nlp density. Define $\Theta_{0,n} = \{\boldsymbol{\beta}_{\mathcal{S}_0} \in \mathbb{R}^{s_0} : \|\widetilde{\boldsymbol{X}}\|/\sqrt{n}\, \|\boldsymbol{\beta}_{\mathcal{S}_0} - \boldsymbol{\beta}_{0,\mathcal{S}_0}\|_1 \le \bar{\varepsilon}_n\}$. There exists a constant $C_1 > 0$ such that*

$$\Pi_\beta \left(n^{-1/2} \|\widetilde{\boldsymbol{X}}\| \|\boldsymbol{\beta} - \boldsymbol{\beta}_0\|_1 \le \bar{\varepsilon}_n \right) \ge \pi_p(s_0) \exp\{-C_1 s_0 \log p\} \tag{S18}$$

Proof. If $s_0 = 0$, the result is immediate. Hence assume $s_0 \ge 1$ and let $h = \min\{1, \sqrt{n}\, \bar{\varepsilon}_n / (s_0 \tau \|\widetilde{\boldsymbol{X}}\|)\}$. If $|\beta_j - \beta_{0j}| \le h\tau$ for all $j \in \mathcal{S}_0$, then $\|\boldsymbol{\beta}_{\mathcal{S}_0} - \boldsymbol{\beta}_{0,\mathcal{S}_0}\|_1 \le s_0 h \tau \le \sqrt{n}\, \bar{\varepsilon}_n / \|\widetilde{\boldsymbol{X}}\|$. Below we split arguments for the *exp*, *poly*, and *nlp* slab densities.
exp slab: Write $\tau = \tau^{(T)}$. By Lemma S4 and $h \le 1$, we have the following local lower bound on the *exp* density. For every $j \in \mathcal{S}_0$,

$$\int_{\beta_{0j} - h\tau}^{\beta_{0j} + h\tau} \pi_A(u)\, du \ge 2 c_g h \big(2 + |\beta_{0j}|/\tau\big)^{-1/2} \exp\{-|\beta_{0j}|/\tau - 1\}$$

$$\ge ch \exp\{-C|\beta_{0j}|/\tau\}.$$

Therefore,

$$\int_{\Theta_{0,n}} \pi_{A,\mathcal{S}_0}(\boldsymbol{\beta}_{\mathcal{S}_0})\, d\boldsymbol{\beta}_{\mathcal{S}_0} \geq \prod_{j\in\mathcal{S}_0} \int_{\beta_{0j}-h\tau}^{\beta_{0j}+h\tau} \pi_A(u)\, du \geq c^{s_0} h^{s_0} \exp\{-C\tau^{-1}\|\boldsymbol{\beta}_0\|_1\}.$$

To bound $\log(1/h)$, notice that Condition (A3) implies that

$$\|\boldsymbol{X}_j^{(T)}\|_2^2 \geq K_{\text{sep}} \left(\|\boldsymbol{X}_j^{(T)}\|_2^2 + 2\sum_{k=1}^{K} \|\boldsymbol{X}_j^{(k)}\|_2^2 \right),$$
$$\|\widetilde{\boldsymbol{X}}\|^2 \leq \frac{1+K_{\text{sep}}}{2K_{\text{sep}}} \|\boldsymbol{X}^{(T)}\|^2.$$

Together with Condition (P-A2) gives $\tau\|\widetilde{\boldsymbol{X}}\| \lesssim p^{L_1}$. Since $n\bar{\varepsilon}_n^2 \to \infty$ and $s_0 \leq p$, we obtain $1/h \leq \max\{1, Cp^{L_1+1}\}$ for all sufficiently large n, so $\log(1/h) \lesssim \log p$. Combined with Assumption (A2),

$$\begin{aligned} -\log \int_{\Theta_{0,n}} \pi_{A,\mathcal{S}_0}(\boldsymbol{\beta}_{\mathcal{S}_0})\, d\boldsymbol{\beta}_{\mathcal{S}_0} &\leq -s_0 \log c + s_0 \log(1/h) + C\tau^{-1}\|\boldsymbol{\beta}_0\|_1 \\ &\lesssim s_0 \log p. \end{aligned}$$

poly & nlp slab: By Lemma S2,

$$\begin{aligned} \int_{\Theta_{0,n}} \pi_{A,\mathcal{S}_0}(\boldsymbol{\beta}_{\mathcal{S}_0})\, d\boldsymbol{\beta}_{\mathcal{S}_0} &\geq \prod_{j\in\mathcal{S}_0} \int_{\beta_{0j}-h\tau}^{\beta_{0j}+h\tau} g_{m,\tau}(u)\, du \\ &\geq c_m^{s_0} h^{ms_0} \exp\left\{ -C_m \sum_{j\in\mathcal{S}_0} J_{m,\tau}(\beta_{0j}) \right\}. \end{aligned}$$

Since $n\bar{\varepsilon}_n^2 \to \infty$ and $s_0 \leq p$, Condition (P-A2) gives $1/h \leq \max\{1, p^{L_2+1}\}$ for all sufficiently large n, so $\log(1/h) \lesssim \log p$. Moreover, $J_{m,\tau}(\beta_{0j}) = \log\{1 + (|\beta_{0j}|/\tau)^{2m}\} \leq \log(1+p^{2mL_\beta}) \lesssim \log p$ for every $j \in \mathcal{S}_0$. Therefore,

$$-\log \int_{\Theta_{0,n}} \pi_{A,\mathcal{S}_0}(\boldsymbol{\beta}_{\mathcal{S}_0})\, d\boldsymbol{\beta}_{\mathcal{S}_0} \leq -s_0 \log c_m + ms_0 \log(1/h) + C_m \sum_{j\in\mathcal{S}_0} J_{m,\tau}(\beta_{0j}) \lesssim s_0 \log p.$$

Restricting to the true support $\mathcal{S}_0$ and using $\binom{p}{s_0} \leq p^{s_0}$, we obtain

$$\begin{aligned} &\Pi_\beta\left(n^{-1/2}\|\widetilde{\boldsymbol{X}}\| \|\boldsymbol{\beta} - \boldsymbol{\beta}_0\|_1 \leq \bar{\varepsilon}_n \right) \\ &\quad \geq \pi_p(s_0) \binom{p}{s_0}^{-1} \int_{\Theta_{0,n}} \pi_{A,\mathcal{S}_0}(\boldsymbol{\beta}_{\mathcal{S}_0})\, d\boldsymbol{\beta}_{\mathcal{S}_0} \\ &\quad \geq \pi_p(s_0) \exp\{-C_1 s_0 \log p\}, \end{aligned}$$

which proves (S18). □

Together with JG-C2, the bound (S18) supplies the prior concentration required for the denominator bound in Lemma 1 of [5] under the *exp*, *poly*, and *nlp* slabs.

$\boldsymbol{\beta}$ sieve in the proof of Theorem 2. The β sieve required in the proof of Theorem 2 of [5] can be constructed by the same argument as in JG-C5*. Let $\Theta_n = \{\boldsymbol{\beta} \in \mathbb{R}^p : s_\beta \le K_1 s_\star\}$, the posterior mass of Θ_n^c tends to zero by their Theorem 1, using the denominator bound obtained from Lemma S3 and condition JG-C2.

Let $d_{\beta,n}(\boldsymbol{\beta}, \boldsymbol{\beta}') = n^{-1/2}\|\widetilde{\boldsymbol{X}}(\boldsymbol{\beta} - \boldsymbol{\beta}')\|_2$. For the *exp* slab, take $\Theta_n^* = \{\boldsymbol{\beta} \in \Theta_n : \max_j |\beta_j|/\tau^{(T)} \le M_L s_\star \log p\}$, and for the *nlp* and *poly* slabs, take $\Theta_n^* = \{\boldsymbol{\beta} \in \Theta_n : \sum_{j \in \mathcal{S}_\beta} J_{m,\tau^{(T)}}(\beta_j) \le M_J s_\star \log p\}$, where $M_L, M_J > 0$ are sufficiently large constants. These mirror the sieves of $\boldsymbol{b}$ from JG-C5*, and under the prior conditions the same covering number and prior-tail arguments apply to $\boldsymbol{\beta}$. Hence, for each fixed $C > 0$, choosing M_L, M_J sufficiently large yields

$$\begin{gathered} \log N\left(\frac{1}{6n^{3/2}}, \Theta_n^*, d_{\beta,n}\right) \lesssim s_\star \log p, \\ e^{C s_\star \log p} \Pi_\beta(\Theta_n \setminus \Theta_n^*) \longrightarrow 0. \end{gathered} \tag{S19}$$

To connect these covers to the testing argument, observe that

$$\frac{1}{n}\left\|\widetilde{\boldsymbol{X}}(\boldsymbol{\beta} - \boldsymbol{\beta}') + \boldsymbol{Z}(\boldsymbol{b} - \boldsymbol{b}')\right\|_2^2 \le 2d_{\beta,n}^2(\boldsymbol{\beta}, \boldsymbol{\beta}') + 2d_n^2(\boldsymbol{b}, \boldsymbol{b}').$$

These sieve bounds supply the two ingredients needed in the proof of Theorem 2 of [5]. First, the preceding inequality shows that the product covers give sufficiently small neighborhoods for the local tests in their Lemma 2. By (S19) and JG-C5*, the log cover number is $O(s_\star \log p) = O(n\varepsilon_n^2)$, as required in their equation (32). Second, the prior-tail bounds in (S19) and JG-C5* make the posterior contribution from $\Theta_n \setminus \Theta_n^*$ negligible after division by the denominator, whose lower bound follows from Lemma S3 and JG-C2.

S3.2 Proof of Theorem 3.3

S3.2.1 Setup & Notation

Here we work with the full model written as

$$\boldsymbol{Y} = \widetilde{\boldsymbol{X}}\boldsymbol{\beta}_0 + \boldsymbol{Z}\boldsymbol{b}_0 + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim N_n(0, \boldsymbol{I}_n),$$

For convenience, we write $\Pi(\cdot \mid \boldsymbol{Y})$ for $\Pi(\cdot \mid \mathcal{D}_n)$ as the full posterior distribution. In this section, we frequently consider the full model as an augmented regression model, thus we make the following additional definitions. Let $\mathcal{U} = (\mathcal{S}, \mathcal{R})$, $\mathcal{U}_0 = (\mathcal{S}_0, \mathcal{R}_0)$, and $\mathcal{R}_k = \{j : (k, j) \in \mathcal{R}\}$. We also denote $\mathcal{U} \triangle \mathcal{U}_0 = (\mathcal{U} \setminus \mathcal{U}_0) \cup (\mathcal{U}_0 \setminus \mathcal{U})$ as a shorthand operation. For uniformity with the common slab scale τ_B, we denote $\tau_\beta = \tau^{(T)}$. Further, define

$$\begin{aligned} &d_{\mathcal{U}} = |\mathcal{S}| + |\mathcal{R}|, && \boldsymbol{W}_{\mathcal{U}} = [\widetilde{\boldsymbol{X}}_{\mathcal{S}}, \boldsymbol{Z}_{\mathcal{R}}], && \boldsymbol{D}_{\mathcal{U}} = \mathrm{diag}(\tau_\beta \boldsymbol{I}_{|\mathcal{S}|}, \tau_B \boldsymbol{I}_{|\mathcal{R}|}), \\ &\boldsymbol{V}_{\mathcal{U}} = \boldsymbol{W}_{\mathcal{U}}\boldsymbol{D}_{\mathcal{U}}, && \boldsymbol{G}_{\mathcal{U}} = \boldsymbol{V}_{\mathcal{U}}'\boldsymbol{V}_{\mathcal{U}}, && \boldsymbol{P}_{\mathcal{U}} = P_{\mathrm{col}(\boldsymbol{W}_{\mathcal{U}})}, \quad \boldsymbol{M}_{\mathcal{U}} = \boldsymbol{I}_n - \boldsymbol{P}_{\mathcal{U}}. \end{aligned}$$

Let $\boldsymbol{\theta}_0 = (\boldsymbol{\beta}_{0,\mathcal{S}_0}', \boldsymbol{b}_{0,\mathcal{R}_0}')'$, $\boldsymbol{\mu}_0 = \boldsymbol{W}_{\mathcal{U}_0}\boldsymbol{\theta}_0$, and

$$s_{\mathrm{out}} = |\mathcal{S}_0 \setminus \mathcal{S}|, \quad s_{\mathrm{in}} = |\mathcal{S} \setminus \mathcal{S}_0|, \quad r_{\mathrm{out}} = |\mathcal{R}_0 \setminus \mathcal{R}|, \quad r_{\mathrm{in}} = |\mathcal{R} \setminus \mathcal{R}_0|,$$

so $h_{\mathcal{U}} = |\mathcal{U} \triangle \mathcal{U}_0| = s_{\mathrm{out}} + s_{\mathrm{in}} + r_{\mathrm{out}} + r_{\mathrm{in}}$. For fixed $M < \infty$, define

$$\mathcal{L}_M = \{\mathcal{U} : |\mathcal{S}| \le M s_\star,\ |\mathcal{R}_k| \le M s_\star,\ k = 1, \dots, K\}.$$

Let $\pi(\mathcal{U}) = \pi(\mathcal{S}) \prod_{k=1}^K \pi(\mathcal{R}_k)$ be the prior probability of selecting $\mathcal{U}$, and let $m_{\mathcal{U}}(\boldsymbol{Y})$ be the marginal likelihood conditional on $\mathcal{U}$. Lastly, denote $n_{\min} = \min n_k$ and $n_{\max} = \max n_k$ over $k = T, 1, \dots, K$. Throughout this section, $c, C \in (0, \infty)$ denote constants that are allowed to change from line to line for convenience. We write $a \lesssim b$ if $a \le Cb$.

S3.2.2 Supporting lemmas

Lemma S4. *By standard properties of the Bessel function, we have the following bounds:*

$$\begin{aligned} &K_0(z) \le C_0 \log(e/z), && 0 < z \le 1, \\ &K_0(z) \le C_0 z^{-1/2} e^{-z}, && z \ge 1, \\ &K_0(z) \ge c_0 (1+z)^{-1/2} e^{-z}. \end{aligned} \tag{S20}$$

Furthermore, for all $s, h \in \mathbb{R}$,

$$K_0(|s-h|) \lesssim e^{\frac{3}{2}|h|} K_0(|s|) + \log(e/|s-h|)\mathbf{1}\{|s-h| \le 1\}.$$

Proof of last assertion. We split into two cases of whether $|s-h| \le 1$. If $|s-h| > 1$, then

$$\frac{K_0(|s-h|)}{K_0(|s|)} \lesssim e^{|s|-|s-h|}\sqrt{\frac{1+|s|}{|s-h|}} \lesssim e^{|h|+|h|/2}.$$

Since $\sqrt{1+|s|}/\sqrt{|s-h|} \le \sqrt{1+|s-h|+|h|}/\sqrt{|s-h|}$, which is further bounded by a constant multiple of $e^{|h|/2}$. Thus, $K_0(|s-h|) \lesssim K_0(|s|)e^{3|h|/2}$.
If $|s-h| \le 1$, we have that $K_0(|s-h|) \lesssim \log(e/|s-h|)$. Combining both cases proves the result. □

In the following, we use g_τ to denote the *exp* density π_A in Example 2.4 of the main manuscript.

Lemma S5. *Assume Conditions (B1) and (B2) hold. For every fixed $M < \infty$, n, p sufficiently large, it holds that*

$$cp^{2L_3}\frac{n_{\min}}{n_{\max}}\boldsymbol{I}_{d_{\mathcal{U}}} \preceq \boldsymbol{G}_{\mathcal{U}} \preceq Cp^{2L_3}\boldsymbol{I}_{d_{\mathcal{U}}}, \quad \forall\, \mathcal{U} \in \mathcal{L}_M. \tag{S21}$$

The same bounds hold for every Schur complement $\boldsymbol{G}_{\mathcal{J}_1|\mathcal{J}_0}$ of $\boldsymbol{G}_{\mathcal{J}}$, where $\mathcal{J}$ is the union of two sets in $\mathcal{L}_M$. Moreover,

$$\frac{n_{\min}}{s_\star} \longrightarrow \infty, \quad \inf_{\mathcal{U}\in\mathcal{L}_M} \lambda_{\min}(\boldsymbol{G}_{\mathcal{U}}) \gg p, \quad \sup_{\mathcal{U}\in\mathcal{L}_M} d_{\mathcal{U}} = o(p). \tag{S22}$$

Lemma S6. *Assume Conditions (B1) – (B3) hold. For any positive definite $r \times r$ matrix $\boldsymbol{G}$, define*

$$m_{\boldsymbol{G},\tau}(\boldsymbol{x}) = \int_{\mathbb{R}^r} \exp\left\{-\frac{1}{2}(\boldsymbol{u}-\boldsymbol{x})'\boldsymbol{G}(\boldsymbol{u}-\boldsymbol{x})\right\} g_{\tau,r}(\boldsymbol{u})\, d\boldsymbol{u}. \tag{S23}$$

When $r = 0$, $m_{\boldsymbol{G},\tau}$ is interpreted as 1. There exists $C_\ell < \infty$ such that for every fixed $M < \infty$, there exists a (deterministic) sequence $\eta_n \to 0$ and an event Ω_M with $\mathbb{P}_0(\Omega_M) \to 1$ on which

$$\log\frac{m_{\mathcal{U}}(\boldsymbol{Y})}{m_{\mathcal{U}_0}(\boldsymbol{Y})} \le -\frac{1}{4}\|\boldsymbol{M}_{\mathcal{U}}\boldsymbol{\mu}_0\|_2^2 + C_\ell h_{\mathcal{U}}\log p + (L_3-1)(d_{\mathcal{U}_0}-d_{\mathcal{U}})\log p + \eta_n \log p \tag{S24}$$

holds for all $\mathcal{U} \in \mathcal{L}_M$.

Lemma S7. *Under the prior in Example 2.4 with hyperparameters specified as (P-B1), for sufficiently large p we have that*

$$\pi(\mathcal{U}_0) \ge 2^{-(d_{\mathcal{U}_0}+K+1)} c_T^{-s_0} c_B^{-r_0} p^{-a_T s_0 - a_B r_0}. \tag{S25}$$

Further, if $|\mathcal{S}| \vee |\mathcal{S}_0| \vee \max_{1\le k\le K}\{|\mathcal{R}_k| \vee |\mathcal{R}_{0k}|\} \le (M+1)s_\star$, then

$$\frac{\pi(\mathcal{S})}{\pi(\mathcal{S}_0)} \le 2^{s_{\text{out}}}\{(M+1)s_\star\}^{s_{\text{in}}} c_T^{s_{\text{out}}-s_{\text{in}}} p^{a_T(s_{\text{out}}-s_{\text{in}})},$$

$$\prod_{k=1}^{K}\frac{\pi(\mathcal{R}_k)}{\pi(\mathcal{R}_{0k})} \le 2^{r_{\text{out}}}\{(M+1)s_\star\}^{r_{\text{in}}} c_B^{r_{\text{out}}-r_{\text{in}}} p^{a_B(r_{\text{out}}-r_{\text{in}})}.$$

Lemma S8. *There exists $C_E < \infty$ and events $\mathcal{E}_E$ with $\mathbb{P}_0(\mathcal{E}_E) \to 1$ such that*

$$\mathcal{Z}(\boldsymbol{Y}) := \int \frac{L(\boldsymbol{Y};\boldsymbol{\beta},\boldsymbol{b})}{L(\boldsymbol{Y};\boldsymbol{\beta}_0,\boldsymbol{b}_0)}\, d\Pi(\boldsymbol{\beta},\boldsymbol{b}) \ge e^{-C_E s_\star \log p} \tag{S26}$$

on $\mathcal{E}_E$. With $a_m = \min\{a_T - 1, a_B - 1\}$, $M_a = \max\left\{2, \left\lceil \frac{C_E+3}{a_m} \right\rceil\right\}$, the posterior satisfies

$$\mathbb{E}_0\Pi(\mathcal{L}_{M_a}^c \mid \mathcal{D}_n) = o(1). \tag{S27}$$

Lemma S9. *Assume conditions (B3) and (B2) hold. For every fixed $M < \infty$ and every $\mathcal{U} \in \mathcal{L}_M$,*

$$\frac{1}{4}\|\boldsymbol{M}_{\mathcal{U}}\boldsymbol{\mu}_0\|_2^2 \ge \{c_b r_{\text{out}} + c_\beta(s_{\text{out}} - r_{\text{in}})_+\}\log p, \tag{S28}$$

where $c_\beta = \underline{\kappa}B_\beta/4(K+1)$ and $c_b = \underline{\kappa}B_b/4(K+1)$.

S3.2.3 Proof of main theorem

Proof. Choose the following constants: $a_T > \max\{1, C_\ell + 1\}, a_B > a_T + 2C_\ell + 1, c_\beta = \frac{\kappa B_\beta}{4(K+1)} > a_T + C_\ell + L_3 - 1, c_b = \frac{\kappa B_b}{4(K+1)} > a_B + C_\ell + L_3 - 1$.

Let $M = M_a$ as in Lemma S8. On the event Ω_M from Lemma S6, for $\mathcal{U} \in \mathcal{L}_M$ we have,

$$\begin{aligned}\frac{\pi(\mathcal{U})m_{\mathcal{U}}(\boldsymbol{Y})}{\pi(\mathcal{U}_0)m_{\mathcal{U}_0}(\boldsymbol{Y})} &\le \frac{\pi(\mathcal{U})}{\pi(\mathcal{U}_0)}\exp(-\|\boldsymbol{M}_{\mathcal{U}}\mu_0\|_2^2/4 + C_\ell h_{\mathcal{U}}\log p + (L_3-1)(d_{\mathcal{U}_0} - d_{\mathcal{U}})\log p + \eta_n \log p) \\ &\le \frac{\pi(\mathcal{U})}{\pi(\mathcal{U}_0)}\exp\{(-c_b r_{\text{out}} - c_\beta(s_{\text{out}} - r_{\text{in}})_+ \\ &\quad + C_\ell h_{\mathcal{U}} + (L_3-1)(s_{\text{out}} + r_{\text{out}} - s_{\text{in}} - r_{\text{in}}) + \eta_n)\log p).\end{aligned}$$

Where the second line follows from Lemma S9. For fixed $s_{\text{out}}, s_{\text{in}}, r_{\text{out}}, r_{\text{in}}$, let $\mathcal{L}_M(s_{\text{out}}, s_{\text{in}}, r_{\text{out}}, r_{\text{in}})$ denote the corresponding subset of $\mathcal{L}_M$. Its cardinality is bounded by

$$\begin{aligned}&\binom{s_0}{s_{\text{out}}}\binom{p-s_0}{s_{\text{in}}}\binom{r_0}{r_{\text{out}}}\binom{Kp-r_0}{r_{\text{in}}} \\ &\qquad \le s_\star^{s_{\text{out}}+r_{\text{out}}} p^{s_{\text{in}}}(Kp)^{r_{\text{in}}}.\end{aligned}$$

From Lemma S7, we have

$$\frac{\pi(\mathcal{U})}{\pi(\mathcal{U}_0)} \le 2^{s_{\text{out}}+r_{\text{out}}}\{(M+1)s_\star\}^{s_{\text{in}}+r_{\text{in}}} p^{a_T(s_{\text{out}}-s_{\text{in}})+a_B(r_{\text{out}}-r_{\text{in}})} c_T^{s_{\text{out}}-s_{\text{in}}} c_B^{r_{\text{out}}-r_{\text{in}}}.$$

Together with the last display gives

$$\sum_{\mathcal{U}\in\mathcal{L}_M(s_{\text{out}},s_{\text{in}},r_{\text{out}},r_{\text{in}})} \frac{\pi(\mathcal{U})m_{\mathcal{U}}(\boldsymbol{Y})}{\pi(\mathcal{U}_0)m_{\mathcal{U}_0}(\boldsymbol{Y})} \le p^{\eta_n} 2^{s_{\text{out}}+r_{\text{out}}} c_T^{s_{\text{out}}-s_{\text{in}}} c_B^{r_{\text{out}}-r_{\text{in}}}\{(M+1)s_\star\}^{s_{\text{in}}+r_{\text{in}}} s_\star^{s_{\text{out}}+r_{\text{out}}} K^{r_{\text{in}}} p^{\Psi},$$

where

$$\begin{aligned}\Psi = &(-a_T - (L_3-1) + C_\ell + 1)s_{\text{in}} + (a_T + L_3 - 1 + C_\ell)s_{\text{out}} \\ &+ (-a_B - (L_3-1) + C_\ell + 1)r_{\text{in}} + (a_B + L_3 - 1 + C_\ell - c_b)r_{\text{out}} - c_\beta(s_{\text{out}} - r_{\text{in}})_+.\end{aligned}$$

We next verify that $\Psi < 0$ whenever $\mathcal{U} \neq \mathcal{U}_0$. By initial choice of constants, we have $-a_T - (L_3-1) + C_\ell + 1$ and $a_B + L_3 - 1 + C_\ell - c_b$ strictly negative. The remaining terms of Ψ equal

$$\begin{cases}(a_T + L_3 - 1 + C_\ell - c_\beta)(s_{\text{out}} - r_{\text{in}}) + (a_T - a_B + 2C_\ell + 1)r_{\text{in}}, & s_{\text{out}} \ge r_{\text{in}}, \\ (a_T - a_B + 2C_\ell + 1)s_{\text{out}} + (-a_B - (L_3-1) + C_\ell + 1)(r_{\text{in}} - s_{\text{out}}), & s_{\text{out}} < r_{\text{in}}.\end{cases}$$

Indeed each term in both cases above is strictly negative. Defining

$$\begin{aligned}\delta_0 = \min\Big\{&a_T + L_3 - 1 - C_\ell - 1, c_b - a_B - (L_3-1) - C_\ell, c_\beta - a_T - (L_3-1) - C_\ell, \\ &\tfrac{1}{2}(a_B - a_T - 2C_\ell - 1), a_B + L_3 - 1 - C_\ell - 1\Big\} > 0,\end{aligned}$$

we obtain $\Psi \le -\delta_0 h_{\mathcal{U}}$.

It remains to upper bound the factors preceding p^{Ψ}. Assumption (B1) implies $\log\{(M+1)s_\star\} = o(\log p)$. The sequence $\epsilon_n = (\log 2 + |\log c_T| + |\log c_B| + \log K + \log\{(M+1)s_\star\} + \log s_\star)/\log p$ satisfies $\epsilon_n = o(1)$, and

$$2^{s_{\text{out}}+r_{\text{out}}} c_T^{s_{\text{out}}-s_{\text{in}}} c_B^{r_{\text{out}}-r_{\text{in}}} \{(M+1)s_\star\}^{s_{\text{in}}+r_{\text{in}}} s_\star^{s_{\text{out}}+r_{\text{out}}} K^{r_{\text{in}}} \le p^{\epsilon_n h_{\mathcal{U}}}$$

Taking $\delta = \delta_0/2$, for all sufficiently large n,

$$\begin{aligned}\sum_{\substack{\mathcal{U}\in\mathcal{L}_M \\ \mathcal{U}\neq\mathcal{U}_0}} \frac{\pi(\mathcal{U}) m_{\mathcal{U}}(\boldsymbol{Y})}{\pi(\mathcal{U}_0) m_{\mathcal{U}_0}(\boldsymbol{Y})} &\le p^{\eta_n} \sum_{h=1}^{\infty} \binom{h+3}{3} p^{-\delta h} \\ &\le p^{\eta_n} \sum_{h=1}^{\infty} (h+1)^3 p^{-\delta h} \lesssim p^{-\delta+\eta_n} = o(1).\end{aligned}$$

The last inequality follows from the standard power-series bound with $p^{-\delta} \le 1/2$. Consequently, on Ω_M,

$$\Pi(\mathcal{U} \neq \mathcal{U}_0,\ \mathcal{U} \in \mathcal{L}_M \mid \boldsymbol{Y}) \le \sum_{\substack{\mathcal{U}\in\mathcal{L}_M \\ \mathcal{U}\neq\mathcal{U}_0}} \frac{\pi(\mathcal{U}) m_{\mathcal{U}}(\boldsymbol{Y})}{\pi(\mathcal{U}_0) m_{\mathcal{U}_0}(\boldsymbol{Y})} = o(1).$$

Together with Lemma S8, this gives

$$\begin{aligned}\mathbb{E}_0 \Pi\{(\mathcal{S},\mathcal{R}) \neq (\mathcal{S}_0,\mathcal{R}_0) \mid \mathcal{D}_n\} \le{}& \mathbb{E}_0\{\Pi(\mathcal{U}\neq\mathcal{U}_0, \mathcal{U}\in\mathcal{L}_M \mid \boldsymbol{Y}) \mathbf{1}_{\Omega_M}\} \\ &+ \mathbb{P}_0(\Omega_M^c) + \mathbb{E}_0 \Pi(\mathcal{L}_M^c \mid \mathcal{D}_n) = o(1).\end{aligned}$$

□

S3.2.4 Proofs of lemmas

Proof of Lemma S5. Fix $\mathcal{U} = (\mathcal{S},\mathcal{R}) \in \mathcal{L}_M$. For $\boldsymbol{u}, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_K \in \mathbb{R}^p$ with $\operatorname{supp}(\boldsymbol{u}) \subseteq \mathcal{S}$ and $\operatorname{supp}(\boldsymbol{v}_k) \subseteq \mathcal{R}_k$, set $\boldsymbol{z} = (\boldsymbol{u}'_{\mathcal{S}}, \boldsymbol{v}'_{1,\mathcal{R}_1}, \ldots, \boldsymbol{v}'_{K,\mathcal{R}_K})'$. A direct computation gives

$$z' \boldsymbol{G}_{\mathcal{U}} z = \tau_\beta^2 \left\{ \|\boldsymbol{X}^{(T)}\boldsymbol{u}\|_2^2 + \sum_{k=1}^{K} \|\boldsymbol{X}^{(k)}(\boldsymbol{u} + c_\tau \boldsymbol{v}_k)\|_2^2 \right\}. \tag{S29}$$

Note that, by the Cauchy Schwarz inequality

$$\begin{aligned}\|\boldsymbol{u}\|_2^2 + \sum_{k=1}^{K} \|\boldsymbol{u} + c_\tau \boldsymbol{v}_k\|_2^2 &\le (2K+1)\|\boldsymbol{u}\|_2^2 + 2c_\tau^2 \sum_{k=1}^{K} \|\boldsymbol{v}_k\|_2^2, \\ \|\boldsymbol{u}\|_2^2 + \sum_{k=1}^{K} \|\boldsymbol{v}_k\|_2^2 &\le \left(1 + \frac{2K}{c_\tau^2}\right) \|\boldsymbol{u}\|_2^2 + \frac{2}{c_\tau^2} \sum_{k=1}^{K} \|\boldsymbol{u} + c_\tau \boldsymbol{v}_k\|_2^2.\end{aligned}$$

which gives $\|\boldsymbol{u}\|_2^2 + \sum_k \|\boldsymbol{v}_k\|_2^2 \asymp \|\boldsymbol{u}\|_2^2 + \sum_k \|\boldsymbol{u} + c_\tau \boldsymbol{v}_k\|_2^2$. Applying Condition (B2) to (S29) and using $\tau_\beta^2 n_{\min} \asymp p^{2L_3} n_{\min}/n_{\max}$ and $\tau_\beta^2 n_{\max} \lesssim p^{2L_3}$ proves (S21).

Next, we prove the same result for the Schur complement setting. Let $\mathcal{J} = (\mathcal{S}_{\mathcal{J}}, \mathcal{R}_{\mathcal{J}})$ be the union of two supports in $\mathcal{L}_M$, with $\mathcal{R}_{\mathcal{J},k} = \{j : (k,j) \in \mathcal{R}_{\mathcal{J}}\}$. Since $|\mathcal{S}_{\mathcal{J}}| \vee$

$\max_{1\le k\le K}|\mathcal{R}_{\mathcal{J},k}| \le 2Ms_\star$, the preceding argument gives $l_p\boldsymbol{I} \preceq \boldsymbol{G}_{\mathcal{J}} \preceq u_p\boldsymbol{I}$, where $l_p = cp^{2L_3}\frac{n_{\min}}{n_{\max}}$ and $u_p = Cp^{2L_3}$. Let $\mathcal{J}_0 \subset \mathcal{J}$ and $\mathcal{J}_1 = \mathcal{J}\setminus\mathcal{J}_0$, completing the square gives

$$\boldsymbol{x}'\boldsymbol{G}_{\mathcal{J}_1|\mathcal{J}_0}\boldsymbol{x} = \min_{\boldsymbol{y}}(\boldsymbol{y}',\boldsymbol{x}')\boldsymbol{G}_{\mathcal{J}}(\boldsymbol{y}',\boldsymbol{x}')', \quad \boldsymbol{G}_{\mathcal{J}_1|\mathcal{J}_0} = \boldsymbol{G}_{\mathcal{J}_1\mathcal{J}_1} - \boldsymbol{G}_{\mathcal{J}_1\mathcal{J}_0}\boldsymbol{G}_{\mathcal{J}_0\mathcal{J}_0}^{-1}\boldsymbol{G}_{\mathcal{J}_0\mathcal{J}_1}.$$

The lower bound for $\boldsymbol{G}_{\mathcal{J}}$ applies to every $\boldsymbol{y}$, while $\boldsymbol{y}=0$ gives the upper bound. Hence $l_p\boldsymbol{I} \preceq \boldsymbol{G}_{\mathcal{J}_1|\mathcal{J}_0} \preceq u_p\boldsymbol{I}$.

Finally, Condition (B2) implies that for every fixed $C>0$, each $\boldsymbol{X}_{\mathcal{A}}^{(k)}$ with $|\mathcal{A}| = \lfloor Cs_\star\rfloor$ is full rank, thus $n_{\min} \ge \lfloor Cs_\star\rfloor$ eventually, which implies $n_{\min}/s_\star \to \infty$. Together with $\inf_{\mathcal{U}\in\mathcal{L}_M}\lambda_{\min}(\boldsymbol{G}_{\mathcal{U}})/p \ge c\frac{p^{2L_3-1}n_{\min}}{n_{\max}} \ge c\frac{p}{n_{\max}} \longrightarrow \infty$, and $\sup_{\mathcal{U}\in\mathcal{L}_M} d_{\mathcal{U}} \le (K+1)Ms_\star = o(p)$, this proves (S22). □

Proof of Lemma S6. Since $\mathcal{L}_M \subseteq \mathcal{L}_{\max\{M,1\}}$, it suffices to consider $M \ge 1$, in which case $\mathcal{U}_0 \in \mathcal{L}_M$. We use $m_{\boldsymbol{G},1}$ from (S23) for any $\boldsymbol{G}\in\mathbb{S}_d^{++}$. For a fixed set $\mathcal{U}$, its conditional marginal likelihood is

$$m_{\mathcal{U}}(\boldsymbol{Y}) = \int (2\pi)^{-n/2}\exp\{-\tfrac{1}{2}\|\boldsymbol{Y}-\boldsymbol{W}_{\mathcal{U}}\boldsymbol{\theta}_{\mathcal{U}}\|_2^2\}\,d\Pi(\boldsymbol{\theta}_{\mathcal{U}}).$$

Set $\boldsymbol{z} = \boldsymbol{D}_{\mathcal{U}}^{-1}\boldsymbol{\theta}_{\mathcal{U}}$ and define $\widehat{\boldsymbol{\gamma}}_{\mathcal{U}} = \boldsymbol{G}_{\mathcal{U}}^{-1}\boldsymbol{V}_{\mathcal{U}}'\boldsymbol{Y}$. Note that this is well-defined as $\boldsymbol{G}_{\mathcal{U}} = \boldsymbol{V}_{\mathcal{U}}'\boldsymbol{V}_{\mathcal{U}}$ is positive definite by Lemma S5. Also note that $\boldsymbol{V}_{\mathcal{U}}\widehat{\boldsymbol{\gamma}}_{\mathcal{U}} = \boldsymbol{P}_{\mathcal{U}}\boldsymbol{Y}$ and the least-squares term can be decomposed as

$$\begin{aligned}\|\boldsymbol{Y}-\boldsymbol{V}_{\mathcal{U}}\boldsymbol{z}\|_2^2 &= \|\boldsymbol{Y}-\boldsymbol{V}_{\mathcal{U}}\widehat{\boldsymbol{\gamma}}_{\mathcal{U}}+\boldsymbol{V}_{\mathcal{U}}(\widehat{\boldsymbol{\gamma}}_{\mathcal{U}}-\boldsymbol{z})\|_2^2\\ &= \|\boldsymbol{M}_{\mathcal{U}}\boldsymbol{Y}\|_2^2 + (\boldsymbol{z}-\widehat{\boldsymbol{\gamma}}_{\mathcal{U}})'\boldsymbol{G}_{\mathcal{U}}(\boldsymbol{z}-\widehat{\boldsymbol{\gamma}}_{\mathcal{U}}).\end{aligned}$$

Thus the conditional marginal likelihood can be represented as

$$m_{\mathcal{U}}(\boldsymbol{Y}) = (2\pi)^{-n/2}\exp\{-\tfrac{1}{2}\|\boldsymbol{M}_{\mathcal{U}}\boldsymbol{Y}\|_2^2\}m_{\boldsymbol{G}_{\mathcal{U}},1}(\widehat{\boldsymbol{\gamma}}_{\mathcal{U}}). \tag{S30}$$

We first derive a bound on $m_{\boldsymbol{G},1}(\boldsymbol{x})$ that holds uniformly over $\boldsymbol{x}\in\mathbb{R}^d$. Let $q = \frac{\log p+1}{\log p}$, $q' = \log p+1$, such that $q^{-1}+q'^{-1}=1$.
By Hölder's inequality, we have that

$$\begin{aligned}m_{\boldsymbol{G},1}(\boldsymbol{x}) &\le \left\{\int_{\mathbb{R}^d} e^{-q(\boldsymbol{z}-\boldsymbol{x})'\boldsymbol{G}(\boldsymbol{z}-\boldsymbol{x})/2}\,d\boldsymbol{z}\right\}^{1/q}\left\{\int_{\mathbb{R}^d}\prod_{i=1}^d g_1(z_i)^{q'}\,d\boldsymbol{z}\right\}^{1/q'}\\ &= \left(\frac{2\pi}{q}\right)^{d/(2q)}|\boldsymbol{G}|^{-1/(2q)}\|g_1\|_{q'}^d.\end{aligned}$$

For $r\ge 2$, Lemma S4 gives

$$\begin{aligned}\int_{|z_i|\le 1} g_1(z_i)^r\,dz_i &\le 2C_0^r\int_0^1\{\log(e/z_i)\}^r\,dz_i = 2C_0^r\int_0^\infty (1+t)^r e^{-t}\,dt,\\ \int_{|z_i|>1} g_1(z_i)^r\,dz_i &\le 2C_0^r\int_1^\infty z_i^{-r/2}e^{-rz_i}\,dz_i \le \frac{2C_0^r}{r}e^{-r}.\end{aligned}$$

Notice that by convexity of $f(x) = x^r$,

$$\int_0^\infty (1+t)^r e^{-t}\,dt \le \int_0^\infty 2^{r-1}(1+t^r)e^{-t}\,dt \\ = 2^{r-1}(1+\Gamma(r+1)).$$

This then implies that $\|g_1\|_r^r \le C_1^r \Gamma(r+1)$, and since $\Gamma(r+1)^{1/r} \lesssim r$ by Stirling's inequality, we obtain $\|g_1\|_r \lesssim r$ for $r \ge 2$.

Moreover, $|\boldsymbol{G}|^{-1/(2q)} = |\boldsymbol{G}|^{-1/2}|\boldsymbol{G}|^{1/(2q')}$. Consider $\boldsymbol{G} = \boldsymbol{G}_{\mathcal{U}}$, where $\mathcal{U} \in \mathcal{L}_M$, Lemma S5 gives $\lambda_{\max}(\boldsymbol{G}) \le Cp^{2L_3}$, and hence

$$|\boldsymbol{G}|^{1/(2q')} \le (Cp^{2L_3})^{d/[2(\log p+1)]} = \exp(d\frac{\log C + 2L_3 \log p}{2+2\log p}) \le e^{Cd}.$$

Combining all terms and using $q' = \log p + 1$ yields

$$m_{\boldsymbol{G},1}(\boldsymbol{x}) \le |\boldsymbol{G}|^{-1/2} \exp\{Cd \log\log p\}, \quad \boldsymbol{x} \in \mathbb{R}^d. \tag{S31}$$

We next lower bound $m_{\boldsymbol{G}_{\mathcal{U}_0},1}(\widehat{\boldsymbol{\gamma}}_{\mathcal{U}_0})$. First, by Lemma S4 we have the standard lower bound for the slab distribution, $g_1(t) \ge c(1+|t|)^{-1/2}e^{-|t|} \ge ce^{-\frac{3}{2}|t|}$, and thus $\prod_{i=1}^d g_1(u_i) \ge c^d e^{-\frac{3}{2}\|\boldsymbol{u}\|_1}$. Next notice that

$$\begin{aligned} m_{\mathbf{G},1}(\boldsymbol{x}) &= (2\pi)^{d/2} \int \phi_d(\mathbf{G}^{1/2}\boldsymbol{v}) \prod_{i=1}^d g_1(v_i + x_i)\,d\boldsymbol{v} \\ &= (2\pi)^{d/2}|\mathbf{G}|^{-1/2}\mathbb{E}_\xi\left[\prod_{i=1}^d g_1(x_i+\xi_i)\right] \\ &\ge (2\pi)^{d/2}|\mathbf{G}|^{-1/2}\mathbb{E}_\xi\left[c^d e^{-\frac{3}{2}\|\boldsymbol{x}+\boldsymbol{\xi}\|_1}\right] \\ &\ge (c^2\,2\pi)^{d/2}|\mathbf{G}|^{-1/2}e^{-\frac{3}{2}\|\boldsymbol{x}\|_1 - \frac{3}{2}\mathbb{E}_\xi\|\boldsymbol{\xi}\|_1}, \end{aligned}$$

where $\boldsymbol{\xi} \sim N(0, \mathbf{G}^{-1})$. The last line follows from Jensen's inequality and the triangle inequality. Further, standard normality theory gives that

$$\mathbb{E}\|\boldsymbol{\xi}\|_1 = \sqrt{\frac{2}{\pi}}\sum_{i=1}^d \sqrt{(\mathbf{G}^{-1})_{ii}} \le \sqrt{\frac{2}{\pi}} d/\sqrt{\lambda_{\min}(\mathbf{G})}.$$

Consequently, whenever $\lambda_{\min}(\mathbf{G}) \ge 1$,

$$m_{\mathbf{G},1}(\boldsymbol{x}) \ge |\mathbf{G}|^{-1/2}\exp\left(-Cd - \frac{3}{2}\|\boldsymbol{x}\|_1\right).$$

Now we apply this bound to $\boldsymbol{x} = \widehat{\boldsymbol{\gamma}}_{\mathcal{U}_0}$. Define $\boldsymbol{\gamma}_0 = \boldsymbol{D}_{\mathcal{U}_0}^{-1}\boldsymbol{\theta}_0$, then $\widehat{\boldsymbol{\gamma}}_{\mathcal{U}_0}$ can be written as $\boldsymbol{\gamma}_0 + \boldsymbol{\zeta}$, where $\boldsymbol{\zeta} = \boldsymbol{G}_{\mathcal{U}_0}^{-1}\boldsymbol{V}_{\mathcal{U}_0}'\boldsymbol{\varepsilon} \sim N_{d_{\mathcal{U}_0}}(0, \boldsymbol{G}_{\mathcal{U}_0}^{-1})$. By Cauchy–Schwarz and Lemma S5,

$$\begin{aligned} \mathbb{E}_0\|\zeta\|_1 &\le \sum_{i=1}^{d_{\mathcal{U}_0}}\{(\boldsymbol{G}_{\mathcal{U}_0}^{-1})_{ii}\}^{1/2} \le d_{\mathcal{U}_0}\lambda_{\min}(\boldsymbol{G}_{\mathcal{U}_0})^{-1/2} \\ &\lesssim \frac{d_{\mathcal{U}_0}\sqrt{n_{\max}}}{p^{L_3}\sqrt{n_{\min}}} = \frac{d_{\mathcal{U}_0}}{n_{\min}}\frac{\sqrt{n_{\min}n_{\max}}}{p^{L_3}} = o(1). \end{aligned}$$

The last step follows by the fact that $d_{\mathcal{U}_0} \leq (K+1)s_\star$ and $n_{\min}/s_\star \to \infty$ by Lemma S5, while $\sqrt{n_{\min} n_{\max}}/p^{L_3} \leq n_{\max}/p \to 0$. Thus, we can choose a sequence $\delta_n = \{\mathbb{E}_0\|\boldsymbol{\zeta}\|_1\}^{1/2} \to 0$ such that $\mathbb{P}_0(\|\boldsymbol{\zeta}\|_1 \leq \delta_n) \longrightarrow 1$ by Markov's inequality. Thus, on this event, $\|\widehat{\boldsymbol{\gamma}}_{\mathcal{U}_0}\|_1 \leq \|\boldsymbol{\gamma}_0\|_1 + \delta_n$. Finally, since Lemma S5 imply $\lambda_{\min}(\boldsymbol{G}_{\mathcal{U}}) \geq 1$ eventually,

$$m_{\boldsymbol{G}_{\mathcal{U}_0},1}(\widehat{\boldsymbol{\gamma}}_{\mathcal{U}_0}) \geq |\boldsymbol{G}_{\mathcal{U}_0}|^{-1/2} \exp\{-Cd_{\mathcal{U}_0} - \frac{3}{2}\|\gamma_0\|_1 - \frac{3}{2}\delta_n\}. \tag{S32}$$

It remains to compare the residual and determinant terms. Since $\boldsymbol{M}_{\mathcal{U}_0}\boldsymbol{\mu}_0 = 0$, expansion under $\boldsymbol{Y} = \boldsymbol{\mu}_0 + \boldsymbol{\varepsilon}$ gives

$$\begin{aligned} -\frac{1}{2}\{\|\boldsymbol{M}_{\mathcal{U}}\boldsymbol{Y}\|_2^2 - \|\boldsymbol{M}_{\mathcal{U}_0}\boldsymbol{Y}\|_2^2\} = & -\frac{1}{2}\|\boldsymbol{M}_{\mathcal{U}}\boldsymbol{\mu}_0\|_2^2 - (\boldsymbol{M}_{\mathcal{U}}\boldsymbol{\mu}_0)'\boldsymbol{\varepsilon} \\ & + \frac{1}{2}\boldsymbol{\varepsilon}'(\boldsymbol{P}_{\mathcal{U}} - \boldsymbol{P}_{\mathcal{U}_0})\boldsymbol{\varepsilon}. \end{aligned} \tag{S33}$$

Consider a fixed set $\mathcal{U} \neq \mathcal{U}_0$ and let $h = h_{\mathcal{U}}$. Let $x_h \geq h$ be a quantity depending on h to be specified later. Lemma S15 gives with probability at least $1 - e^{-x_h}$,

$$-(\boldsymbol{M}_{\mathcal{U}}\boldsymbol{\mu}_0)'\boldsymbol{\varepsilon} \leq \|\boldsymbol{M}_{\mathcal{U}}\boldsymbol{\mu}_0\|_2\sqrt{2x_h} \leq \frac{1}{4}\|\boldsymbol{M}_{\mathcal{U}}\boldsymbol{\mu}_0\|_2^2 + 2x_h.$$

For the quadratic term, let $\boldsymbol{A}_{\mathcal{U}} = (\boldsymbol{P}_{\mathcal{U}} - \boldsymbol{P}_{\mathcal{U}_0})_+$ denote the positive semidefinite part of $\boldsymbol{P}_{\mathcal{U}} - \boldsymbol{P}_{\mathcal{U}_0}$. Lemma S5, applied to $\mathcal{U} \cup \mathcal{U}_0$, implies that the columns in this union are linearly independent. Hence $\mathrm{col}(\boldsymbol{W}_{\mathcal{U}}) \cap \mathrm{col}(\boldsymbol{W}_{\mathcal{U}_0}) = \mathrm{col}(\boldsymbol{W}_{\mathcal{U}\cap\mathcal{U}_0})$, and the positive eigenspace of $\boldsymbol{P}_{\mathcal{U}} - \boldsymbol{P}_{\mathcal{U}_0}$ has dimension at most $|\mathcal{U} \setminus \mathcal{U}_0| \leq h$. The spectrum of a difference of orthogonal projections lies in $[-1, 1]$. Therefore, we have the following facts:

$$\mathrm{rank}(\boldsymbol{A}_{\mathcal{U}}) \leq h, \quad \|\boldsymbol{A}_{\mathcal{U}}\|_{\mathrm{sp}} \leq 1, \quad \mathrm{tr}(\boldsymbol{A}_{\mathcal{U}}) \leq h, \quad \mathrm{tr}(\boldsymbol{A}_{\mathcal{U}}^2) \leq h.$$

Moreover, $\boldsymbol{\varepsilon}'(\boldsymbol{P}_{\mathcal{U}} - \boldsymbol{P}_{\mathcal{U}_0})\boldsymbol{\varepsilon} \leq \boldsymbol{\varepsilon}'\boldsymbol{A}_{\mathcal{U}}\boldsymbol{\varepsilon}$. By Lemma S15, the following holds with probability at least $1 - e^{-x_h}$,

$$\boldsymbol{\varepsilon}'(\boldsymbol{P}_{\mathcal{U}} - \boldsymbol{P}_{\mathcal{U}_0})\boldsymbol{\varepsilon} \leq h + 2\sqrt{hx_h} + 2x_h \leq 5x_h.$$

Let Ω'_M be the event on which the two preceding stochastic bounds hold for every $\mathcal{U} \in \mathcal{L}_M$. Since there are at most $\binom{(K+1)p}{h}$ sets $\mathcal{U}$ such that $|\mathcal{U} \triangle \mathcal{U}_0| = h$, and setting $x_h = 2h\log p + 2h\log\{e(K+1)\}$,

$$\binom{(K+1)p}{h} e^{-x_h} \leq \left\{\frac{e(K+1)p}{h}\right\}^h p^{-2h}\{e(K+1)\}^{-2h} \leq p^{-h},$$

Then $2\sum_{h=1}^{(K+1)p}\binom{(K+1)p}{h}e^{-x_h} \leq 2\sum_{h=1}^{\infty} p^{-h} = o(1)$ and we have that $\mathbb{P}_0(\Omega'_M) \to 1$. The case $h_{\mathcal{U}} = 0$ is $\mathcal{U} = \mathcal{U}_0$ and is immediate.

We now examine the term $|G_{\mathcal{U}}|$. Let $C = \mathcal{U} \cap \mathcal{U}_0$, $A = \mathcal{U} \setminus C$, and $B = \mathcal{U}_0 \setminus C$. Schur factorization gives $|\boldsymbol{G}_{\mathcal{U}}| = |\boldsymbol{G}_C||\boldsymbol{G}_{A|C}|$ and $|\boldsymbol{G}_{\mathcal{U}_0}| = |\boldsymbol{G}_C||\boldsymbol{G}_{B|C}|$, and Lemma S5 bounds every eigenvalue of both $\boldsymbol{G}_{A|C}, \boldsymbol{G}_{B|C}$ between $l_p = cp^{2L_3}n_{\min}/n_{\max}$ and $u_p = Cp^{2L_3}$. Since $cp^2 n_{\min}/n_{\max} \to \infty$, this implies $l_p \geq p^{2(L_3-1)}$ eventually, hence

$$\begin{aligned} \frac{1}{2}\log\frac{|\boldsymbol{G}_{\mathcal{U}_0}|}{|\boldsymbol{G}_{\mathcal{U}}|} &= \frac{1}{2}\{\log|\boldsymbol{G}_{B|C}| - \log|\boldsymbol{G}_{A|C}|\} \\ &\leq L_3|B|\log p - (L_3-1)|A|\log p + C|B| \\ &\leq h_{\mathcal{U}}\log p + (L_3-1)(d_{\mathcal{U}_0} - d_{\mathcal{U}})\log p + Ch_{\mathcal{U}}. \end{aligned} \tag{S34}$$

On $\Omega_M = \Omega'_M \cap \{\|\zeta\|_1 \le \delta_n\}$, combine (S30), (S31), (S32), (S33), and (S34). For $h_{\mathcal{U}} \ge 1$, this gives

$$\begin{aligned}\log \frac{m_{\mathcal{U}}(\boldsymbol{Y})}{m_{\mathcal{U}_0}(\boldsymbol{Y})} &\le -\frac{1}{4}\|\boldsymbol{M}_{\mathcal{U}}\boldsymbol{\mu}_0\|_2^2 + \frac{9}{2}x_{h_{\mathcal{U}}} + h_{\mathcal{U}}\log p + (L_3-1)(d_{\mathcal{U}_0}-d_{\mathcal{U}})\log p \\ &\quad + C(h_{\mathcal{U}} + d_{\mathcal{U}}\log\log p + d_{\mathcal{U}_0}) + \frac{3}{2}\|\gamma_0\|_1 + \frac{3}{2}\delta_n \\ &\le -\frac{1}{4}\|\boldsymbol{M}_{\mathcal{U}}\boldsymbol{\mu}_0\|_2^2 + C_\ell h_{\mathcal{U}}\log p + (L_3-1)(d_{\mathcal{U}_0}-d_{\mathcal{U}})\log p \\ &\quad + C\{h_{\mathcal{U}} + d_{\mathcal{U}}\log\log p + d_{\mathcal{U}_0} + \|\gamma_0\|_1 + 1\}.\end{aligned}$$

The same conclusion is immediate for $\mathcal{U} = \mathcal{U}_0$. Define $\eta_n \log p = C\{s_\star \log\log p + \|\gamma_0\|_1 + 1\}$. Assumptions (B1) and (B3), together with the fact that $\mathcal{U} \in \mathcal{L}_M$ imply $\eta_n \to 0$, while $\mathbb{P}_0(\Omega_M) \to 1$. This proves (S24). □

Proof of Lemma S7. For any $\mathcal{A} \subseteq [p]$ such that $|\mathcal{A}| = r$, by standard Beta-Binomial properties,

$$\pi(\mathcal{A}) = cp^a B(r+1, p-r+cp^a).$$

Here we use $\pi(\mathcal{A}), \pi(r)$ interchangeably. The Beta function identity then gives

$$\frac{\pi(r+1)}{\pi(r)} = \frac{r+1}{p-r-1+cp^a}, \quad \frac{\pi(r-1)}{\pi(r)} = \frac{p-r+cp^a}{r}. \tag{S35}$$

Since $a > 1$, for large enough p, $p \le cp^a$. Thus $\pi(0) = cp^a/(p+cp^a) \ge 1/2$ and

$$\frac{\pi(r)}{\pi(r-1)} = \frac{r}{p-r+cp^a} \ge \frac{1}{2cp^a}.$$

Iteration gives $\pi(r) \ge 2^{-(r+1)}c^{-r}p^{-ar}$. Applying this to $\mathcal{U}_0$ proves (S25).

For $\mathcal{A}, \mathcal{A}_0 \subset [p]$, let $a_0 = |\mathcal{A}_0|$, $d_{\text{out}} = |\mathcal{A}_0 \backslash \mathcal{A}|$, and $d_{\text{in}} = |\mathcal{A} \backslash \mathcal{A}_0|$. If $|\mathcal{A}| \vee |\mathcal{A}_0| \le (M+1)s_\star$, first removing the d_{out} omitted variables and then adding the d_{in} new variables gives

$$\begin{aligned}\frac{\pi(\mathcal{A})}{\pi(\mathcal{A}_0)} &= \left\{\prod_{\ell=0}^{d_{\text{out}}-1}\frac{\pi(a_0-\ell-1)}{\pi(a_0-\ell)}\right\}\left\{\prod_{\ell=0}^{d_{\text{in}}-1}\frac{\pi(a_0-d_{\text{out}}+\ell+1)}{\pi(a_0-d_{\text{out}}+\ell)}\right\} \\ &\le (2cp^a)^{d_{\text{out}}}\left(\frac{(M+1)s_\star}{cp^a}\right)^{d_{\text{in}}} = 2^{d_{\text{out}}}\{(M+1)s_\star\}^{d_{\text{in}}}c^{d_{\text{out}}-d_{\text{in}}}p^{a(d_{\text{out}}-d_{\text{in}})}.\end{aligned} \tag{S36}$$

Applying (S36) with $(c,a) = (c_T, a_T)$ to $(\mathcal{S}, \mathcal{S}_0)$, and with $(c,a) = (c_B, a_B)$ to each $(\mathcal{R}_k, \mathcal{R}_{0k})$, proves the remaining bounds. □

Proof of Lemma S8. We first bound from below the posterior denominator $\mathcal{Z}(\boldsymbol{Y})$, then use the exponentially small prior probability of oversized models to show that their posterior probability vanishes in expectation. Retaining only $\mathcal{U}_0$ in the prior mixture and using (S30),

$$\begin{aligned}\mathcal{Z}(\boldsymbol{Y}) &\ge \pi(\mathcal{U}_0)\frac{m_{\mathcal{U}_0}(\boldsymbol{Y})}{L(\boldsymbol{Y};\boldsymbol{\beta}_0,\boldsymbol{b}_0)} \\ &= \pi(\mathcal{U}_0)\exp\{\tfrac{1}{2}\boldsymbol{\varepsilon}'\boldsymbol{P}_{\mathcal{U}_0}\boldsymbol{\varepsilon}\}m_{\boldsymbol{G}_{\mathcal{U}_0},1}(\boldsymbol{\gamma}_0+\boldsymbol{\zeta}) \\ &\ge \pi(\mathcal{U}_0)m_{\boldsymbol{G}_{\mathcal{U}_0},1}(\boldsymbol{\gamma}_0+\boldsymbol{\zeta}).\end{aligned}$$

By (S32), on an event with probability tending to one,

$$m_{\boldsymbol{G}_{\mathcal{U}_0},1}(\boldsymbol{\gamma}_0+\boldsymbol{\zeta}) \geq |\boldsymbol{G}_{\mathcal{U}_0}|^{-1/2}\exp\{-Cd_{\mathcal{U}_0}-\frac{3}{2}\|\boldsymbol{\gamma}_0\|_1-o(1)\}.$$

Lemma S5 also gives

$$|\boldsymbol{G}_{\mathcal{U}_0}|^{-1/2} \geq \exp\{-L_3 d_{\mathcal{U}_0}\log p - Cd_{\mathcal{U}_0}\}.$$

Together with (S25) and condition (B3),

$$\log \mathcal{Z}(\boldsymbol{Y}) \geq -(a_T s_0 + a_B r_0 + Cd_{\mathcal{U}_0})\log p - Cd_{\mathcal{U}_0} - \frac{3}{2}\|\boldsymbol{\gamma}_0\|_1 - C.$$

Conditions (B1) and (B3) imply (S26) for some $C_E<\infty$ on events $\mathcal{E}_E$ with $\mathbb{P}_0(\mathcal{E}_E)\to 1$. Let $\pi_a(r)=\binom{p}{r}\pi(r)$. For $a>1$, (S35) gives

$$\frac{\pi_a(r+1)}{\pi_a(r)} = \frac{p-r}{p-r-1+cp^a} \leq c^{-1}p^{1-a} =: \rho_a.$$

Thus $\pi_a(r)\leq \rho_a^r$, and, since $\rho_a\leq 1/2$ eventually,

$$\sum_{r=\lfloor t\rfloor+1}^{p}\pi_a(r) \leq 2\big(c^{-1}p^{-(a-1)}\big)^{\lfloor t\rfloor+1} = 2p^{-(a-1)\lfloor t\rfloor}c^{-(\lfloor t\rfloor+1)}p^{-(a-1)}$$

for all $t>0$. Setting $t=M_a s_\star$, condition (B1) implies $M_a s_\star = o(\log p)$, and consequently,

$$\begin{aligned}\sum_{\mathcal{U}\in\mathcal{L}_{M_a}^c}\pi(\mathcal{U}) &\leq \sum_{r>M_a s_\star}\pi_{a_T}(r) + \sum_{k=1}^{K}\sum_{r>M_a s_\star}\pi_{a_B}(r)\\ &\leq 2(K+1)\exp\{-a_m M_a s_\star \log p\}. \end{aligned} \tag{S37}$$

For every fixed $\mathcal{U}$, by Fubini's theorem we have

$$\begin{aligned}\mathbb{E}_0\frac{m_{\mathcal{U}}(\boldsymbol{Y})}{L(\boldsymbol{Y};\boldsymbol{\beta}_0,\boldsymbol{b}_0)} &= \int_{\mathbb{R}^n} m_{\mathcal{U}}(\boldsymbol{y})\,d\boldsymbol{y}\\ &= \int\left\{\int_{\mathbb{R}^n} L(\boldsymbol{y};\boldsymbol{\theta}_{\mathcal{U}})\,d\boldsymbol{y}\right\}d\Pi_{\mathcal{U}}(\boldsymbol{\theta}_{\mathcal{U}}) = 1.\end{aligned} \tag{S38}$$

For any collection $\mathcal{V}$ of supports, (S26) and (S38) imply

$$\begin{aligned}\mathbb{E}_0\{\Pi(\mathcal{V}\mid\boldsymbol{Y})\mathbf{1}_{\mathcal{E}_E}\} &\leq e^{C_E s_\star\log p}\sum_{\mathcal{U}\in\mathcal{V}}\pi(\mathcal{U})\mathbb{E}_0\frac{m_{\mathcal{U}}(\boldsymbol{Y})}{L(\boldsymbol{Y};\boldsymbol{\beta}_0,\boldsymbol{b}_0)}\\ &= e^{C_E s_\star\log p}\sum_{\mathcal{U}\in\mathcal{V}}\pi(\mathcal{U}).\end{aligned}$$

Taking $\mathcal{V}=\mathcal{L}_{M_a}^c$, (S37) and $a_m M_a - C_E \geq 3$ give

$$\mathbb{E}_0\{\Pi(\mathcal{L}_{M_a}^c\mid\boldsymbol{Y})\mathbf{1}_{\mathcal{E}_E}\} \leq 2(K+1)e^{-3s_\star\log p} = o(1).$$

Since $\mathbb{P}_0(\mathcal{E}_E^c)=o(1)$, this completes the proof of (S27). □

Proof of Lemma S9. Fix $\mathcal{U} \in \mathcal{L}_M$ and any $\boldsymbol{\theta}_\mathcal{U}$ supported on $\mathcal{U}$. Set $\beta_j = 0$ for $j \notin \mathcal{S}$ and $b_j^{(k)} = 0$ for $j \notin \mathcal{R}_k$, and let $a_j = \beta_j - \beta_{0j}$, $c_{kj} = b_j^{(k)} - b_{0j}^{(k)}$, and $E_j = n_T a_j^2 + \sum_{k=1}^K n_k (a_j + c_{kj})^2$. Since $\mathrm{supp}(\boldsymbol{a}) \subset \mathcal{S} \cup \mathcal{S}_0, \mathrm{supp}(\boldsymbol{a} + \boldsymbol{c}_k) \subset \mathcal{S} \cup \mathcal{S}_0 \cup \mathcal{R}_k \cup \mathcal{R}_{0k}$, and the two sets on the right have sizes at most $(2M+2)s_\star$, Condition (B2) gives

$$\begin{aligned} \|\boldsymbol{\mu}_0 - \boldsymbol{W}_\mathcal{U} \boldsymbol{\theta}_\mathcal{U}\|_2^2 &= \|\boldsymbol{X}^{(T)} \boldsymbol{a}\|_2^2 + \sum_{k=1}^K \|\boldsymbol{X}^{(k)} (\boldsymbol{a} + \boldsymbol{c}_k)\|_2^2 \\ &\geq \underline{\kappa} \left\{ n_T \|\boldsymbol{a}\|_2^2 + \sum_{k=1}^K n_k \|\boldsymbol{a} + \boldsymbol{c}_k\|_2^2 \right\} = \underline{\kappa} \sum_{j=1}^p E_j. \end{aligned} \tag{S39}$$

If $(k,j) \in \mathcal{R}_0 \setminus \mathcal{R}$, then $c_{kj} = -b_{0j}^{(k)}$, and completing the square gives

$$\begin{aligned} n_T a_j^2 + n_k \{a_j - b_{0j}^{(k)}\}^2 &= (n_T + n_k) \left\{ a_j - \frac{n_k}{n_T + n_k} b_{0j}^{(k)} \right\}^2 + \frac{n_T n_k}{n_T + n_k} \{b_{0j}^{(k)}\}^2 \\ &\geq \frac{n_T n_k}{n_T + n_k} \{b_{0j}^{(k)}\}^2. \end{aligned} \tag{S40}$$

Let $\mathcal{O}_\mathcal{U} = \{j \in \mathcal{S}_0 \setminus \mathcal{S} : \{k : (k,j) \in \mathcal{R} \setminus \mathcal{R}_0\} = \varnothing\}$. Each $j \in (\mathcal{S}_0 \setminus \mathcal{S}) \setminus \mathcal{O}_\mathcal{U}$ has at least one pair $(k,j) \in \mathcal{R} \setminus \mathcal{R}_0$, therefore

$$s_{\text{out}} - |\mathcal{O}_\mathcal{U}| \leq |\mathcal{R} \setminus \mathcal{R}_0| = r_{\text{in}}, \quad |\mathcal{O}_\mathcal{U}| \geq (s_{\text{out}} - r_{\text{in}})_+. \tag{S41}$$

For $j \in \mathcal{O}_\mathcal{U}$, $a_j = -\beta_{0j}$. If $(k,j) \notin \mathcal{R}_0$, then the definition of $\mathcal{O}_\mathcal{U}$ also gives $(k,j) \notin \mathcal{R}$, and so $c_{kj} = 0$. Therefore

$$n_T a_j^2 + \sum_{k:(k,j) \notin \mathcal{R}_0} n_k (a_j + c_{kj})^2 = n_j^{\text{eff}} \beta_{0j}^2, \quad j \in \mathcal{O}_\mathcal{U}. \tag{S42}$$

For each j, sum (S40) over k with $(k,j) \in \mathcal{R}_0 \setminus \mathcal{R}$, and include (S42) when $j \in \mathcal{O}_\mathcal{U}$. Since each of these at most $K+1$ left-hand sides is bounded by E_j, we obtain

$$(K+1) E_j \geq \mathbf{1}\{j \in \mathcal{O}_\mathcal{U}\} n_j^{\text{eff}} \beta_{0j}^2 + \sum_{k:(k,j) \in \mathcal{R}_0 \setminus \mathcal{R}} \frac{n_T n_k}{n_T + n_k} \{b_{0j}^{(k)}\}^2.$$

Summing over j, using (S41), and applying Condition (B3),

$$\sum_{j=1}^p E_j \geq \frac{B_\beta (s_{\text{out}} - r_{\text{in}})_+ + B_b r_{\text{out}}}{K+1} \log p.$$

Combining this inequality with (S39), taking the infimum over $\boldsymbol{\theta}_\mathcal{U}$, and dividing by four gives (S28). □

S3.3 Proof of Theorem 3.4

S3.3.1 Settings & Notation

With Theorem 3.3, it suffices to prove the contraction rates conditional on $\mathcal{R} = \mathcal{R}_0$, since $\Pi(\boldsymbol{\beta} \in \mathcal{A} \mid \boldsymbol{Y}) \le \Pi(\mathcal{A} \mid \mathcal{R}_0, \boldsymbol{Y}) + \Pi(\mathcal{R} \neq \mathcal{R}_0 \mid \boldsymbol{Y})$. We thus work below with the conditional posterior $\Pi_0(\mathcal{A} \mid \boldsymbol{Y}) := \Pi(\boldsymbol{\beta} \in \mathcal{A} \mid \boldsymbol{Y}, \mathcal{R} = \mathcal{R}_0)$. Similar to Section S3.2, we define $\boldsymbol{P}_0$ as the orthogonal projection onto the column space of $\boldsymbol{Z}_{\mathcal{R}_0}$ and $\boldsymbol{M}_0 = \boldsymbol{I}_n - \boldsymbol{P}_0$. Further, we introduce some more quantities,

$$\boldsymbol{A}_0 = (\boldsymbol{Z}_{\mathcal{R}_0}'\boldsymbol{Z}_{\mathcal{R}_0})^{-1}\boldsymbol{Z}_{\mathcal{R}_0}'\widetilde{\boldsymbol{X}}, \quad h_0(\boldsymbol{\beta}) = \boldsymbol{A}_0(\boldsymbol{\beta} - \boldsymbol{\beta}_0), \text{ and } Q_0(\boldsymbol{\beta}) = \|\boldsymbol{M}_0\widetilde{\boldsymbol{X}}(\boldsymbol{\beta} - \boldsymbol{\beta}_0)\|_2^2.$$

If $r_0 = 0$, all formulae below are interpreted with $\boldsymbol{Z}_{\mathcal{R}_0}$ empty and $\boldsymbol{P}_0 = 0$. For $\mathcal{T} \subset [p]$, define the effective-sample-size matrix

$$D_0(\mathcal{T}) = \operatorname{diag}\{n_j^{\text{eff}} : j \in \mathcal{T}\}, \quad \omega_A(C_{\mathcal{T}}) = \sup_{\mathcal{T}:|\mathcal{T}|\le C_{\mathcal{T}} s_0} \sup_{\boldsymbol{u}\neq 0} \frac{\|\boldsymbol{A}_{0,\mathcal{T}}\boldsymbol{u}\|_1}{\boldsymbol{u}' D_0(\mathcal{T})\boldsymbol{u}^{1/2}}.$$

For $\boldsymbol{h} \in \mathbb{R}^{r_0}$, define the shifted nuisance marginal likelihood and its marginal likelihood ratio by

$$I_Y(\boldsymbol{h}) = \int_{\mathbb{R}^{r_0}} p(\boldsymbol{Y}|\boldsymbol{\beta}_0, \boldsymbol{t}) g_{\tau_B, r_0}(\boldsymbol{t} - \boldsymbol{h})\, d\boldsymbol{t}, \quad S_Y(\boldsymbol{h}) = \frac{I_Y(\boldsymbol{h})}{I_Y(0)}.$$

Similar to previous sections, we use $c, C \in (0, \infty)$ to denote constants that are allowed to change from line to line.

S3.3.2 Supporting lemmas

The proof of Theorem 3.4 relies on the conditions (C1)–(C4) below, which are formulated for the posterior contraction argument. The following lemma shows that these conditions follow from Conditions (B1)–(B3) under the stated prior specifications.

Lemma S10. *Assume (B1)–(B3). Then, for every fixed $C_{\mathcal{T}} < \infty$, the following conditions hold for all n sufficiently large.*

(C1) For every $\mathcal{S} \subset [p]$ with $|\mathcal{S}| \le C_{\mathcal{T}} s_0$ and every $\boldsymbol{u} \in \mathbb{R}^{|\mathcal{S}|}$,

$$\underline{\kappa}\, \boldsymbol{u}' D_0(\mathcal{S})\boldsymbol{u} \;\le\; \|\boldsymbol{M}_0\widetilde{\boldsymbol{X}}_{\mathcal{S}}\boldsymbol{u}\|_2^2 \;\le\; \overline{\kappa}\, \boldsymbol{u}' D_0(\mathcal{S})\boldsymbol{u}; \tag{S43}$$

(C2) Denote $\log_+(x) = \max\{\log x, 0\}$,

$$\max_{1\le j\le p} \log_+\Big(\tau_\beta \sqrt{n_j^{\text{eff}}}/\log p\Big) \lesssim \log p, \tag{S44}$$

$$\|\boldsymbol{\beta}_{0,\mathcal{S}_0}\|_1/\tau_\beta \lesssim s_0 \log p. \tag{S45}$$

(C3) Under (P-B1),

$$-\log \pi_p(s_0) \lesssim s_0 \log p. \tag{S46}$$

(C4) Let $\boldsymbol{G}_0 = \boldsymbol{Z}_{\mathcal{R}_0}'\boldsymbol{Z}_{\mathcal{R}_0}$ and define $L_Z = \sum_{j=1}^{r_0} \log\Big[C_{sh} \log\{e + \tau_B\sqrt{(\boldsymbol{G}_0)_{jj}}\}\Big]$, where C_{sh} is the constant from Lemma S11. If $r_0 > 0$, then

$$L_Z \lesssim s_0 \log p, \tag{S47}$$

$$\tau_B^{-2}\, \omega_A(C_{\mathcal{T}})^2 \lesssim s_0 \log p. \tag{S48}$$

Lemma S11. *Let $m_{\boldsymbol{G},\tau}(\boldsymbol{x})$ be as defined in (S23). There exist constants $C_{sh} < \infty$ and $a_{sh} < \infty$ such that, for all $r \geq 1$, $\tau > 0$, and $\boldsymbol{x}, \boldsymbol{h} \in \mathbb{R}^r$,*

$$\frac{m_{\boldsymbol{G},\tau}(\boldsymbol{x}-\boldsymbol{h})}{m_{\boldsymbol{G},\tau}(\boldsymbol{x})} \leq \exp\left\{\sum_{\ell=1}^{r} \log\left[C_{sh}\log(e+\tau\sqrt{\boldsymbol{G}_{\ell\ell}})\right] + a_{sh}\tau^{-1}\|\boldsymbol{h}\|_1\right\}. \tag{S49}$$

The reverse lower bound also holds:

$$\frac{m_{\boldsymbol{G},\tau}(\boldsymbol{x}-\boldsymbol{h})}{m_{\boldsymbol{G},\tau}(\boldsymbol{x})} \geq \exp\left\{-\sum_{\ell=1}^{r} \log\left[C_{sh}\log(e+\tau\sqrt{\boldsymbol{G}_{\ell\ell}})\right] - a_{sh}\tau^{-1}\|\boldsymbol{h}\|_1\right\}. \tag{S50}$$

Lemma S12. *Under Conditions (C1) – (C3), there exists a measurable set $\mathcal{L}_0$ such that*

$$\mathcal{L}_0 \subset \{\boldsymbol{\beta} : \mathcal{S}_{\boldsymbol{\beta}} = \mathcal{S}_0\},\ \sup_{\boldsymbol{\beta}\in\mathcal{L}_0} Q_0(\boldsymbol{\beta}) \lesssim s_0 \log p,\ \text{and}\ -\log \Pi_\beta(\mathcal{L}_0) \lesssim s_0 \log p.$$

Lemma S13. *Fix $C_{\mathcal{T}} < \infty$ and suppose Condition (C1) holds. Then there exists an event $\Omega_{\mathrm{sc}}(C_{\mathcal{T}})$ with $\mathbb{P}_0\{\Omega_{\mathrm{sc}}(C_{\mathcal{T}})\} \to 1$ on which*

$$\left|\boldsymbol{\varepsilon}'\boldsymbol{M}_0\widetilde{\boldsymbol{X}}_{\mathcal{T}}\boldsymbol{u}\right| \leq \frac{1}{16}\|\boldsymbol{M}_0\widetilde{\boldsymbol{X}}_{\mathcal{T}}\boldsymbol{u}\|_2^2 + Cs_0\log p \tag{S51}$$

for all $\mathcal{T} \subset [p]$, $|\mathcal{T}| \leq C_{\mathcal{T}} s_0$ and all $\boldsymbol{u} \in \mathbb{R}^{|\mathcal{T}|}$.

Lemma S14. *Assume Conditions (C1) and (C4) hold. Define $\Psi(h) = L_Z + a_{sh}\tau_B^{-1}\|h\|_1$ for $h \in \mathbb{R}^{r_0}$, where a_{sh} is the constant from Lemma S11. Then*

$$\exp\{-\Psi(h)\} \leq S_Y(h) \leq \exp\{\Psi(h)\}, \qquad h \in \mathbb{R}^{r_0}, \tag{S52}$$

$$\Psi\{h_0(\boldsymbol{\beta})\} \leq \frac{1}{16}Q_0(\boldsymbol{\beta}) + Cs_0\log p, \qquad |\mathcal{S}_{\boldsymbol{\beta}} \cup \mathcal{S}_0| \leq C_{\mathcal{T}} s_0. \tag{S53}$$

S3.3.3 Proof of main theorem

Proof. From the definitions above, we see that $\boldsymbol{Z}_{\mathcal{R}_0}h_0(\boldsymbol{\beta}) = \boldsymbol{P}_0\widetilde{\boldsymbol{X}}(\boldsymbol{\beta}-\boldsymbol{\beta}_0)$. Denote $\tilde{\boldsymbol{b}}_{\mathcal{R}_0}(\boldsymbol{\beta},\boldsymbol{b}) \coloneqq \boldsymbol{b}_{\mathcal{R}_0} + h_0(\boldsymbol{\beta})$ for every $\boldsymbol{b}_{\mathcal{R}_0} \in \mathbb{R}^{r_0}$, we have

$$\widetilde{\boldsymbol{X}}\boldsymbol{\beta} + \boldsymbol{Z}_{\mathcal{R}_0}\boldsymbol{b}_{\mathcal{R}_0} - (\widetilde{\boldsymbol{X}}\boldsymbol{\beta}_0 + \boldsymbol{Z}_{\mathcal{R}_0}\tilde{\boldsymbol{b}}_{\mathcal{R}_0}) = \boldsymbol{M}_0\widetilde{\boldsymbol{X}}(\boldsymbol{\beta}-\boldsymbol{\beta}_0).$$

Under the true model $\boldsymbol{Y} = \widetilde{\boldsymbol{X}}\boldsymbol{\beta}_0 + \boldsymbol{Z}_{\mathcal{R}_0}\boldsymbol{b}_{0,\mathcal{R}_0} + \boldsymbol{\varepsilon}$,

$$p(\boldsymbol{Y} \mid \boldsymbol{\beta}, \boldsymbol{b}_{\mathcal{R}_0}) = \Lambda_0(\boldsymbol{\beta})\, p(\boldsymbol{Y} \mid \boldsymbol{\beta}_0, \tilde{\boldsymbol{b}}_{\mathcal{R}_0}),$$

where the likelihood ratio is free of $\boldsymbol{b}$ and is given by

$$\Lambda_0(\boldsymbol{\beta}) = \exp\left\{\boldsymbol{\varepsilon}'\boldsymbol{M}_0\widetilde{\boldsymbol{X}}(\boldsymbol{\beta}-\boldsymbol{\beta}_0) - \frac{1}{2}Q_0(\boldsymbol{\beta})\right\}. \tag{S54}$$

Denote the posterior event $\mathcal{A}$ as

$$\Pi_0(\mathcal{A} \mid \boldsymbol{Y}) = \frac{N(\mathcal{A};\boldsymbol{Y})}{D(\boldsymbol{Y})} \coloneqq \frac{\int_{\mathcal{A}}\int p(\boldsymbol{Y}|\boldsymbol{\beta},\boldsymbol{b}_{\mathcal{R}_0})g_{\tau_B,r_0}(\boldsymbol{b}_{\mathcal{R}_0})\,d\boldsymbol{b}_{\mathcal{R}_0}\,\Pi_\beta(d\boldsymbol{\beta})}{\int\int p(\boldsymbol{Y}|\boldsymbol{\beta},\boldsymbol{b}_{\mathcal{R}_0})g_{\tau_B,r_0}(\boldsymbol{b}_{\mathcal{R}_0})\,d\boldsymbol{b}_{\mathcal{R}_0}\,\Pi_\beta(d\boldsymbol{\beta})}.$$

We first lower bound $D(\boldsymbol{Y})$. Followed by the change of variables $\boldsymbol{t} = \boldsymbol{b}_{\mathcal{R}_0} + h_0(\boldsymbol{\beta})$,

$$\begin{aligned} D(\boldsymbol{Y}) &= \int \int p(\boldsymbol{Y}|\boldsymbol{\beta}, \boldsymbol{b}_{\mathcal{R}_0}) g_{\tau_B, r_0}(\boldsymbol{b}_{\mathcal{R}_0})\, d\boldsymbol{b}_{\mathcal{R}_0}\, \Pi_\beta(d\boldsymbol{\beta}) \\ &= \int \Lambda_0(\boldsymbol{\beta}) I_Y\{h_0(\boldsymbol{\beta})\}\, \Pi_\beta(d\boldsymbol{\beta}) \\ &\geq I_Y(0) \int_{\mathcal{L}_0} \Lambda_0(\boldsymbol{\beta}) S_Y\{h_0(\boldsymbol{\beta})\}\, \Pi_\beta(d\boldsymbol{\beta}), \end{aligned} \tag{S55}$$

Apply Lemma S13 with $C_{\mathcal{T}} = 1$. On $\Omega_{\text{sc}}(1)$, for $\boldsymbol{\beta} \in \mathcal{L}_0$, Lemma S13 and (S54) give

$$\log \Lambda_0(\boldsymbol{\beta}) \geq -\left(\frac{1}{2} + \frac{1}{16}\right) Q_0(\boldsymbol{\beta}) - C s_0 \log p = -\frac{9}{16} Q_0(\boldsymbol{\beta}) - C s_0 \log p.$$

Since $\mathcal{S}_{\boldsymbol{\beta}} = \mathcal{S}_0$ on $\mathcal{L}_0$, (S52) and (S53) give

$$\log S_Y\{h_0(\boldsymbol{\beta})\} \geq -\Psi\{h_0(\boldsymbol{\beta})\} \geq -\frac{1}{16} Q_0(\boldsymbol{\beta}) - C s_0 \log p.$$

Together with $Q_0(\boldsymbol{\beta}) \lesssim s_0 \log p$ on $\mathcal{L}_0$, (S55) yields

$$\begin{aligned} D(\boldsymbol{Y}) &\geq I_Y(0) \exp\{-C s_0 \log p\} \Pi_\beta(\mathcal{L}_0) \\ &\geq I_Y(0) \exp\{-C_d s_0 \log p\}. \end{aligned} \tag{S56}$$

Next, we upper bound $N(\mathcal{A}; \boldsymbol{Y})$. Similar to $D(\boldsymbol{Y})$,

$$N(\mathcal{A}; \boldsymbol{Y}) = I_Y(0) \int_{\mathcal{A}} \Lambda_0(\boldsymbol{\beta}) S_Y\{h_0(\boldsymbol{\beta})\}\, \Pi_\beta(d\boldsymbol{\beta}).$$

For any event $\mathcal{A}$, Lemma S14 (S52) gives

$$N(\mathcal{A}; \boldsymbol{Y}) \leq I_Y(0) \int_{\mathcal{A}} \Lambda_0(\boldsymbol{\beta}) \exp\{\Psi(h_0(\boldsymbol{\beta}))\}\, \Pi_\beta(d\boldsymbol{\beta}).$$

Combining this with (S56), on $\Omega_{\text{sc}}(1)$, gives the posterior-ratio inequality

$$\Pi_0(\mathcal{A} \mid \boldsymbol{Y}) \leq \exp\{C_d s_0 \log p\} \int_{\mathcal{A}} \Lambda_0(\boldsymbol{\beta}) \exp\{\Psi(h_0(\boldsymbol{\beta}))\}\, \Pi_\beta(d\boldsymbol{\beta}). \tag{S57}$$

Next, note that controlling the estimated dimension follows straightforwardly from Theorem 3.3. Fix any $M_4 \geq 1$ and define $\mathcal{B}_{\text{dim}}(M_4) = \{\boldsymbol{\beta} : s_{\boldsymbol{\beta}} > M_4 s_0\}$, we have that

$$\mathbb{E}_0 \Pi_0(\mathcal{B}_{\text{dim}}(M_4) \mid \boldsymbol{Y}) = o(1). \tag{S58}$$

Set $C_{\mathcal{T}} = M_4 + 1$ and

$$\Omega_\star = \Omega_{\text{sc}}(1) \cap \Omega_{\text{sc}}(C_{\mathcal{T}}).$$

Then $\mathbb{P}_0(\Omega_\star) \to 1$. Next define $\mathcal{B}_{\text{pred}}(M_5) = \{\boldsymbol{\beta} : s_{\boldsymbol{\beta}} \leq M_4 s_0,\ Q_0(\boldsymbol{\beta}) > M_5 s_0 \log p\}$. For $\boldsymbol{\beta} \in \mathcal{B}_{\text{pred}}(M_5)$, set $\mathcal{T} = \mathcal{S}_{\boldsymbol{\beta}} \cup \mathcal{S}_0$. Then $|\mathcal{T}| \leq (M_4 + 1) s_0 = C_{\mathcal{T}} s_0$. On $\Omega_\star$, Lemma S13 gives

$$\boldsymbol{\varepsilon}' \boldsymbol{M}_0 \widetilde{\boldsymbol{X}} (\boldsymbol{\beta} - \boldsymbol{\beta}_0) \leq \frac{1}{16} Q_0(\boldsymbol{\beta}) + C s_0 \log p.$$

Together with Lemma S14 (S53), this yields

$$\log\left[\Lambda_0(\boldsymbol{\beta})\exp\{\Psi(h_0(\boldsymbol{\beta}))\}\right] \le \left(\frac{1}{16}-\frac{1}{2}+\frac{1}{16}\right)Q_0(\boldsymbol{\beta}) + Cs_0\log p$$
$$= -\frac{3}{8}Q_0(\boldsymbol{\beta}) + Cs_0\log p.$$

Substituting the above into (S57), and using $\Pi_\beta(\mathcal{B}_{\text{pred}}(M_5)) \le 1$, gives on $\Omega_\star$

$$\begin{aligned}
&\Pi_0(\mathcal{B}_{\text{pred}}(M_5) \mid \boldsymbol{Y}) \\
&\le \exp\{C_d s_0 \log p\} \int_{\mathcal{B}_{\text{pred}}(M_5)} \exp\left\{-\frac{3}{8}Q_0(\boldsymbol{\beta}) + Cs_0\log p\right\} \Pi_\beta(d\boldsymbol{\beta}) \\
&\le \exp\left\{\left(C_4 - \frac{3}{8}M_5\right) s_0 \log p\right\}. \qquad \text{(S59)}
\end{aligned}$$

Choose $M_5 > 8(C_4+1)/3$. Combining (S59) with $\mathbb{P}_0(\Omega_\star^c) \to 0$ gives

$$\mathbb{E}_0\Pi_0(\mathcal{B}_{\text{pred}}(M_5) \mid \boldsymbol{Y}) \to 0.$$

The event $\{Q_0(\boldsymbol{\beta}) > M_5 s_0 \log p\}$ is contained in $\mathcal{B}_{\dim}(M_4) \cup \mathcal{B}_{\text{pred}}(M_5)$, thus (S58) and (S59) give

$$\mathbb{E}_0\Pi_0\{Q_0(\boldsymbol{\beta}) > M_5 s_0 \log p \mid \boldsymbol{Y}\} = o(1). \qquad \text{(S60)}$$

It remains to convert the prediction loss to the transfer-weighted active-coordinate loss. On $\{s_{\boldsymbol{\beta}} \le M_4 s_0\}$, again let $\mathcal{T} = \mathcal{S}_{\boldsymbol{\beta}} \cup \mathcal{S}_0$. Assumption (C1) gives

$$\begin{aligned}
Q_0(\boldsymbol{\beta}) &= \|\boldsymbol{M}_0\widetilde{\boldsymbol{X}}_{\mathcal{T}}(\boldsymbol{\beta}_{\mathcal{T}} - \boldsymbol{\beta}_{0,\mathcal{T}})\|_2^2 \ge \underline{\kappa}(\boldsymbol{\beta}_{\mathcal{T}} - \boldsymbol{\beta}_{0,\mathcal{T}})' D_0(\mathcal{T})(\boldsymbol{\beta}_{\mathcal{T}} - \boldsymbol{\beta}_{0,\mathcal{T}}) \\
&\ge \underline{\kappa} \sum_{j\in\mathcal{S}_0} n_j^{\text{eff}}(\beta_j - \beta_{0j})^2. \qquad \text{(S61)}
\end{aligned}$$

Set $M_3 = M_5/\underline{\kappa}$. If the weighted loss in (22) exceeds $M_3 s_0 \log p$ and $s_{\boldsymbol{\beta}} \le M_4 s_0$, then (S61) implies $Q_0(\boldsymbol{\beta}) > M_5 s_0 \log p$. Together with (S58) and (S60) proves (22). Finally, on $\{\mathcal{S}_{\boldsymbol{\beta}} = \mathcal{S}_0\}$,

$$\|\boldsymbol{\beta} - \boldsymbol{\beta}_0\|_2^2 \le \frac{1}{\min_{j\in\mathcal{S}_0} n_j^{\text{eff}}} \sum_{j\in\mathcal{S}_0} n_j^{\text{eff}}(\beta_j - \beta_{0j})^2,$$

so Theorem 3.3, together with (22), proves (23). The ℓ_1 bound follows from the usual Cauchy Schwarz inequality. □

S3.3.4 Proofs of lemmas

Proof of Lemma S10. Proof of (C1). First, note that $\boldsymbol{M}_0 = \text{diag}\{\boldsymbol{I}_{n_T}, \boldsymbol{M}^{(1)}, \ldots, \boldsymbol{M}^{(K)}\}$ where $\boldsymbol{M}^{(k)} = \boldsymbol{I} - \boldsymbol{P}_{\boldsymbol{X}^{(k)}_{\mathcal{R}_{0k}}}$. Consider any $\mathcal{S}$ such that $|\mathcal{S}| \le C_{\mathcal{T}} s_0$. Let $\mathcal{S}_k = \mathcal{S} \setminus \mathcal{R}_{0k}$, we have $\boldsymbol{M}^{(k)}\boldsymbol{X}^{(k)}_{\mathcal{S}}\boldsymbol{u} = \boldsymbol{M}^{(k)}\boldsymbol{X}^{(k)}_{\mathcal{S}_k}\boldsymbol{u}_{\mathcal{S}_k}$ by definition. It follows that $\|\boldsymbol{M}^{(k)}\boldsymbol{X}^{(k)}_{\mathcal{S}_k}\boldsymbol{u}_{\mathcal{S}_k}\|_2^2 \le \|\boldsymbol{X}^{(k)}_{\mathcal{S}_k}\boldsymbol{u}_{\mathcal{S}_k}\|_2^2 \le \overline{\kappa}\, n_k \|\boldsymbol{u}_{\mathcal{S}_k}\|_2^2$ by condition (B2). Summing over all K source studies and using $n_j^{\text{eff}} = n_T + \sum_{k:\,(k,j)\notin\mathcal{R}_0} n_k$, we have the following upper bound

$$\|\boldsymbol{M}_0\widetilde{\boldsymbol{X}}_{\mathcal{S}}\boldsymbol{u}\|_2^2 \le \overline{\kappa}\Big\{n_T\|\boldsymbol{u}\|_2^2 + \sum_{k=1}^K n_k\|\boldsymbol{u}_{\mathcal{S}_k}\|_2^2\Big\} = \overline{\kappa}\,\boldsymbol{u}' D_0(\mathcal{S})\boldsymbol{u}.$$

For the lower bound, since $\mathcal{S}_k$ and $\mathcal{R}_{0k}$ are disjoint,

$$\|\boldsymbol{M}^{(k)}\boldsymbol{X}^{(k)}_{\mathcal{S}_k}\boldsymbol{u}_{\mathcal{S}_k}\|_2^2 = \min_{\boldsymbol{v}\in\mathbb{R}^{r_{0k}}} \|X^{(k)}_{\mathcal{S}_k}u_{\mathcal{S}_k} - X^{(k)}_{\mathcal{R}_{0k}}v\|_2^2 \geq \underline{\kappa}\, n_k \min_{\boldsymbol{v}} \left(\|\boldsymbol{u}_{\mathcal{S}_k}\|_2^2 + \|\boldsymbol{v}\|_2^2\right) = \underline{\kappa}\, n_k \|\boldsymbol{u}_{\mathcal{S}_k}\|_2^2,$$

Summing over blocks gives the left inequality in (S43).
Proof of (C2). Condition (B2) implies $\underline{\kappa}\, n_k \leq \|\boldsymbol{X}^{(k)}\|^2 \leq \overline{\kappa}\, n_k$ for every source study, hence $\|\widetilde{\boldsymbol{X}}\|_\star \geq (\underline{\kappa}\, n_{\max})^{1/2}$. Since $n_j^{\text{eff}} \leq n \leq (K+1)n_{\max}$, (P-B2) gives

$$\tau_\beta\sqrt{n_j^{\text{eff}}} \leq \frac{p^{L_3}}{c_\tau\|\widetilde{\boldsymbol{X}}\|_\star}\sqrt{(K+1)n_{\max}} \leq c_\tau^{-1}p^{L_3}\left(\frac{K+1}{\underline{\kappa}}\right)^{1/2},$$

so $\log_+\left(\tau_\beta\sqrt{n_j^{\text{eff}}}/\log p\right) \lesssim \log p$ for sufficiently large p, which proves (S44). (S45) is implied by (B3).
Proof of (C3). This is the display following (P-B1) (also see Example 2.2 of [2]; Section 2.2, equation (3) of [5]) and follows directly from Lemma S7.
Proof of (C4). Notice that from definition, $(\boldsymbol{G}_0)_{jj} \leq \|\widetilde{\boldsymbol{X}}\|_\star^2$ for $j = 1, \ldots, r_0$. By (P-B2) $\tau_B\sqrt{(\boldsymbol{G}_0)_{\ell\ell}} \leq p^{L_3}$ and since we assumed $p \geq e^2$,

$$L_Z \leq r_0 \log\{2L_3C_{sh}\log p\} \lesssim s_\star \log\log p = o(\log p)$$

by (B1); since $s_0 \geq 1$, this proves (S47).

For (S48), fix $\mathcal{T}$ with $|\mathcal{T}| \leq C_{\mathcal{T}}s_0$ and $\boldsymbol{u} \neq 0$. Focusing on a particular block $\mathcal{R}_{0k}$ of $\boldsymbol{A}_0$, let $\boldsymbol{P}^{(k)}$ denote the projection onto $col(\boldsymbol{X}^{(k)}_{\mathcal{R}_{0k}})$, we have that $(\boldsymbol{A}_{0,\mathcal{T}}\boldsymbol{u})_{\mathcal{R}_{0k}} = (\boldsymbol{G}_0^{(k)})^{-1}\boldsymbol{X}^{(k)\prime}_{\mathcal{R}_{0k}}\boldsymbol{X}^{(k)}_{\mathcal{T}}\boldsymbol{u}$, and

$$\left\|(\boldsymbol{G}_0^{(k)})^{-1}\boldsymbol{X}^{(k)\prime}_{\mathcal{R}_{0k}}\boldsymbol{X}^{(k)}_{\mathcal{T}}\boldsymbol{u}\right\|_2 \leq (\underline{\kappa}\, n_k)^{-1/2}\left\|\boldsymbol{P}^{(k)}\boldsymbol{X}^{(k)}_{\mathcal{T}}\boldsymbol{u}\right\|_2 \leq \left(\frac{\overline{\kappa}}{\underline{\kappa}}\right)^{1/2}\|\boldsymbol{u}\|_2,$$

by (B2) applied twice. Converting each block to the ℓ_1 norm and applying the Cauchy–Schwarz inequality,

$$\begin{aligned}\|\boldsymbol{A}_{0,\mathcal{T}}\boldsymbol{u}\|_1 &\leq \sum_{k=1}^K (r_{0k})^{1/2}\left(\frac{\overline{\kappa}}{\underline{\kappa}}\right)^{1/2}\|\boldsymbol{u}\|_2 \\ &\leq \left(\frac{K\overline{\kappa}\, r_0}{\underline{\kappa}}\right)^{1/2}\|\boldsymbol{u}\|_2 \leq \left(\frac{K\overline{\kappa}\, r_0}{\underline{\kappa}\, n_T}\right)^{1/2}\{\boldsymbol{u}' D_0(\mathcal{T})\boldsymbol{u}\}^{1/2},\end{aligned}$$

using $n_j^{\text{eff}} \geq n_T$. Hence $\omega_A(C_{\mathcal{T}})^2 \leq K(\overline{\kappa}/\underline{\kappa})\, r_0/n_T$ and, since $\tau_B^{-2} = \|\widetilde{\boldsymbol{X}}\|_\star^2/p^{2L_3} \leq \overline{\kappa}(K+1)\, n_{\max}/p^{2L_3}$,

$$\tau_B^{-2}\,\omega_A(C_{\mathcal{T}})^2 \leq K(K+1)\frac{\overline{\kappa}^2}{\underline{\kappa}}\frac{r_0 n_{\max}}{p^{2L_3}n_T} = o(1),$$

because $r_0 \leq s_\star = o(p)$ by (B1) and $n_{\max} = o(p)$, which proves (S48). □

Proof of Lemma S11. We first show that the conclusion holds for the univariate case. For $\alpha > 0$, define $M_\alpha(z) = \int_{\mathbb{R}} e^{-\alpha^2(t-z)^2/2}K_0(|t|)\, dt$. We show there is a finite constant C such that

$$M_\alpha(z-h) \leq C\log(e+\alpha)e^{C|h|}M_\alpha(z), \quad z, h \in \mathbb{R}. \tag{S62}$$

By substituting $s = t + h$ and Lemma S4,

$$\begin{aligned} M_\alpha(z-h) &= \int_{\mathbb{R}} e^{-\alpha^2(s-z)^2/2} K_0(|s-h|)\, ds \\ &\le \int_{\mathbb{R}} e^{-\alpha^2(s-z)^2/2} C e^{C|h|} K_0(|s|)\, ds + \int_{|s-h|\le 1} e^{-\alpha^2(s-z)^2/2} C \log(e/|s-h|)\, ds \end{aligned}$$

The first term integrates to $Ce^{C|h|} M_\alpha(z)$. The second term follows, after setting $u = s - h$,

$$C \int_{-1}^{1} e^{-\alpha^2(u-(z-h))^2/2} \log(e/|u|)\, du \le C \log(e+\alpha) \int_{-1}^{1} e^{-\alpha^2(u-(z-h))^2/2}\, du$$

by Lemma S16. On the other hand, by Lemma S4,

$$\begin{aligned} M_\alpha(z) &\ge \int_{-1}^{1} e^{-\alpha^2(u-(z-h))^2/2} K_0(|h+u|)\, du \\ &\gtrsim e^{-C|h|} \int_{-1}^{1} e^{-\alpha^2(u-(z-h))^2/2}\, du. \end{aligned}$$

Thus the second term is also bounded by a constant multiple of $\log(e+\alpha) e^{C|h|} M_\alpha(z)$. Since $\log(e+\alpha) \ge 1$, this proves (S62).
Now define $m_{a,\tau}(y) = \int_{\mathbb{R}} e^{-a(u-y)^2/2} g_\tau(u)\, du$. With $u = \tau s$, $z = y/\tau$, $r = h/\tau$, and $\alpha = \tau\sqrt{a}$, one has $m_{a,\tau}(y) = \pi^{-1} M_\alpha(z)$. Therefore (S62) gives

$$m_{a,\tau}(y-h) \le C \log(e + \tau\sqrt{a}) \exp\{C|h|/\tau\} m_{a,\tau}(y). \tag{S63}$$

Next we extend the result to $r > 1$. For any $1 \le \ell \le r$, notice that

$$(\boldsymbol{u}-\boldsymbol{x})' \boldsymbol{G} (\boldsymbol{u}-\boldsymbol{x}) = \boldsymbol{G}_{\ell\ell} \{\boldsymbol{u}_\ell - \mu_\ell(\boldsymbol{u}_{-\ell}, \boldsymbol{x})\}^2 + R_\ell(\boldsymbol{u}_{-\ell}, \boldsymbol{x}_{-\ell}),$$

where $\mu_\ell(\boldsymbol{u}_{-\ell}, \boldsymbol{x}) = \boldsymbol{x}_\ell - \boldsymbol{G}_{\ell\ell}^{-1} \boldsymbol{G}_{\ell,-\ell}(\boldsymbol{u}_{-\ell} - \boldsymbol{x}_{-\ell})$ and R_ℓ is a residual term free of $\boldsymbol{u}_\ell$. Thus,

$$m_{\boldsymbol{G},\tau}(\boldsymbol{x}) = \int_{\mathbb{R}^{r-1}} e^{-R_\ell(\boldsymbol{u}_{-\ell}, \boldsymbol{x}_{-\ell})/2}\, m_{\boldsymbol{G}_{\ell\ell},\tau}\{\mu_\ell(\boldsymbol{u}_{-\ell}, \boldsymbol{x})\} \prod_{j \ne \ell} g_\tau(u_j)\, d\boldsymbol{u}_{-\ell}.$$

Now replace $\boldsymbol{x}$ by $\boldsymbol{x} - h_\ell \boldsymbol{e}_\ell$, where $\boldsymbol{e}_\ell$ is the ℓ-th coordinate vector. Since R_ℓ is unchanged and $\mu_\ell(\boldsymbol{u}_{-\ell}, \boldsymbol{x} - h_\ell e_\ell) = \mu_\ell(\boldsymbol{u}_{-\ell}, \boldsymbol{x}) - h_\ell$, applying (S63) with $a = \boldsymbol{G}_{\ell\ell}$, $y = \mu_\ell(\boldsymbol{u}_{-\ell}, \boldsymbol{x})$, and $h = h_\ell$, gives

$$m_{\boldsymbol{G}_{\ell\ell},\tau}\{\mu_\ell(\boldsymbol{u}_{-\ell}, \boldsymbol{x}) - h_\ell\} \le C \log(e + \tau\sqrt{\boldsymbol{G}_{\ell\ell}}) \exp\{C|h_\ell|/\tau\} m_{\boldsymbol{G}_{\ell\ell},\tau}\{\mu_\ell(\boldsymbol{u}_{-\ell}, \boldsymbol{x})\}$$

and thus the one-coordinate bound

$$m_{\boldsymbol{G},\tau}(\boldsymbol{x} - h_\ell e_\ell) \le K_\ell\, m_{\boldsymbol{G},\tau}(\boldsymbol{x}),$$

where $K_\ell \coloneqq C \log(e + \tau\sqrt{\boldsymbol{G}_{\ell\ell}}) \exp\{C|h_\ell|/\tau\}$. Finally, define $\boldsymbol{x}^{(0)} \coloneqq \boldsymbol{x}$, $\boldsymbol{x}^{(k)} \coloneqq \boldsymbol{x} - \sum_{j=1}^{k} h_j e_j$, $1 \le k \le r$. Applying the univariate bound successively gives

$$m_{\boldsymbol{G},\tau}(\boldsymbol{x}^{(k)}) = m_{\boldsymbol{G},\tau}(\boldsymbol{x}^{(k-1)} - h_k e_k) \le K_k\, m_{\boldsymbol{G},\tau}(\boldsymbol{x}^{(k-1)}), \quad k = 1, \ldots, r.$$

Taking the product over $k = 1, \ldots, r$, we obtain

$$m_{\boldsymbol{G},\tau}(\boldsymbol{x} - \boldsymbol{h}) = m_{\boldsymbol{G},\tau}(\boldsymbol{x}^{(r)}) \leq \left(\prod_{k=1}^{r} K_k \right) m_{\boldsymbol{G},\tau}(\boldsymbol{x}).$$

Thus

$$\frac{m_{\boldsymbol{G},\tau}(\boldsymbol{x} - \boldsymbol{h})}{m_{\boldsymbol{G},\tau}(\boldsymbol{x})} \leq \exp\left\{ \sum_{k=1}^{r} \log\left[C_{sh} \log(e + \tau\sqrt{\boldsymbol{G}_{kk}}) \right] + a_{sh}\tau^{-1}\|\boldsymbol{h}\|_1 \right\}$$

for some constants C_{sh}, a_{sh}. This proves (S49). The lower bound follows from the upper bound applied with base point $\boldsymbol{x} - \boldsymbol{h}$ and shift $-\boldsymbol{h}$. Taking reciprocals then gives (S50). □

Proof of Lemma S12. The standard limiting forms of the bessel function K_0 at zero and at infinity imply the uniform lower bound

$$g_\tau(x) \gtrsim \tau^{-1}(1 + |x|/\tau)^{-1/2} \exp\{-|x|/\tau\}, \quad x \in \mathbb{R},\ \tau > 0. \tag{S64}$$

This follows from the lower bound in Lemma S4 above.
Define $\rho_j := (\log p / n_j^{\text{eff}})^{1/2} \wedge \tau_\beta$, $j \in \mathcal{S}_0$. Define the set

$$\mathcal{L}_0 = \left\{ \boldsymbol{\beta} : \boldsymbol{\beta}_{\mathcal{S}_0^c} = 0,\ \beta_j = \beta_{0j} + \operatorname{sign}(\beta_{0j}) t_j,\ 0 < t_j \leq \rho_j,\ j \in \mathcal{S}_0 \right\}.$$

For $\boldsymbol{\beta} \in \mathcal{L}_0$, let $u_j := \beta_j - \beta_{0j}$. Since $|u_j| \leq \rho_j \leq \sqrt{\log p / n_j^{\text{eff}}}$, assumption (C1) implies

$$Q_0(\boldsymbol{\beta}) \leq \overline{\kappa} \sum_{j \in \mathcal{S}_0} n_j^{\text{eff}} u_j^2 \lesssim s_0 \log p.$$

It remains to lower-bound $\Pi_\beta(\mathcal{L}_0)$. Since $\mathcal{L}_0$ uses exactly the support $\mathcal{S}_0$,

$$\Pi_\beta(\mathcal{L}_0) = \pi_p(s_0) \binom{p}{s_0}^{-1} \prod_{j \in \mathcal{S}_0} \int_0^{\rho_j} g_{\tau_\beta}\{\beta_{0j} + \operatorname{sign}(\beta_{0j}) t\}\, dt. \tag{S65}$$

For $0 < t \leq \rho_j$, $|\beta_{0j} + \operatorname{sign}(\beta_{0j}) t| = |\beta_{0j}| + t$. Applying (S64),

$$\int_0^{\rho_j} g_{\tau_\beta}\{\beta_{0j} + \operatorname{sign}(\beta_{0j}) t\}\, dt \gtrsim \rho_j \tau_\beta^{-1} (2 + |\beta_{0j}|/\tau_\beta)^{-1/2} \exp\{-|\beta_{0j}|/\tau_\beta - 1\}. \tag{S66}$$

Assumption (C2) (S44) implies

$$\log\left(\frac{\rho_j}{\tau_\beta} \right) \geq -C \log p, \quad j \in \mathcal{S}_0.$$

If $\rho_j = \tau_\beta$, this is immediate. Otherwise, $\log(\rho_j/\tau_\beta) = -\log\{\tau_\beta \sqrt{n_j^{\text{eff}}}/\sqrt{\log p}\}$, so the same bound follows from (S44). Also,

$$\sum_{j \in \mathcal{S}_0} \log(2 + |\beta_{0j}|/\tau_\beta) \leq s_0 \log 2 + \tau_\beta^{-1} \|\boldsymbol{\beta}_{0,\mathcal{S}_0}\|_1 \lesssim s_0 \log p$$

by assumption (C2), (S45) and $p \geq e^2$. Combining (S46), $\binom{p}{s_0} \leq p^{s_0}$, (S65), and (S66) gives $-\log \Pi_\beta(\mathcal{L}_0) \lesssim s_0 \log p$, which completes the proof. □

Proof of Lemma S13. For any $\mathcal{T}$ such that $|\mathcal{T}| \le C_{\mathcal{T}} s_0$, define $\boldsymbol{g}_{\mathcal{T}} = D_0(\mathcal{T})^{-1/2}\widetilde{\boldsymbol{X}}_{\mathcal{T}}'\boldsymbol{M}_0\boldsymbol{\varepsilon}$. Note that $\boldsymbol{g}_{\mathcal{T}} \sim N(0, \boldsymbol{\Sigma}_{\mathcal{T}})$ where $\boldsymbol{\Sigma}_{\mathcal{T}} = D_0(\mathcal{T})^{-1/2}\widetilde{\boldsymbol{X}}_{\mathcal{T}}'\boldsymbol{M}_0\widetilde{\boldsymbol{X}}_{\mathcal{T}}D_0(\mathcal{T})^{-1/2}$. By assumption (C1), $\boldsymbol{u}'\boldsymbol{\Sigma}_{\mathcal{T}}\boldsymbol{u} \le \overline{\kappa}\|\boldsymbol{u}\|_2^2$ and thus $\lambda_{max}(\boldsymbol{\Sigma}_{\mathcal{T}}) \le \overline{\kappa}$.

From standard properties of the Gaussian distribution, $\|\boldsymbol{g}_{\mathcal{T}}\|_2^2 \overset{d}{=} \sum_{i=1}^{|\mathcal{T}|}\lambda_i z_i^2$ where $z_i \overset{iid}{\sim} N(0,1)$ and λ_i are eigenvalues of $\boldsymbol{\Sigma}_{\mathcal{T}}$. For $0 < t \le 1/(4\overline{\kappa})$,

$$\begin{aligned}\mathbb{E}_0[\exp(t\|\boldsymbol{g}_{\mathcal{T}}\|_2^2)] &= \prod_{i=1}^{|\mathcal{T}|}\mathbb{E}_0[\exp(t\lambda_i z_i^2)] \\ &= \prod_{i=1}^{|\mathcal{T}|}(1-2t\lambda_i)^{-1/2} \le (1-2t\overline{\kappa})^{-|\mathcal{T}|/2}\end{aligned}$$

Setting $t = 1/(4\overline{\kappa})$ and using Markov's inequality, $\mathbb{P}_0(\|\boldsymbol{g}_{\mathcal{T}}\|_2^2 > x) \le 2^{|\mathcal{T}|/2}\exp(-x/(4\overline{\kappa}))$. The number of sets $\mathcal{T}$ with $|\mathcal{T}| \le C_{\mathcal{T}} s_0$ satisfies $\sum_{q=0}^{C_{\mathcal{T}} s_0}\binom{p}{q} \le \exp(Cs_0\log p)$. Consequently, the union bound gives, with probability tending to one,

$$\sup_{\mathcal{T}:|\mathcal{T}|\le C_{\mathcal{T}} s_0}\|\boldsymbol{g}_{\mathcal{T}}\|_2^2 \lesssim s_0\log p.$$

Denote this event by $\Omega_{\text{sc}}(C_{\mathcal{T}})$. On this event, for any $\boldsymbol{u} \in \mathbb{R}^{|\mathcal{T}|}$, we have

$$\begin{aligned}|\boldsymbol{\varepsilon}'\boldsymbol{M}_0\widetilde{\boldsymbol{X}}_{\mathcal{T}}\boldsymbol{u}| &= |\boldsymbol{g}_{\mathcal{T}}'D_0(\mathcal{T})^{1/2}\boldsymbol{u}| \\ &\le \|\boldsymbol{g}_{\mathcal{T}}\|_2(\boldsymbol{u}'D_0(\mathcal{T})\boldsymbol{u})^{1/2} \\ &\le \|\boldsymbol{g}_{\mathcal{T}}\|_2\underline{\kappa}^{-1/2}\|\boldsymbol{M}_0\widetilde{\boldsymbol{X}}_{\mathcal{T}}\boldsymbol{u}\|_2 \\ &\le \frac{1}{16}\|\boldsymbol{M}_0\widetilde{\boldsymbol{X}}_{\mathcal{T}}\boldsymbol{u}\|_2^2 + Cs_0\log p\end{aligned}$$

where the second, third, and fourth lines follow from the Cauchy–Schwarz inequality, assumption (C1), and Young's inequality, respectively. This proves (S51). □

Proof of Lemma S14. If $r_0 = 0$, then $I_{\boldsymbol{y}}(\boldsymbol{h}) = I_{\boldsymbol{y}}(0)$, $S_Y(\boldsymbol{h}) = 1$, $h_0(\boldsymbol{\beta}) = 0$, and the claims are immediate. Assume $r_0 \ge 1$. Define $\widehat{\boldsymbol{b}}_{\mathcal{R}_0} := (\boldsymbol{Z}_{\mathcal{R}_0}^\top\boldsymbol{Z}_{\mathcal{R}_0})^{-1}\boldsymbol{Z}_{\mathcal{R}_0}^\top(\boldsymbol{y} - \widetilde{\boldsymbol{X}}\boldsymbol{\beta}_0)$, then

$$p(\boldsymbol{y} \mid \boldsymbol{\beta}_0, \boldsymbol{t}) = c(\boldsymbol{y})\exp\left\{-\frac{1}{2}(\boldsymbol{t} - \widehat{\boldsymbol{b}}_{\mathcal{R}_0})^\top\boldsymbol{G}_0(\boldsymbol{t} - \widehat{\boldsymbol{b}}_{\mathcal{R}_0})\right\},$$

where $c(\boldsymbol{y})$ does not depend on $\boldsymbol{t}$. Thus, $I_{\boldsymbol{y}}(\boldsymbol{h}) = c(\boldsymbol{y})m_{\boldsymbol{G}_0,\tau_B}(\widehat{\boldsymbol{b}}_{\mathcal{R}_0} - \boldsymbol{h})$, $I_{\boldsymbol{y}}(0) = c(\boldsymbol{y})m_{\boldsymbol{G}_0,\tau_B}(\widehat{\boldsymbol{b}}_{\mathcal{R}_0})$. Lemma S11 gives

$$\exp\{-\Psi(\boldsymbol{h})\} \le S_Y(\boldsymbol{h}) \le \exp\{\Psi(\boldsymbol{h})\},$$

which proves (S52).

Next, suppose $|\mathcal{S}_{\boldsymbol{\beta}} \cup \mathcal{S}_0| \le C_{\mathcal{T}} s_0$, and let $\mathcal{T} = \mathcal{S}_{\boldsymbol{\beta}} \cup \mathcal{S}_0$, $\boldsymbol{u} = (\boldsymbol{\beta}_{\mathcal{T}} - \boldsymbol{\beta}_{0,\mathcal{T}})$. By the definition of $\omega_A(C_{\mathcal{T}})$ and assumption (C1),

$$\|h_0(\boldsymbol{\beta})\|_1 \le \omega_A(C_{\mathcal{T}})\{\boldsymbol{u}^\top D_0(\mathcal{T})\boldsymbol{u}\}^{1/2} \le \omega_A(C_{\mathcal{T}})\underline{\kappa}^{-1/2}Q_0(\boldsymbol{\beta})^{1/2}.$$

Young's inequality therefore gives

$$a_{sh}\tau_B^{-1}\|h_0(\boldsymbol{\beta})\|_1 \le \frac{1}{16}Q_0(\boldsymbol{\beta}) + \frac{4a_{sh}^2}{\underline{\kappa}}\tau_B^{-2}\omega_A(C_{\mathcal{T}})^2.$$

By (S48), the second term is bounded by $Cs_0\log p$. Combining this with (S47) proves (S53). □

S3.4 Auxiliary results

Lemma S15. *Let $\boldsymbol{\varepsilon} \sim N_n(0, \boldsymbol{I}_n)$. For every $\boldsymbol{v} \in \mathbb{R}^n$, every positive semidefinite $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, and every $x > 0$,*

$$\mathbb{P}\{-\boldsymbol{v}'\boldsymbol{\varepsilon} > \|\boldsymbol{v}\|_2\sqrt{2x}\} \le e^{-x}, \tag{S67}$$

$$\mathbb{P}\{\boldsymbol{\varepsilon}'\boldsymbol{A}\boldsymbol{\varepsilon} > \operatorname{tr}(\boldsymbol{A}) + 2\sqrt{\operatorname{tr}(\boldsymbol{A}^2)x} + 2\|\boldsymbol{A}\|_{\mathrm{sp}}x\} \le e^{-x}. \tag{S68}$$

Proof. The first assertion is the Gaussian Chernoff bound. The second is Proposition 1 of Hsu, Kakade and Zhang [4]. □

Lemma S16. *Let $\ell(x) = \log(e/x)$ for $0 < x \le 1$. There exists a constant $C_1 < \infty$ such that, for every $\alpha > 0$ and $\mu \in \mathbb{R}$,*

$$\int_{-1}^{1} e^{-\alpha^2(x-\mu)^2/2}\ell(|x|)\,dx \le C_1 \log(e+\alpha)\int_{-1}^{1} e^{-\alpha^2(x-\mu)^2/2}\,dx. \tag{S69}$$

Proof. By symmetry we consider $\mu \ge 0$. For $x \in [0,1]$, define the measure $d\nu_\alpha(x) = e^{-\alpha^2x^2/2}dx$. By observing that

$$\int_{-1}^{1} e^{-\alpha^2(x-\mu)^2/2}dx = \int_{0}^{1} e^{-\alpha^2(x-\mu)^2/2}dx + \int_{0}^{1} e^{-\alpha^2(-x-\mu)^2/2}dx,$$

we have the following ratio:

$$\Lambda(\alpha,\mu) := \frac{\int_{-1}^{1} e^{-\alpha^2(x-\mu)^2/2}\ell(|x|)\,dx}{\int_{-1}^{1} e^{-\alpha^2(x-\mu)^2/2}\,dx} = \frac{\int_0^1 \cosh(\alpha^2\mu x)\ell(x)\,d\nu_\alpha(x)}{\int_0^1 \cosh(\alpha^2\mu x)\,d\nu_\alpha(x)}.$$

Since $\cosh(\alpha^2\mu x)$ is increasing and $\ell(x)$ is decreasing in x, the covariance inequality implies that

$$\Lambda(\alpha,\mu) \le \frac{\int_0^1 \ell(x)\,d\nu_\alpha(x)}{\int_0^1 d\nu_\alpha(x)} = \frac{\int_0^1 \log(e/x)e^{-\alpha^2x^2/2}\,dx}{\int_0^1 e^{-\alpha^2x^2/2}\,dx} =: R_\alpha.$$

We show that $R_\alpha \le C_1\log(e+\alpha)$ for some constant C_1.
Case 1: $0 < \alpha \le 1$. Then

$$\int_0^1 e^{-\alpha^2x^2/2}\,dx \ge \int_0^1 e^{-1/2}\,dx = e^{-1/2}, \quad \int_0^1 \log(e/x)e^{-\alpha^2x^2/2}\,dx \le \int_0^1 \log(e/x)\,dx = 2.$$

Hence $R_\alpha \le 2e^{1/2}$. Since $\log(e+\alpha) \ge 1$, this gives $R_\alpha \le C\log(e+\alpha)$ for $0 < \alpha \le 1$.
Case 2: $\alpha > 1$. The denominator satisfies

$$\int_0^1 e^{-\alpha^2x^2/2}\,dx \ge \int_0^{1/\alpha} e^{-\alpha^2x^2/2}\,dx \ge e^{-1/2}\alpha^{-1}.$$

For the numerator, split the integral as

$$\int_0^1 \log(e/x)e^{-\alpha^2x^2/2}\,dx = \int_0^{1/\alpha} \log(e/x)e^{-\alpha^2x^2/2}\,dx + \int_{1/\alpha}^1 \log(e/x)e^{-\alpha^2x^2/2}\,dx.$$

The first term is less than $\int_0^{1/\alpha} \log(e/x)\,dx$, which integrates to $\alpha^{-1}\log(e\alpha) + \alpha^{-1}$ and is bounded by $C\alpha^{-1}\log(e\alpha)$. For the second term, since $x \geq 1/\alpha$, we have $\log(e/x) \leq \log(e\alpha)$. Therefore, with $t = \alpha x$, we have that

$$\int_{1/\alpha}^{1} \log(e/x) e^{-\alpha^2 x^2/2}\,dx \leq \alpha^{-1}\log(e\alpha) \int_1^{\alpha} e^{-t^2/2}\,dt \leq C\alpha^{-1}\log(e\alpha).$$

Collecting terms and combine the denominator yields $R_\alpha \leq C' \log(e\alpha)$. The result follows from the fact that $\log(e\alpha) \leq C\log(e+\alpha)$ for $\alpha > 1$. □

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